Chapter 6

Inversion

The aim of this chapter is to illustrate the basic theory for the solution of the inverse problem associated with the propagation of surface waves. This is the final step of data interpretation when surface wave testing is used for site characterization. It involves a mathematical operation, called *inversion*, by which the experimental dispersion and/or attenuation curve is processed to obtain the unknown profile of shear wave velocity and/or shear damping ratio at the site.

The theory of inversion is a subject of fundamental importance in geophysics as well as in applied science and engineering because it is concerned with the inference of parameters characterizing a physical system or even a time-dependent process from a set of experimental measurements. Sophisticated imaging techniques used in medicine such as computerized X-ray tomography (CT), positron emission tomography (PET), and magnetic resonance imaging (MRI) are based on exploiting powerful algorithms for the solution of complicated nonlinear inverse problems. Also, the capabilities of nondestructive testing methods that are widespread in several areas of engineering are founded on the solution of inverse problems. Furthermore, inverse problems are a subject of great interest to mathematicians who developed a formal theory based on functional analysis and integral equations. Several monographs are dedicated to this subject (e.g., Menke 1989; Groetsch 1993; Parker 1994; Engl et al. 1996; Aster et al. 2005; Tarantola 2005; Kirsch 2011) with content and flavor that may be quite different depending on the background and the adopted perspective of the author(s).

The main focus of this chapter is the application of discrete inverse theory to surface wave measurements for near-surface site characterization. Background material on nomenclature, classification, and solution strategies of linear and nonlinear inverse problems is provided before showing applications specifically devoted to surface wave inversion. The chapter is subdivided into five main sections.

Section 6.1 illustrates some basic features and conceptual issues related to the theory of inverse problems as applied to geophysics and then

contextualizes them with reference to the parameter identification problem associated with surface waves. The inherent ill-posedness of inverse problems is discussed in conjunction with the strategies that may be adopted for its mitigation.

Section 6.2 reviews forward modeling associated with the propagation of surface waves and highlights its importance in the solution of the corresponding inverse problem. One-dimensional (1D), layered versus continuous models are discussed jointly with the discretization scheme adopted to represent the spatial variability of model parameters.

Section 6.3 illustrates a few algorithms commonly used to solve geophysical inverse problems starting with the steady-state Rayleigh method (SSRM), an empirical procedure adopted in the early days of surface wave testing. The SSRM was the precursor of modern spectral analyses of surface waves (SASW) and multichannel analyses of surface waves (MASW) testing.

Section 6.4 is the core of the chapter, and it is devoted to illustrating some of the details of the standard theory for the solution of linear and nonlinear inverse problems associated with surface waves that are relevant for the inversion of the experimental attenuation and dispersion curve, respectively. After reviewing a few definitions on the goodness of data fitting, the elegant and efficient formalism of the Moore–Penrose generalized inverse matrix is introduced to systematically solve the linear inverse problem. This allows us to address the solution of all categories of linear inverse problems, namely the under-, over-, and mixed-determined problems. Mitigation of ill-posedness is then discussed through the introduction of zeroth and higher-order Tikhonov regularization methods. Treatment and solution of nonlinear inverse problems are illustrated in conjunction with the uncoupled and joint inversion of the Rayleigh dispersion curve. Several classes of inversion algorithms are subsequently presented with an in-depth discussion of Occam's algorithm. The section ends with a brief discussion of the importance of a priori information in surface wave inversion.

Section 6.5 is concerned with the assessment of the influence that experimental errors in surface wave measurement have on the inversion process. The sources of uncertainty in determining the experimental dispersion and attenuation curves are examined, followed by an application of the first-order second-moment (FOSM) reliability method to determine how this uncertainty of the experimental data is projected, via the inversion algorithm, into uncertainty of model parameters, which are the shear wave velocity and shear damping ratio profiles at a site. The section concludes with a brief discussion on the importance that awareness of the inherent trade-off between model resolution and uncertainty has when solving an inverse problem, specifically that associated with surface waves, and is followed by an outline on the use of the Bayesian approach as an alternative method (with respect to the classical frequentist approach) to solve inverse problems.

6.1 CONCEPTUAL ISSUES

6.1.1 Forward and inverse problems in geophysics

Given the set of medium parameters $\{\lambda = \lambda(x_2), \mu = \mu(x_2), \rho = \rho(x_2)\}$ defining the physical and mechanical properties of a site and their variability with depth (see Figure 2.19), the problem of determining the dispersion and attenuation curves $V_R(\omega)$ and $\alpha_R(\omega)$ associated with that site is often referred to as the Rayleigh *direct* or *forward* problem. Conversely, if $V_R(\omega)$ and $\alpha_R(\omega)$ are given, then the problem of determining the unknown medium parameters $\{\lambda = \lambda(x_2), \mu = \mu(x_2), \rho = \rho(x_2)\}$ defines the Rayleigh *backward* or *inverse* problem.

If the aforementioned Rayleigh forward problem is interpreted as a *deterministic–mechanistic process*, the dispersion and attenuation curves can be viewed as special types of "response functions" of a physical system to a given "excitation." The system is represented by the soil deposit, whereas the excitation may be small perturbations of the initial equilibrium conditions of the system (free-vibration problem) or a source with a certain geometry and time variation (e.g., vertical, time-harmonic point or line force, Dirac-type impulse). In this interpretation, direct problems are concerned with determining the effects (i.e., the response function) induced on a physical system by certain causes (i.e., the excitation). In inverse problems, the roles of causes and effects are reversed, and the objective is to determine the causes that generated the observed effects (Engl 1993; Groetsch 1993).

If the deterministic–mechanistic process is viewed as a formal mapping $G{:}G(m) = d$, the forward problem corresponds to the problem of computing d from knowing G and m. In this mapping, G is a *mathematical operator* representing the model of the deterministic process, m is the excitation, and d is the response. Numerical simulations are nothing but approximate solutions of forward mathematical problems.[01] As for the inverse problem, there are two options: the first is the *causative problem* (inverse problem of category 1), which corresponds to determining m given G and d, and the second is the *model-identification problem* (inverse problem of category 2), where the objective is to compute G from knowing m and d. Figure 6.1 shows these alternatives graphically.

The inverse problem associated with the propagation of surface waves belongs to the inverse problem of category 2. *Any* problem concerned with geophysical prospecting and seismological studies of the Earth's structure where the objective is to determine some type of geological information about

[01] In the theory of linear systems, the forward problem is described by the *convolution* operator between an *input* signal, $m(t)$, and the *impulse response* function $G(\delta(t))$, which is the response of the system to a Dirac delta excitation $\delta(t)$. The symbol $d(t)$ denotes the *output* signal. In the frequency domain, the impulse response function $G(\delta(t))$ becomes the *transfer* function of the system.

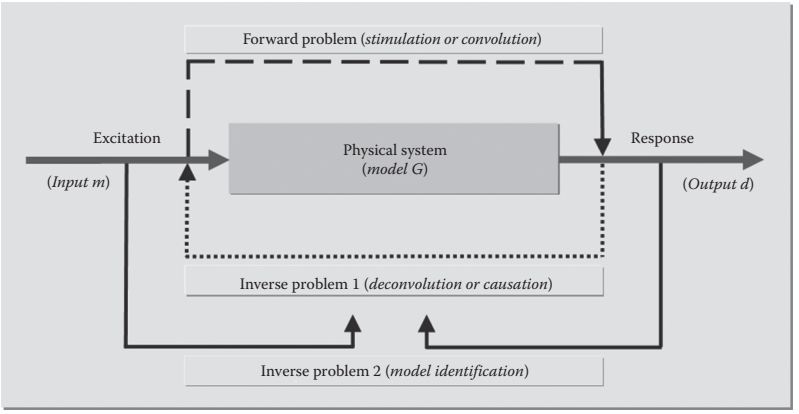

Figure 6.1 Solution of forward and inverse problems associated with a physical system. The problem is described through a deterministic process *G* formalized by the mapping *G*: *G*(*m*) = *d*, where *G* is the mathematical operator, *m* is the excitation, and *d* is the response. (Modified from Lai, C. G., *Surface Waves in Geomechanics*: *Direct and Inverse Modelling for Soil and Rocks,* Springer-Verlag, New York, 2005.)

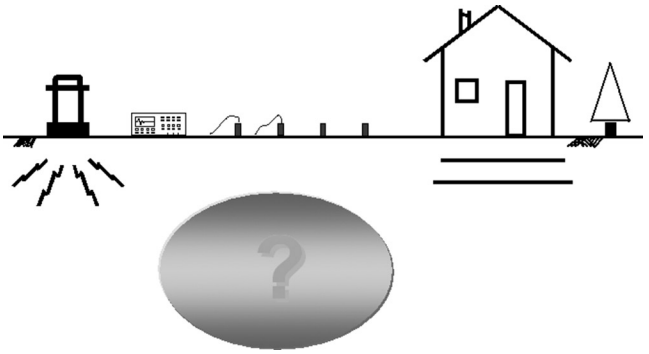

Figure 6.2 Essence of the geophysical inverse problem. Find one or more unknown geometrical (e.g., shape and size of a buried anomaly or the spatial variability of the contacts among different geological units) and/or mechanical (e.g., shear and constrained moduli) fields in the interior of the ground surface from measurements at its boundary and/or along boreholes (cross-well tomography).

the Earth's interior from measurements at its surface (Figure 6.2) belongs to the same category. Category 2 also includes the geodetic inverse problem of determining the shape of the Earth from gravimetric measurements.

Instead, the problem of retrieving information on source parameters of an earthquake from the analysis of recorded seismograms data belongs to

inverse problems of category 1, which is also called the category of *deconvolution problems*. Determining the input signal of a seismometer from the knowledge of the output signal and the seismometer response is another example of a deconvolution problem.

Model-identification is often denoted interchangeably as a *parameter-identification* or *parameter-estimation problem* because the model G of a physical system or a process[02] is usually defined a priori from the laws of physics. Consequently, model-identification actually becomes parameter-identification because the original objective has been converted to a problem of determining a discrete set of model parameters.[03]

In recent years, several important inverse problems arising in modern science and engineering have spurred a renewed interest for inverse problem theory, the developments and ramifications of which, even in the restricted field of geophysical prospecting, have been spectacular. With regard to surface waves, solution of the associated parameter identification problem lies at the very core of the success achieved by sophisticated techniques used in nondestructive testing and in geophysical–geotechnical site characterization, which are in essence powerful solvers of inverse problems.

6.1.2 Ill-posedness of inverse problems

Inverse problems are generally known to be *ill-posed* or *unstable*, which is perhaps their most relevant feature if compared with the *well-posedness* of the corresponding forward problems (Engl 1993; Kirsch 2011). According to Hadamard (1923), a mathematical problem is said to be *well-posed* or *stable* if it satisfies the following three conditions:[04]

1. For all admissible data, a solution exists (*existence*).
2. For all admissible data, the solution is unique (*uniqueness*).
3. The solution depends continuously on the data (*stability*).

If any of these three conditions fails to hold, Hadamard called the problem *ill-posed*.[05] In reality, the *Hadamard's postulates* of well-posedness apply

[02] Typically, the process is represented by a system of partial differential equations, integral equations, and/or integro-differential equations and their associated initial and/or boundary conditions. Sometimes G is an algebraic operator that may be reduced to a matrix.

[03] In general, these are the constant or variable coefficients of the aforementioned differential/integral equations.

[04] A precise definition of *well-posedness* should also specify the functional space in which the solution is supposed to exist and the restrictions that a given set of data must satisfy to be considered admissible.

[05] At the origin of the Hadamard's definition of *well-posedness* is the idea that a mathematical model of a physical problem *must always* be well-posed. It was later recognized, however, that there are physical phenomena that do not necessarily satisfy this requirement (e.g., quantum physics).

to both *forward* and *inverse* problems. However, it is only in recent years that the importance of *instability* in the solution of forward problems has been widely recognized in studies of nonlinear dynamics and chaos theory[06] where small perturbations in the initial data produce unpredictable changes in the solution (Parker 1994).

In inverse problems, the two conditions that are most often violated are *uniqueness* and *stability*.[07] Of particular relevance in the solution of parameter identification problems is violation of uniqueness that is the existence for a given problem and set of data of more than one solution. In the deterministic–mechanistic interpretation of a forward problem given in Section 6.1.1 (Figure 6.1), this would correspond to a physical process where distinct causes may yield the same effect, a situation that is not so uncommon in physics and engineering. For instance, given a distribution of mass inside the Earth, one can uniquely predict the gravity field around the planet (forward problem). However, there are several different distributions of mass in the Earth that give exactly the same gravity field. Thus, the problem of inferring the mass distribution from the observation of the gravity field (inverse problem) has multiple solutions (Tarantola 2005).

An example drawn from structural engineering is that of a loaded, elastic beam. Although for a given loading configuration a unique deflected shape of the beam can be predicted (forward problem), the same deflected shape may be obtained from different loading patterns. Hence, the problem of determining the load configuration corresponding to a deflected shape (inverse problem) is *ill-posed* because the solution is not unique.

For the surface wave inverse problem, nonuniqeness implies that a given experimental dispersion (or attenuation) curve may correspond to more than one shear wave velocity (or material damping ratio) profile of the soil deposit.

From a mathematical point of view, nonuniqeness in the solution of an inverse problem is caused by a lack of sufficient information to constrain the solution. Alternatively, the available information is not completely independent, at least in certain regions of the domain of definition of the solution[08] (solution space).

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[06] A classical example is the *butterfly paradigm* where a butterfly flapping its wings in New York generates a hurricane in San Francisco! Even the messing up of in an electronic, typewritten document produced by the introduction of an extra blank character is an example of *instability* of a forward problem; a small perturbation (the extra blank character) may have "catastrophic" consequences in the document formatting with page jumps, spoiling of figures, associated captions, and so on.

[07] This happens particularly in nonlinear problems such as the inverse problem associated with the propagation of surface waves, which will be discussed later in this chapter.

[08] A quotation attributed to Lanczos (1961) states that "A lack of information in the solution of a problem cannot be remedied by any mathematical trickery."

Two strategies can be used to enforce uniqueness in the solution of an inverse problem. The first is to add *a priori information* about the solution of the problem. For the parameter identification problem of surface waves, this might be, for instance, independent knowledge of model parameters in one or more layers (obtained, for example, from certain geotechnical tests) or the layer thickness obtained from borehole logs. Mass density and Poisson ratio are model parameters usually assumed to be known in advance because of the weak sensitivity of the dispersion (and attenuation) curve with respect to variations of these parameters (Nazarian 1984).

Another strategy to enforce uniqueness is to constrain the solution to satisfy certain requirements such as *smoothness* and *bounds*. In some cases, adding a priori information to the solution may be regarded as a constraint. An obvious example is that of requiring the model parameter to vary within a prescribed range or to be nonnegative (e.g., material damping ratio). There are, however, constraints of a different nature because they enforce features of *global* behavior to the solution.

In discussing strategies to enforce uniqueness, it should be remarked that the available methods are relatively simple for ideal, error-free measurements; however, the situation is more complicated for data containing bias and random errors, as will be discussed in Section 6.5.

Violation of the *stability* condition in the Hadamard's definition of well-posedness is also an important concern in the solution of forward and inverse problems. For instance, the solution of the *Fredholm integral equation of the first-kind* is very sensitive to small perturbations of the initial data caused by noise and unavoidable measurement errors, which may result in very large changes in the solution (Groetsch 1993). It is important to emphasize that the instability of these problems is a feature inherent to their nature and have nothing to do with a particular type of numerical algorithm used to solve them. For *linear* inverse problems with discrete and continuous linear operators, a *stability analysis* is usually carried out through the method of *singular-value decomposition*[09] (Strang 1988) as will be shown in detail in Section 6.4.

Very unstable parameter identification problems can be solved using mathematical techniques known as *regularization methods* that approximate the ill-posed problem with a parameter-dependent family of neighboring well-posed problems (Tikhonov and Arsenin 1977). Because some of these regularization methods admit a variational formulation (e.g., Tikhonov regularization) where the objective is the minimization of appropriate functionals, they can also be applied successfully for the solution of nonlinear inverse problems. Sections 6.4.2 and 6.4.3 will discuss Tikhonov regularization methods in some detail.

[09] Using this method, the *smallest singular value* controls the amplification of the perturbation errors, whereas the rate of decay of the singular values arranged in order of decreasing magnitude is used as a measure to quantify the *degree of instability* of a given inverse problem.

6.1.3 Inversion strategies: Local versus global methods

Solving the parameter identification problem associated with surface wave motion using the dispersion data as response functions is equivalent to solving an *inverse eigenvalue problem* or an *inverse spectral problem*[10] (Kirsch 2011). In fact, the objective is to determine certain coefficients of the two sets of first-order, linear ordinary differential equations (see Equations 2.70 and 2.71) from the knowledge of their eigenvalues ω/V_R (or ω/V_L).

In practice, a parameter identification problem is usually solved by converting it into a *parameter optimization* problem, the solution of which is then found from the stationary condition of an unconstrained or constrained functional (Parker 1994). In general, the techniques used to solve nonlinear optimization problems such as the inversion of an experimental dispersion curve in surface wave testing can be broadly divided into *global-search* (GS) and *local-search* (LS) methods. The functional of a nonlinear optimization problem will have, in fact, several stationary points in the solution space, and a question arises about finding the global extremum. LS procedures are iterative schemes that, starting from an initial guess of the solution, generate a sequence of improved approximations converging under suitable conditions to the solution. Most LS methods are calculus-based techniques that linearize a nonlinear functional at each iteration until a stationary point is reached. These techniques require the functional to be sufficiently smooth so that its *Fréchet*[11] derivatives (with respect to model parameters) exist and are continuous. Furthermore, even if all the smoothness requirements for the functional are satisfied, the sequence of approximations of the solution is guaranteed to converge only if the initial guess is sufficiently close to the solution. However, the most important limitation of LS procedures is that even when they succeed in finding a stationary point, there are no simple means to determine whether it is a local or a global stationary point in the solution space.

This dilemma is addressed by GS procedures, which are optimization techniques where the search for a global stationary point is conducted through an exploration of the entire solution space. This can be done either systematically by defining a grid or randomly as in Monte Carlo simulations. Other GS methods include simulated annealing and genetic inversion. Usually GS techniques are computationally more expensive than LS methods; however, they are more robust and reliable in finding the global extremum in the solution space. A further discussion on the differences between LS and GS techniques is reported in Section 6.4.3.

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[10] The study of *inverse eigenvalue problems* is a standard topic in inverse theory. A classical paper on the subject is Kac (1966).

[11] The Fréchet derivative is defined on Banach spaces, and it is used to define the derivative of a functional needed for the solution of an optimization problem using the calculus of variations.

With regard to the parameter identification problem associated with surface waves, ordinarily the strategies adopted for the inversion of surface wave data belong to the category of LS procedures using modal or apparent dispersion and attenuation curves as response functions. The dispersion curves are usually computed from measurements of surface wave phase velocity, although group velocities have also been used. Typically, the dispersion curves are calculated with respect to the fundamental mode of propagation, which implicitly assumes that this mode governs the experimental dispersion curve measured at the site. The approach is substantially correct only if the fundamental mode is effectively dominating the measured seismograms such as what happens in *normally dispersive* soil deposits where the mechanical impedance increases regularly with depth (see Chapter 2). However, there are circumstances in which the experimental dispersion curve also reflects the contribution of higher modes. In this case, what is really measured in the field is not a "modal" but rather an "apparent" dispersion curve or a combination of both (see Section 2.4). The shape of these experimental curves is strongly influenced by the superposition of various modes of propagation. It should be pointed out, however, that the nature of the measured dispersion curve depends not only on the characteristics of the subsurface, but it is also affected by the configuration of the receivers adopted during acquisition (see Chapter 3). Typical examples where the distinction between modal and apparent dispersion curves is relevant include *inversely dispersive* soil deposits, which are media where the mechanical impedance varies irregularly and/or abruptly with depth (e.g., soil profiles characterized by the presence of a top stiff layer). In these situations, the solution of the inverse problem must properly take into account modal superposition. O'Neill (2004) shows an example of the use of such a type of inversion algorithm.

Dispersion and attenuation curves each represent one possible type of response function; however, there are other choices, in principle. In the frequency domain, these include displacement amplitudes and phase spectra. In the time domain, they might be the seismograms. The ability to successfully invert the response functions measured at the free surface of the ground to determine reliable estimates of model parameters in the interior depends to a significant degree on the response function selected to describe the medium response to a dynamic excitation. Factors governing the selection of appropriate response functions include the ability to experimentally measure the field variables, the capability to solve the corresponding inverse problem,[12] and the information content and sensitivity associated with the selected response function with respect to the desired model parameters.

[12] This also depends on whether an analytical or a numerical procedure is used to compute the *Jacobian* of the response function with respect to model parameters.

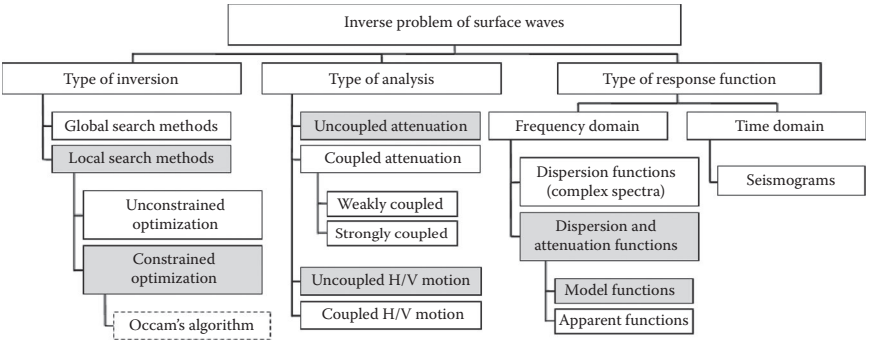

Figure 6.3 Algorithms for the solution of the parameter identification problem associated with surface waves. The shaded boxes indicate the most commonly used methods in geophysical–geotechnical prospecting to date. (Modified from Lai, C. G., Simultaneous inversion of Rayleigh phase velocity and attenuation for near-surface site characterization, PhD Thesis, Georgia Institute of Technology, 1998.)

Figure 6.3 shows a sketch illustrating the combination of some of the possible algorithms that could be conceived for the solution of the parameter identification problem associated with surface waves.

Coupled and uncoupled H/V motion denotes the kinematics of surface wave motion considered in the inversion analysis. Usually, only the vertical component of Rayleigh particle motion is taken into account. Love waves or horizontal components of Rayleigh waves are rarely used even if it would be desirable to increase the information content of measured surface wave data (Tokimatsu 1995; Strobbia 2003).

6.2 FORWARD MODELING

In geophysics and other applied sciences, the ability to solve a parameter-identification problem often relies on the capacity to actually solve the corresponding forward problem. This is particularly true in the case of nonlinear problems such as the inversion of Rayleigh or Love dispersion curves. Solution of the forward problem in turn implies a definition of a mathematical model of the physical system under study (Figure 6.1). In surface wave modeling for near-surface site characterization, the issues to be addressed in defining the mathematical model are, at a minimum, the following:

- Geometrical modeling of the subsurface
- Numerical modeling of the subsurface
- Constitutive modeling of geomaterials

Geometrical modeling of the subsurface is concerned with the idealization related to the spatial variability of the physical properties of the medium. If the soil deposit is assumed laterally homogeneous and the physical properties are considered to vary only in the direction of gravity, then the geometrical model may be unidimensional. Two-dimensional (2D) or three-dimensional (3D) models are mathematical idealizations in which the properties are also allowed to vary in the horizontal direction. Unidimensional modeling is the standard in surface wave testing, although 2D profiling has already been proposed in advanced applications.

Numerical modeling of the subsurface refers to the discretization scheme adopted for the representation of the spatial variability of the field variables. In unidimensional models, a vertically inhomogeneous medium is usually idealized by a stack of homogeneous layers, representing different geological units, overlying a homogeneous half-space. This is a *discontinuous* discretization scheme in contrast to a subsurface model in which the material properties vary continuously with depth (Rix and Lai 2014)[13] (Figure 6.4).

In surface wave testing, the choices for geometrical and numerical modeling have important implications in the ability to successfully invert dispersion and attenuation data. Selection of a model that is not consistent with the real subsurface conditions may lead to grossly erroneous results. For instance, the adoption of a unidimensional geometrical model at a site characterized by strong lateral variability in physical and mechanical properties would make the interpretation of surface wave measurements severely biased.

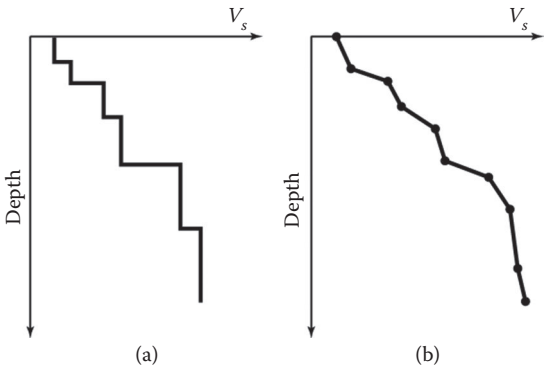

Figure 6.4 Subsurface numerical modeling: (a) layered (i.e., discontinuous) model and (b) continuous model.

[13] In the practical implementation of this numerical scheme, the medium is discretized by a stack of homogenous layers of arbitrary small thickness and the physical and mechanical properties within the individual layers such as V_S and V_P are typically assumed to vary *linearly*.

Similarly, the use of a continuous model (Figure 6.4) at a testing site characterized by strong contrasts in mechanical impedance among different layers would be inappropriate. On the contrary, in a homogenous soil deposit, the shear wave velocity at a site is expected to increase monotonically with depth because of the influence of overburden pressure. In such situations, surface wave measurements should be more appropriately interpreted adopting a subsurface model in which the material properties vary continuously with depth.

Constitutive modeling of geomaterials is related to the assumptions adopted to describe the mechanical response of soils and rocks subjected to low amplitude, dynamic excitations. As discussed in Chapter 2, linear elasticity and linear viscoelasticity have proved to be effective in describing phenomena of wave propagation in geomaterials at low strain levels. As such, they are the two most commonly adopted constitutive models.

Poroelasticity and Biot theory, which explicitly recognize the multicomponent nature of saturated geomaterials, are rarely used because the solution of the forward problem is considerably more complicated. However, special care should be taken when using one-constituent theories to interpret surface wave measurements at sites characterized by the presence of the water table. In fact, wave propagation phenomena associated with geophysical testing induce an undrained response in saturated materials due to the rate of application of the seismic excitation compared with their hydraulic conductivity. Correspondingly, in soil constitutive modeling, the values assumed for the Poisson ratio should correctly reflect the real position of the water table at the testing site because this soil parameter is markedly different in dry and water-saturated geomaterials under undrained loading (Foti and Strobbia 2002).

As mentioned earlier, the solution of a parameter identification problem is founded on the ability to solve the corresponding forward problem, which practically means to synthetically reproduce a set of experimental data associated with a particular mathematical model. In *discrete* inverse theory, the mathematical model is assumed to depend on a certain, finite number of unknown model parameters, the determination of which is the objective of the inversion algorithm (Menke 1989).

The model parameters and the set of experimental data may be conveniently represented by vectors of adequate size **m** and **d**, respectively. In the most general case, the formal relation between data and model parameters is implicit; however, it is often possible to uncouple data **d** from model parameters **m** and write the forward problem as follows

$$\mathbf{G}(\mathbf{m}) = \mathbf{d} \tag{6.1}$$

where **G** is a nonlinear, vector-valued function operator reflecting the type of relationship existing in a particular mathematical model between input data and model parameters. For the forward problem associated with

surface waves, the model may be a linear elastic, multilayered medium. In this case, the model parameters are the velocity of propagation of S-waves of the individual soil layers.[14] The set of experimental data is represented by a discretized dispersion curve describing the frequency dependence of Rayleigh (or Love) phase (or group) velocity (modal or apparent) over a finite frequency range. Equation 6.1 can then be rewritten by making the meaning of vectors **m** and **d** explicit, as follows

$$\mathbf{G}(\mathbf{V}_S) = \mathbf{V}_{R/L} \tag{6.2}$$

where $\mathbf{V}_S = [(V_S)_1, (V_S)_2, \ldots (V_S)_i \ldots (V_S)_{nl}]$ is the vector comprising the shear wave velocities of the individual layers and $\mathbf{V}_{R/L} = [(V_{R/L})_1, (V_{R/L})_2, \ldots (V_{R/L})_j \ldots (V_{R/L})_{nf}]$ is the discretized Rayleigh (or Love) dispersion function. Finally, **G** is a nonlinear vector-valued function of $\mathbf{V}_S$. Section 6.4.3 will show an explicit representation of this vector function. The symbols *nl* and *nf* denote (respectively) the number of layers (including the half-space) of the stratified medium and the set of experimental frequencies at which the Rayleigh (or Love) phase (or group) velocity (modal or apparent) has been measured. Equation 6.2 represents the formal statement of the Rayleigh (or Love) *forward* problem in linear elastic media. As was demonstrated in Chapter 2, this equation is clearly nonlinear. If the medium is viscoelastic, Equation 6.2 still holds; however, the vectors $\mathbf{V}_S$ and $\mathbf{V}_{R/L}$ become complex valued (see Section 6.4.3).

Under the assumption of weak dissipation, a procedure can be established to determine the transversal (or shear) material damping ratio profile $D_S(x_2)$ from the inversion of an experimental attenuation curve $\alpha_R(\omega)$. Details of the procedure are reported in Chapter 2, specifically Equation 2.133. As noted in Section 2.5.3, an important feature of this procedure is that it is *linear*. This is in contrast with the problem of obtaining the $V_S(x_2)$ profile from the inversion of $V_{R/L}(\omega)$, which is highly nonlinear. The statement of the corresponding Rayleigh (or Love) *forward* problem for determining material damping ratio can therefore be written as follows

$$\mathbf{G} \cdot \mathbf{D}_S = \boldsymbol{\alpha}_{R/L} \tag{6.3}$$

where $\mathbf{D}_S = [(D_S)_1, (D_S)_2, \ldots (D_S)_i \ldots (D_S)_{nl}]$ is the vector comprising the shear damping ratio of the individual strata and $\boldsymbol{\alpha}_{R/L} = [(\alpha_{R/L})_1, (\alpha_{R/L})_2, \ldots (\alpha_{R/L})_j \ldots (\alpha_{R/L})_{nf}]$ is the discretized Rayleigh (or Love) attenuation function. Finally, **G** is an *nf* by *nl* matrix composed by terms that include the partial derivatives of the modal (or apparent) phase velocity of surface waves with

[14] In principle, additional or alternative sets of model parameters may also be the *velocity of propagation* of *P-waves*, the *mass density*, and most importantly, *the thickness* of the layers. If the medium is assumed to be viscoelastic, further parameters may include the longitudinal and transversal *material* damping ratio of the strata.

respect to the model parameters $(V_P)_i$ and $(V_s)_i$ of the various layers of the soil deposit. The precise definition is given by Equation 2.133. (The "dot" operator at the left-hand side of Equation 6.3 denotes matrix multiplication.)

Summing up, the *forward problems* that can be conceived in surface wave testing may be of *three* types:

I. The problem of predicting the Rayleigh (or Love) dispersion vector function $\mathbf{V}_{R/L} = [(V_{R/L})_1, (V_{R/L})_2, \ldots (V_{R/L})_j \ldots (V_{R/L})_{nf}]$ by knowing the profile of shear wave velocity $\mathbf{V}_S = [(V_S)_1, (V_S)_2, \ldots (V_S)_i \ldots (V_S)_{nl}]$ of the multilayered medium.
II. The problem of predicting the Rayleigh (or Love) attenuation vector function $\boldsymbol{\alpha}_{R/L} = [(\alpha_{R/L})_1, (\alpha_{R/L})_2, \ldots (\alpha_{R/L})_j \ldots (\alpha_{R/L})_{nf}]$ by knowing the profile of shear damping ratio $\mathbf{D}_S = [(D_S)_1, (D_S)_2, \ldots (D_S)_i \ldots (D_S)_{nl}]$ of the multilayered medium.
III. The problem of predicting the complex-valued Rayleigh (or Love) dispersion vector function $\mathbf{V}^*_{R/L} = \left[\left(V^*_{R/L}\right)_1, \left(V^*_{R/L}\right)_2, \ldots \left(V^*_{R/L}\right)_j \ldots \left(V^*_{R/L}\right)_{nf}\right]$ by knowing the profile of complex-valued shear wave velocity $\mathbf{V}^*_S = \left[\left(V^*_S\right)_1, \left(V^*_S\right)_2, \ldots \left(V^*_S\right)_i \ldots \left(V^*_S\right)_{nl}\right]$ of the multilayered medium. The complex-valued shear wave velocity can be constructed from the profile of shear wave velocity and shear damping ratio using the formalism of complex variable theory (Equation 2.125). In a similar fashion, the Rayleigh (or Love) dispersion and the attenuation functions can be recovered from the corresponding complex-valued dispersion vector functions (Equation 2.124).

Problems I and III are nonlinear forward problems,[15] whereas problem II is linear. Consequently, in investigating the characteristics of the corresponding inverse problems and of the algorithms for their solution, it is necessary to investigate the features of both the *linear* and the *nonlinear* inverse problem.

6.3 SURFACE WAVE INVERSION BY EMPIRICAL METHODS

A simple, empirical procedure for estimating the shear wave velocity profile directly from the experimental dispersion curve was introduced in the early applications of surface wave testing for near-surface site characterization

[15] Problem III is further complicated by the complex nature of the mapping (Equation 6.2). It turns out, however, that owing to the elegance and efficiency of complex variable theory, the complications that arise are not severe. This result carries over even in the solution of the corresponding inverse problem as will be shown in Section 6.4.3.

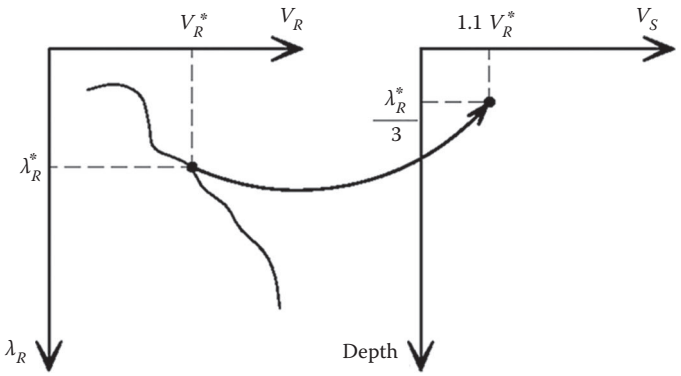

Figure 6.5 Empirical inversion procedure introduced in the early applications of surface wave testing in a test known as the Steady-State Rayleigh Method. (From Foti, S., Multi-station methods for geotechnical characterisation using surface waves, PhD Dissertation, Politecnico di Torino, 2000.)

(Jones 1958). At that time, the test was called the SSRM. Despite the inversion procedure adopted in the SSRM was very rough, it allowed a rapid, preliminary estimation of the shear wave velocity profile at a site. The essence of the empirical interpretation of surface wave data used in the SSRM will be briefly reviewed.

As discussed in Chapter 2, the ground motion induced by a traveling surface wave is confined in the uppermost part of the soil deposit (i.e., the *skin depth*). It can be assumed that most of the strain energy associated with surface wave motion is confined within a depth of about a wavelength from the ground surface (Achenbach 1984). At the same time, in a homogeneous half-space, the Rayleigh phase velocity assumes values that are close to the shear wave velocity of the medium (see Figure 2.15). A rough estimate is $V_S \approx 1.1 \cdot V_R$. To a first approximation, this estimate of shear wave velocity may be considered representative of the value of V_S at a depth equal to one-half/one-third of the wavelength. This interpretation may be viewed as a mapping from the $\{V_R, \lambda_R\}$ domain to the $\{V_S, \text{depth}\}$ range (Figure 6.5).

A rough estimate of the shear wave velocity profile at a site may be determined by repeating the procedure for the whole set of experimental data.

6.3.1 Numerical example

Figure 6.6 shows the results obtained with the empirical inversion of the SSRM for three different synthetic profiles of soil deposits. The Figure clearly shows that the method gives a reasonable estimate of the shear wave velocity profile only for case A, which represents a *normally* dispersive medium with a gradual increase of V_S with depth. This estimate can be considered

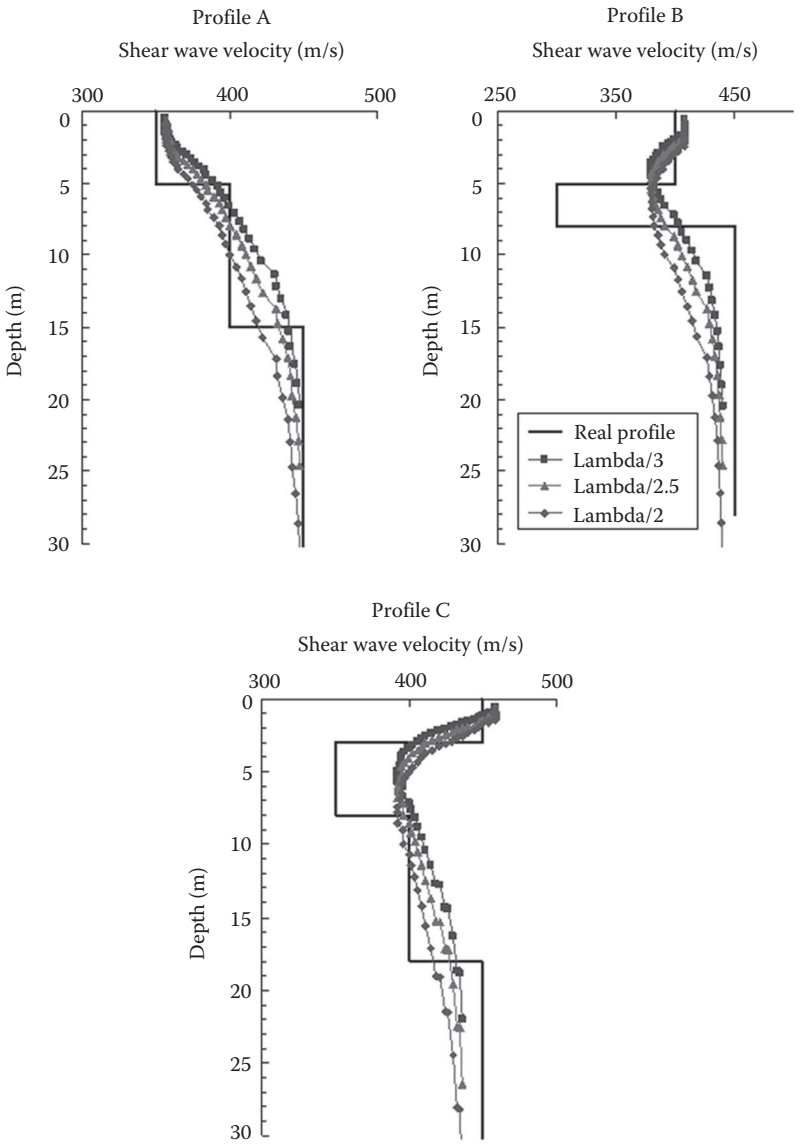

Figure 6.6 Application of the empirical inversion procedure of the SSRM to estimate the shear wave velocity at a site. (From Foti, S., Multi-station methods for geotechnical characterisation using surface waves, PhD Dissertation, Politecnico di Torino, 2000.)

acceptable in some applications without the need for implementing a rigorous inversion procedure. Moreover, it can be used as a good first guess in an iterative inversion algorithm. However, Figure 6.6 shows that the procedure is not acceptable in *inversely* dispersive soil profiles (cases B and C). Yet, it is remarked that the change of curvature exhibited by the V_S profile caused by the presence of a soft, intermediate layer (case B) or alternatively by a stiff top layer (case C) is still captured by the method.

6.3.2 Manual inversion

Empirical methods for surface wave inversion also include *trial-and-error procedures* that are in essence manual approaches to the inversion of the experimental dispersion curve. In these methods, the model parameters are successively adjusted in order to obtain a theoretical dispersion curve that matches the experimental data as best as possible. Their implementation requires the availability of an algorithm for the solution of the forward problem. In practice, a limited number of forward simulations are performed with the model parameters adjusted each time in an attempt to visually minimize the error misfit between the numerical and the experimental dispersion curves. The procedure is strongly subjective (i.e., operator dependent) and requires a certain degree of experience in order to achieve acceptable results in a reasonable amount of time.

Sometimes trial-and-error procedures represent the only viable approach in the inversion of "pathological" dispersion curves. In such situations, automatic algorithms may, in fact, get stuck and fail to converge due to instabilities in the calculation of the Jacobian needed to carry out the inversion, particularly if the latter is not performed analytically but using a discrete, finite difference scheme (Section 6.4.3).

6.4 SURFACE WAVE INVERSION BY ANALYTICAL METHODS

6.4.1 Measures of fitting goodness

A linear regression is the simplest parameter identification problem that can be conceived. This consists of finding a slope and an intercept of a straight line passing through a set of experimental data. Because there are only two model parameters (i.e., the slope and the intercept), a minimum of two measurements are needed to solve the problem. However, in practice, there is a redundancy of experimental data and the linear regression problem is *overdetermined,* which means that no exact solution can be found and the interpolation problem is converted into a *fitting* problem. The number of data is, in fact, greater than the number of model parameters, and the solution can be found only in an *approximate* sense.

The standard approach for solving this problem is that of the *least-squares method*, which consists of finding that particular value of the slope and intercept of the straight line that minimizes the sum of the squares of the individual errors. The latter are defined by the difference between the measured data and the data predicted by the linear model. The linear regression problem is then converted into a *parameter optimization problem* of finding a set of model parameters that minimize the overall misfit between the measured data and the predictions of the model represented by a straight line. If Equation 6.1 is specialized for a linear model, it becomes

$$\mathbf{G} \cdot \mathbf{m} = \mathbf{d} \tag{6.4}$$

In this case, the vector-valued function operator **G** degenerates into a matrix, and the forward problem is represented by a linear system of algebraic equations. The error misfit (or prediction error) may then be written as follows

$$Er = \sum_{i=1}^{N} \left[d_i - \sum_{j=1}^{M} \left(G_{ij} m_j \right) \right]^2 \tag{6.5}$$

where N is the number of experimental data and G_{ij} are the elements of an N by M matrix (in this particular case, $M = 2$) where the first column is constituted of ones and the second column of the N values of the independent variable x_i at which the experimental data are measured. Finally, in this case, m_j is the component of an M-dimensional vector containing the intercept and slope of the linear model. Equation 6.5 may be interpreted as the definition of the Euclidean *norm* of a vector with components $\left[d_i - \sum_{j=1}^{M} \left(G_{ij} m_j \right) \right]$ and $i = 1, N$. So the error misfit Er may be viewed as the square of a Euclidean *distance*, and the method of least squares may be viewed as the procedure of finding the slope and the intercept of the straight line that minimizes a specific measure of distance between the measured data and the model predictions. The solution to this problem is provided by the following equation (Menke 1989)

$$\mathbf{m} = (\mathbf{G}^T\mathbf{G})^{-1}\, \mathbf{G}^T\mathbf{d} \tag{6.6}$$

Fitting a straight line is a particular type of linear regression problem characterized by only two model parameters. By allowing vector **m** to have an arbitrary finite size M, Equation 6.6 still represents the solution

of an *over determined,* linear, discrete, inverse problem by means of the least-squares method. The issue of *stability* of the solution provided by Equation 6.6 will be investigated in Section 6.4.2.

The least-squares solution of the linear regression problem is by no means unique. Other solutions are possible introducing alternative definitions of error misfit and thus of Equation 6.5. Recalling that the L_p norm of a vector with components $\left[d_i - \sum_{j=1}^{M}(G_{ij}m_j)\right]$ is given by the relation

$$\left\| d_i - \sum_{j=1}^{M}(G_{ij}m_j) \right\|_p = \left\{ \sum_{i=1}^{N} \left[d_i - \sum_{j=1}^{M}(G_{ij}m_j) \right]^p \right\}^{\frac{1}{p}} \tag{6.7}$$

it is possible to introduce other linear models fitting a dataset in the sense of being defined by an error misfit different from the L_2 norm. Figure 6.7 shows an example of a set of measurements fitted alternatively by using the L_2 and L_1 norms. The line denoted by L_p norm corresponds to the expected trend with higher values of p. As can be seen from the figure, the dataset includes the presence of an *outlier* that is a datum that is severely discordant with the rest of the measurements; thus, it is likely to be affected

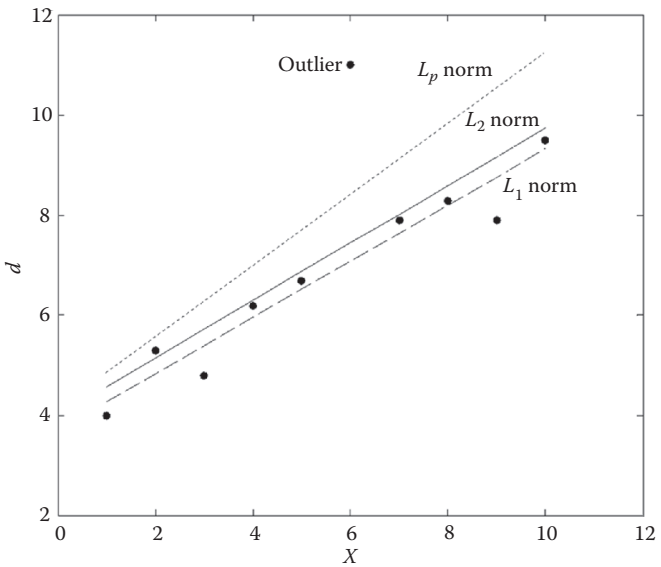

Figure 6.7 Linear regression using different measures of error misfit.

by a gross error. According to Equation 6.7, the higher the order of the norm, the greater is the weight attributed to large errors. Therefore, larger-order norms are suitable for adoption in experimental data characterized by great precision because the fact that an individual datum falls distant from a general trend is important. On the contrary, lower-order norms should be used if the data are characterized by a large uncertainty about the model, and the presence of few outliers should not affect the goodness of the fitting that much. Lower-order norms tend to give similar weight to errors of different sizes, and in this sense, they are less sensitive to the presence of outliers (Menke 1989). Figure 6.7 clearly shows this. Parameter estimation algorithms of this type are said to be *robust* (Claerbout and Muir 1973). Solution of the linear regression problem using the L_1 norm is far more involved than the standard least-squares problem. For instance, it can be carried out using the iteratively reweighted least-squares (IRLS) algorithm (Barrowdale and Roberts 1974). In general, this complexity also holds when using norms of an order higher than $p = 2$.

According to the *maximum likelihood principle*, it can be shown that the solution to the linear regression problem under the L_2 norm corresponds to the most likely solution if the data measurements are normally distributed, that is if they follow Gaussian statistics.

Likewise, by the same principle, it is possible to demonstrate that the L_1 norm solution of the linear regression problem represents the maximum likelihood estimator for data characterized by errors that are distributed like a double-sided exponential distribution (Menke 1989).

6.4.2 Linear inverse problem

6.4.2.1 Singular-value decomposition and Moore–Penrose generalized inverse

As mentioned in Section 6.2, linear inverse modeling occurs in surface wave testing in connection with the problem of determining the material damping ratio profile from the inversion of an experimental attenuation curve as illustrated by Equation 6.3. If the problem is fully *overdetermined,* solution of this equation using the least-squares method is provided by Equation 6.6 after making the following correspondence

$$\mathbf{m} = \mathbf{D}_S; \quad \mathbf{d} = \boldsymbol{\alpha}_{R/L} \tag{6.8}$$

Overdetermined problems typically occur when the number N of measured data is greater than the number M of model parameters, which in this case is equal to nl, the number of layers of the soil deposit (including the half-space). Instead, the model parameters are the values of the shear damping ratio of the individual layers.

In general, if Problem 6.4 is *even-determined*, *underdetermined,* or *mixed-determined*, the solution given by Equation 6.6 may no longer make sense. Indeed, if the problem is underdetermined and thus $N < M$, there are more unknowns than available equations and the problem will no longer have a unique solution. As a matter of fact, it will have infinite solutions. Given this state of affairs, it would be desirable to generalize Equation 6.6 so that it would be applicable to all possible circumstances and not only to overdetermined problems. An efficient and convenient way to obtain this generalization is through the concept of the *Moore–Penrose pseudo- or generalized inverse matrix* (Moore 1920; Penrose 1955), which formally extends the application of the rules of ordinary matrix algebra to the solution of the problem of finding $\mathbf{m}$ from the inversion of Equation 6.4.

The definition of the Moore–Penrose pseudoinverse of a matrix $\mathbf{G}$ requires a prior introduction of the *singular-value decomposition* (SVD), which is a particular type of eigenvalue factorization of a matrix (Strang 1988). Any N by M matrix $\mathbf{G}$ can be decomposed into the product of the following three matrices

$$\mathbf{G} = \mathbf{Q}_1 \Sigma \mathbf{Q}_2^T \tag{6.9}$$

where $\mathbf{Q}_1$ is an N by N square orthogonal matrix with columns that are unit basis vectors spanning the *data space* and $\mathbf{Q}_2$ is an M by M square orthogonal matrix with columns that are unit basis vectors spanning the *model parameter space*.[16] Finally, $\mathbf{\Sigma}$ is an N by M diagonal matrix whose nonnegative diagonal elements, which are called *singular values*.

The SVD generalizes the standard eigenvalue–eigenvector factorization $\mathbf{Q\Lambda Q}^T$ of a symmetric square matrix to rectangular matrices.[17] There are plenty of algorithms to efficiently compute the SVD of a matrix (Golub and Van Loan 1996). However, although numerically not very efficient, it is insightful to perform the SVD of matrix $\mathbf{G}$ by solving two standard eigenvalue problems for matrices $\mathbf{GG}^T$ and $\mathbf{G}^T\mathbf{G}$, respectively. It turns out that the eigenvectors of $\mathbf{GG}^T$ form the columns of matrix $\mathbf{Q}_1$, whereas matrix $\mathbf{Q}_2$ is formed by the eigenvectors of $\mathbf{G}^T\mathbf{G}$. Finally, the singular values of matrix $\mathbf{\Sigma}$ are obtained from the square roots of the nonzero eigenvalues of both $\mathbf{GG}^T$ and $\mathbf{G}^T\mathbf{G}$ (Strang 1988). The singular values of matrix $\mathbf{\Sigma}$ are usually arranged in order of decreasing magnitude. As will be shown in the next section, they play an important role in the construction of the Moore–Penrose generalized inverse matrix and also in controlling the stability of the inversion algorithm.

[16] To understand the rationale for these denominations of vector spaces, recall that in the discussion here M and N represent the number of model parameters and experimental data, respectively.

[17] In this case, $\mathbf{Q}$ is the matrix of eigenvectors and $\mathbf{\Lambda}$ is a diagonal matrix containing the eigenvalues.

If q is the *rank* of matrix **G**, this number also coincides with the number of the first nonzero (i.e., positive) singular values of matrix $\mathbf{\Sigma}$, and the construction of the SVD of matrix **G** simplifies as follows (Menke 1989)

$$\mathbf{G} = \left(\mathbf{Q}_1\right)_q \Sigma_q \left(\mathbf{Q}_2^T\right)_q \tag{6.10}$$

where matrices $(\mathbf{Q}_1)_q$ and $(\mathbf{Q}_2)_q$ denote the first q columns of matrices $\mathbf{Q}_1$ and $\mathbf{Q}_2$, respectively. Once constructed, the SVD of matrix **G** can be used to compute the M by N Moore–Penrose generalized inverse matrix $\mathbf{G}^{-g}$ as follows

$$\mathbf{G}^{-g} = \left(\mathbf{Q}_2\right)_q \Sigma_q^{-1} \left(\mathbf{Q}_1^T\right)_q \tag{6.11}$$

Finally, the solution of the problem represented by Equation 6.4 is given by

$$\mathbf{m} = \mathbf{G}^{-g}\mathbf{d} = \left(\mathbf{Q}_2\right)_q \Sigma_q^{-1} \left(\mathbf{Q}_1^T\right)_q \mathbf{d} \tag{6.12}$$

where **m** and **d** may be interpreted in the sense of Equation 6.8. Equation 6.12 is often denoted as the *natural* solution of Equation 6.4. The Moore–Penrose generalized inverse $\mathbf{G}^{-g}$ formally acts on Equation 6.4 as if the latter would correspond to an even-determined problem with a square matrix **G**. However, Equation 6.12 represents the solution to Equation 6.4 for the most general mixed-determined problem; furthermore, it can be demonstrated that matrix $\mathbf{G}^{-g}$, and thus **m**, always exists (Aster et al. 2005). This is in contrast with the validity of Equation 6.6.

The following special cases are now examined:

- Problem 6.4 is *even-determined*, that is, $q = N = M$. Data space and model parameter space have the same dimension, thus $\mathbf{G}^{-g}$ is the inverse matrix of **G** in the ordinary sense of matrix algebra. The solution in this case is exact and unique.
- Problem 6.4 is *overdetermined,* that is, $q \leq N > M$. This means that the rank of matrix **G** coincides with the dimension of data space that is greater than the dimension of model parameter space. It can be shown that, in this case, Equation 6.12 coincides with Equation 6.6 (i.e., with the least-squares solution).
- Problem 6.4 is *underdetermined,* that is, $q \leq N < M$. This means that there are more unknowns than equations; thus, the solution is not unique and the problem is ill-posed. It can be shown that, in this case, Equation 6.12 provides a least-squares, minimum L_2 norm solution for **m**. The constraint of *minimum length* can be regarded as

supplementing Problem 6.4 with a priori information that is needed to single out one of the infinitely many solutions characterizing an underdetermined inverse problem. The minimum length solution does not always make sense physically, particularly in surface wave inversion. The next sections will further expand this concept of adding a priori information to find the solution of an ill-posed inverse problem.

In real practice, there are problems that are neither fully overdetermined nor fully underdetermined, and they are called *mixed-determined* parameter-estimation problems (Menke 1989). In geophysical prospecting and in surface wave testing, in particular, mixed-determined problems are quite common. They occur when certain portions of the domain to be investigated are fully illuminated by the seismic rays while others are left at "dark" (Figure 6.8).

In the attenuation measurement problem represented by Equation 6.3, this situation occurs if one attempts to determine the shear damping ratio of deep layers that have not been sampled by surface waves. The reason for this may be either because not enough energy at low frequencies has been generated by the seismic source or because at deep layers the motion was completely attenuated. A combination of both phenomena is of course another possibility. On the contrary, shallower layers are typically *overilluminated* due to the superposition of traveling surface waves having a broad frequency band.

The difficulties discussed here for attenuation measurements occur also in dispersion measurements with the additional complication that now the inverse problem is nonlinear, as is shown by Equation 6.2. To eliminate these inconveniences, in the forward models (Equations 6.2 and 6.3), including layers with unknown model parameters (i.e., shear wave velocity and material damping ratio) should be avoided at depths not reached by the passage of surface waves. This can be easily done by computing the skin

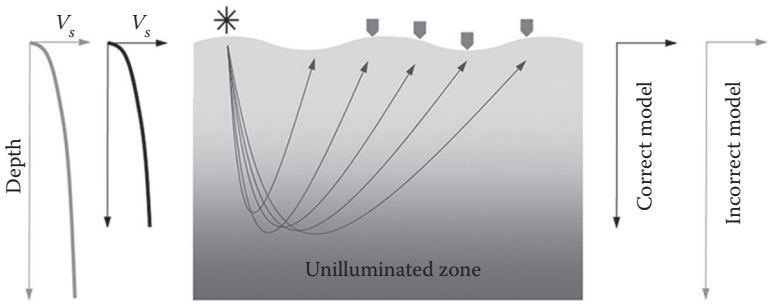

Figure 6.8 Example in geophysical prospecting of a mixed-determined problem. The shallower top layers are overdetermined because they are traversed by several seismic rays. The bottom layers are underdetermined because they are not traversed by any seismic ray. A correct model should take this into account by selecting a proper investigation depth when performing the inversion.

depth (Chapter 2) at the site under investigation from the maximum wavelength associated with the measured dispersion and attenuation curves.

The Moore–Penrose generalized inverse $\mathbf{G}^{-g}$ and thus Equation 6.12 make perfect sense in underdetermined as well as mixed-determined problems by providing the minimum length, least-squares solution to Problem 6.4 while properly accommodating for the various combinations of the rank of **G** and the size of data and model parameter spaces. However, these situations of under- and mixed-determinacy should be prevented because the solution provided by Equation 6.12, even though mathematically correct, may have undesirable features caused by the minimum length constraint appearing as a priori information to solve an otherwise indeterminate problem.

When designing a geophysical experiment and setting up the forward model for the inversion of measured data (Figure 6.8), every effort should be made to achieve redundancy and homogeneity of information throughout the spatial domain of interest so that Problem 6.4 always turns out to be overdetermined with full rank of matrix **G**.[18]

6.4.2.2 Instability of the solution and condition number

As stated in Section 6.1.2, according to Hadamard (1923), a mathematical problem is said to be *stable* if the solution depends continuously on the data. Existence and uniqueness are the other two conditions required by Hadamard to define a mathematical problem as well-posed. In relation to the solution of Equation 6.3, and more in general, to Equation 6.4, uniqueness is violated in underdetermined and mixed-determined problems. The strategy adopted by the Moore–Penrose generalized inverse $\mathbf{G}^{-g}$ to resolve this difficulty and thus enforce uniqueness in the solution represented by Equation 6.12, has been to inject a priori information into Problem 6.4 by means of the minimum length constraint.[19] This approach of introducing a priori information into the solution is by no means unique. In fact, the next section will illustrate a rather general alternative approach, which falls under the name of *Tikhonov regularization methods*.

To address the issue of *stability* of the solution, it is convenient to rewrite Equation 6.12 as follows to make the presence of the singular values explicit (Aster et al. 2005)

$$\mathbf{m} = \left(\mathbf{Q}_2\right)_q \Sigma_q^{-1} \left(\mathbf{Q}_1^T\right)_q \mathbf{d} = \sum_{i=1}^{q} \frac{\left[\left(\mathbf{Q}_1^T\right)\right]_i \mathbf{d}}{s_i} \left[\left(\mathbf{Q}_2\right)\right]_i \tag{6.13}$$

where s_i is the ith singular value of matrix $\mathbf{\Sigma}$ and $[(\mathbf{Q}_1)]_i$ and $[(\mathbf{Q}_2)]_i$ denotes the ith column of matrix $\mathbf{Q}_1$ and $\mathbf{Q}_2$ respectively. The range of singular

[18] If N is the dimension of data space and $q = rank(\mathbf{G})$, then **G** is full rank if $q = N$.

[19] The solution obtained with this method is called the *natural* solution of Problem 6.4.

values is often called the *singular-value spectrum*, and it is important because it controls the stability of the solution of Equation 6.4 given by Equation 6.13. This equation shows that the reciprocal of the singular values s_i play the role of the coefficients of the series expansion, and the largest coefficients of the series are those associated with the smallest s_i. Thus, the solution **m** in Equation 6.13 is controlled de facto by the smallest singular values of matrix **G**. In a problem characterized by a singular-value spectrum with very small s_i, if the true data measurements $\mathbf{d}_{\text{true}} = \mathbf{d}$ are affected by large noise, say $\mathbf{d}_{meas} = \mathbf{d}_{true} + \boldsymbol{\delta}$, where $\boldsymbol{\delta}$ is a noise vector such that $\|\boldsymbol{\delta}\|_2 \approx \|\mathbf{d}_{true}\|_2$, then the solution **m** from Equation 6.13 will contain the noise terms amplified by a factor proportional to $1/s_i$.

It should be noted that the *ill-posedness*[20] and thus the instability of the solution to Problem 6.4 explicitly exhibited by Equation 6.13 is entirely independent from the accuracy of the measured data **d**; rather it is a property of matrix **G** and thus of the setting and design of the geometry of the experiment. A measure of the instability of the solution **m** is represented by the *condition number*, which is defined as (Aster et al. 2005)

$$cond(\mathbf{G}) = \frac{s_1}{s_k} \tag{6.14}$$

where s_1 is the largest singular value of matrix **G** and s_k is the singular value associated with $min(N, M)$ that is the minimum value between the size of data and model parameter space. The larger the condition number of a **G** matrix, the more ill-conditioned is the problem associated with that matrix, which means the more sensitive the solution **m** will be to a small amount of noise in the data.

If **G** is not a full rank matrix,[21] then the condition number becomes infinite because s_k in Equation 6.14 would be equal to zero. However, in Equation 6.13, the solution **m** involves only the first nonzero singular values, which are q in number. Yet, in realistic inverse problems, there might be numerical situations where the singular values are very small but nevertheless different from zero. These singular values are the cause of severe instability in the evaluation of Equation 6.13.

A technique that can be used to mitigate this problem is to set a cutoff size by which singular values smaller than this cutoff are assumed equal to zero (Menke 1989). This operation will result in a truncation of the series represented by Equation 6.13 and as such acts like a *regularization* of the ill-posed Problem 6.4. However, dropping terms with small singular values from Equation 6.13 will determine a modification of the solution **m**; thus, it will worsen the fitting of **m** through the data **d**. It can be shown that

[20] In discrete linear problems, the *ill-posedness* is sometimes denoted as *ill-conditioning*.
[21] If **G** is of full rank, then $k = q = N$.

truncation of the SVD also worsens the *data and model resolution* of the problem, and this issue will be further discussed in the following section.

6.4.2.3 Tikhonov regularization methods

The SVD of matrix **G** is allowed to define a least-squares solution of Equation 6.4, which is of general validity. Simultaneously, this representation highlights the instability of this solution when the singular values s_i are very small. A "work-around" alleviation of this difficulty was the truncation of the terms in Equation 6.13 involving the smallest singular values. However, a more effective and systematic method to mitigate the ill-posedness of linear inverse problems is represented by the so-called *Tikhonov regularization* (Tikhonov and Arsenin 1977). This technique also has the advantage of very effectively addressing the nonuniqeness of the solution, which is another side of the ill-posedness of parameter identification problems.

The *zeroth-order Tikhonov regularization* consists of solving Equation 6.4 by seeking the vector of model parameter **m** and satisfying the following optimization condition

$$min \quad \left\{ \left\| \mathbf{Gm} - \mathbf{d} \right\|_2^2 + \mu^2 \left\| \mathbf{m} \right\|_2^2 \right\} \tag{6.15}$$

where μ is a *regularization parameter* also called the *Lagrange multiplier*. The solution obtained by enforcing this condition corresponds to determining the value of **m** that minimizes a combination of the *prediction error*, represented by $\left\|\mathbf{Gm} - \mathbf{d}\right\|_2^2$, and the *solution length*, represented by $\left\|\mathbf{m}\right\|_2^2$. The possible under- and/or mixed-determinacy, and therefore the lack of uniqueness in the solution of Problem 6.4, is tackled by imposing not only a reduction of the error misfit but also a property of the solution that is represented by its norm. This operation is equivalent to introducing *a priori* information to constrain the solution. In this sense, it generalizes the least-squares, minimum L_2 norm given by Equation 6.12 for underdetermined problems by allowing it to control the relative importance given to the error misfit and the simplicity of the solution represented by its L_2 norm.

As a matter of fact, the measure by which the goodness of fitting is traded in exchange for solution uniqueness is controlled by the regularization parameter μ. By setting μ = 0, Equation 6.15 reduces to the same optimization condition of the standard least-squares problem whose solution is given by Equation 6.6. This solution minimizes the prediction error; however, it has all the flaws discussed earlier concerning the lack of uniqueness in under determined and mixed-determined problems. On the contrary, increasing the value of μ will significantly mitigate the ill-posedness of Problem 6.4 at the expense, however, of a simultaneous increase in the error misfit.

In practice, the choice of the optimum value to be assigned to μ is far from obvious. It is always characterized by some degree of subjectivity, although this will certainly depend on the characteristics of the specific problem at hand. An effort should be made to achieve a trade-off between two conflicting features that it is hoped the solution **m** will have: (1) the ability to fit the experimental data with the smallest possible error misfit and (2) the ability to hold a small Euclidean norm, which is a surrogate of solution simplicity.

Implementation of Equation 6.15 requires finding the minimum of an unconstrained functional; it can be achieved through the standard rules of the calculus of variations. Omitting the details, the result is (Menke 1989)

$$\mathbf{m} = (\mathbf{G}^T\mathbf{G} + \mu^2\mathbf{1})^{-1}\,\mathbf{G}^T\mathbf{d} \tag{6.16}$$

where **1** is the M by M identity matrix. Equation 6.16 is often denoted as the *damped least-squares* solution of Problem 6.4 because its potential underdeterminacy (partial or total) has been damped out. To obtain more insight into the features of the solution, it is convenient to rewrite this equation as follows using the SVD (Aster et al. 2005)

$$\mathbf{m} = \sum_{i=1}^{k} \frac{s_i^2}{s_i^2 + \mu^2} \cdot \frac{\left[\left(\mathbf{Q}_1^T\right)\right]_i \mathbf{d}}{s_i} \left[\left(\mathbf{Q}_2\right)\right]_i \tag{6.17}$$

where $k = min(N, M)$ is the minimum between the size of data space and model parameter space. It is important to note that, with this definition, all the singular values of matrix **G** are included—possibly even the very small ones.

Equations 6.17 and 6.13 differ from each other because of the presence of the *filter factors* $\frac{s_i^2}{s_i^2 + \mu^2}$ in Equation 6.17. These factors play the role of *damping* the negative influence of very small singular values that otherwise, if present, could cause a blowup of some of the terms of the expansion, as is discussed in the previous section. Clearly, from the filter factors, it is seen that the larger the regularization parameter, the greater is the damping effect and vice versa. Overall, Tikhonov regularization has restored uniqueness and stability to the solution, two properties jeopardized by the ill-posedness of Problem 6.4.

Zeroth-order Tikhonov regularization is based on minimizing a functional involving the error misfit and the square of the Euclidean norm $\mathbf{m}^T\mathbf{m}$ that was taken as an a priori measure of solution simplicity. Other measures of simplicity are possible, and in some problems they result more adequate.

Higher-order Tikhonov regularization methods are based on assuming the Euclidean norm of the first-, second-, and higher-order derivatives of **m** as a measure of solution simplicity (Aster et al. 2005). For discrete model parameters, the first and the second derivatives of **m** may be approximated by using finite difference schemes as follows

$$\begin{cases} R_1 = \|\mathbf{Lm}\|_2^2 = (\mathbf{Lm})^T \cdot (\mathbf{Lm}) \\ R_2 = \|\mathbf{L}(\mathbf{Lm})\|_2^2 = (\mathbf{L}^2\mathbf{m})^T \cdot (\mathbf{L}^2\mathbf{m}) \end{cases} \tag{6.18}$$

where **L** is an M by M real-valued matrix representing the two-point, central finite difference operator, and it is given by

$$\mathbf{L} = \begin{bmatrix} 0 & \dots & & \\ -1 & 1 & & 0 \\ \dots & -1 & 1 & \\ & 0 & -1 & 1 \end{bmatrix} \tag{6.19}$$

The scalars R_1 and R_2 in Equation 6.18 are two different definitions of the *roughness* of solution **m**, whereas **L** in Equation 6.19 is called the *roughening matrix*. With reference to Equation 6.3, roughness of **m** would correspond to a scalar measure of the irregularity of the shear damping ratio profile with depth. If, from geological information, the geotechnical parameters at a site are expected to vary smoothly with depth, it would make sense to solve Equation 6.3 by enforcing the minimization of roughness of the damping ratio profile.[22] In this way, solutions that satisfy the criterion of producing a small error misfit but that are too irregular will be automatically rejected by the algorithm. This is an effective way to introduce a priori information when solving a parameter identification problem. It can easily be shown that, for a continuously varying medium, the two definitions of R_1 and R_2 given by Equation 6.18 correspond to the integral over depth of the square of the first and the second derivative, respectively, of model parameter with respect to depth.

The solution to Equation 6.4 using the zeroth-order Tikhonov regularization method was found by enforcing Equation 6.15, which corresponds to determining the value of **m** that minimizes the linear functional $\|\mathbf{Gm}-\mathbf{d}\|_2^2 + \mu^2\|\mathbf{m}\|_2^2$. In higher-order Tikhonov regularization, this condition is replaced by

$$min \quad \|\mathbf{Gm}-\mathbf{d}\|_2^2 + \mu^2\|\mathbf{L}^n\mathbf{m}\|_2^2 \tag{6.20}$$

[22] Minimization of roughness of the shear damping ratio depth profile would correspond to maximization of its opposite, which is *flatness* or *smoothness*.

where $\mathbf{L}^n$ is the *nth-order roughening matrix*. For $n = 1,2$, the previous definitions of first- and second-order roughening matrices are recovered. If $n = 0$, $\mathbf{L}^0 = \mathbf{1}$. Then the procedure to solve Equation 6.20 becomes analogous to that used for the solution of Equation 6.15. Higher-order Tikhonov regularization methods are frequently used in geosciences because in several circumstances physical properties and geophysical and geotechnical parameters may be assumed to vary regularly with depth from the ground surface. Section 6.4.3 will describe in detail the application of first- and second-order Tikhonov regularization for the inversion of an experimental dispersion and attenuation curve to obtain the unknown shear wave velocity and shear damping ratio profiles at a site. The procedure is also known in the literature as the *Occam's algorithm*, and it will be applied to solve Equation 6.2, which is a *nonlinear* parameter identification problem.

6.4.2.4 Other regularization methods

Ill-posedness of inverse problems may be dealt with using other types of regularization methods. The objective is always the same: introduce a priori information to mitigate the instability and nonuniqeness features of the solution. Alternatives to Tikhonov regularization include *bounds constraint methods* and *total variation regularization* (Aster et al. 2005). Bounds constraint methods are based on using prior knowledge of the allowable range of variation of model parameters. For instance, with reference to Equation 6.3, damping ratio in soil layers must be nonnegative, and this poses a constraint on the lower bound that this model parameter can assume such that $\mathbf{D}_S \geq \mathbf{0}$. Constraint least-squares problems can effectively be solved by the *method of Lagrange* multipliers (Logan 2006). An efficient algorithm for determining the nonnegative least-squares solution is provided by Lawson and Hanson (1974).

The *total variation* regularization method is similar to first-order Tikhonov regularization in that it enforces a minimization condition similar to Equation 6.20 with $n = 1$ and the roughening matrix $\mathbf{L}$ given by Equation 6.19. However, in Equation 6.20, this method replaces the Euclidean norm with the L_1 norm. This substitution produces the effect of nonpenalizing discontinuous model parameters such as it happens with the standard Tikhonov regularization, which, in fact, favors smooth spatial variations of model parameters. Thus, the total variation method is appropriate in all those circumstances in which sharp variations and discontinuities in the model parameters are expected. One example could be represented by the presence of geological formations that have very different mechanical properties and that are separated by an abrupt interface. Like the standard Tikhonov algorithm, the total variation method still regularizes the solution of the inverse Problem 6.4; however, it performs this operation without rejecting models with possible sharp spatial variations

in the model parameters. Determining **m** from Problem 6.4 by using the total variation method is not trivial because of the nondifferentiability of the L_1 norm in Equation 6.20. Yet, special algorithms have been purposely developed to overcome this difficulty (Boyd and Vandenberghe 2004).

6.4.2.5 Accuracy and resolution

Assuming the experimental data in Problem 6.4 are exempt from noise, a natural question arises about the goodness of the estimated model parameters. Let $\mathbf{G} \cdot (\mathbf{m})_{true} = (\mathbf{d})_{exp}$ denote the relation between the error-free experimental data and the *true* model parameters. Introducing this relation into the expression of the estimated model parameters $(\mathbf{m})_{est} = \mathbf{G}^{-g}(\mathbf{d})_{exp}$ yields

$$(\mathbf{m})_{est} = \mathbf{G}^{-g}\,(\mathbf{d})_{exp} = \mathbf{G}^{-g}\mathbf{G} \cdot (\mathbf{m})_{true} = \mathbf{R}_m(\mathbf{m})_{true} \tag{6.21}$$

where $\mathbf{R}_m = \mathbf{G}^{-g}\mathbf{G}$ identifies the M by M *model resolution matrix*. This array characterizes the bias introduced by the particular generalized inverse matrix that has been adopted to solve Problem 6.4. If $\mathbf{R}_m$ is equal to the identity matrix, then the model parameters are estimated exactly. It can be shown that in a fully overdetermined problem, $\mathbf{R}_m = \mathbf{1}$ (Menke 1989). In general, however, the model resolution matrix is different from the identity matrix. This is certainly true if **G** is not a full rank matrix that is $q < N$ (Aster et al. 2005). The more the diagonal elements of $\mathbf{R}_m$ are close to one, the more well resolved are the model parameters by the algorithm. Conversely, the smaller the diagonal elements, the more poorly resolved are the model parameters.

Another query that may be asked after solving Problem 6.4 is how well the estimated model parameters fit the experimental data. In other words, it is of interest to evaluate $(\mathbf{d})_{pre} = \mathbf{G} \cdot (\mathbf{m})_{est}$ using $(\mathbf{m})_{est} = \mathbf{G}^{-g}(\mathbf{d})_{exp}$. The result is

$$(\mathbf{d})_{pre} = \mathbf{G} \cdot (\mathbf{m})_{est} = \mathbf{G} \cdot \mathbf{G}^{-g}(\mathbf{d})_{exp} = \mathbf{R}_d\,(\mathbf{d})_{exp} \tag{6.22}$$

$\mathbf{R}_d$ is an N by N array denoted as the *data resolution matrix*. It describes the goodness of data fitting by the model. Having $\mathbf{R}_d$ equal to the identity matrix corresponds to zero prediction error, which means that the experimental data are predicted exactly. This is what happens if Problem 6.4 is fully underdetermined. In most circumstances, however, the data resolution matrix differs from the identity matrix. Thus, the prediction error is not equal to zero, and the experimental data are not perfectly resolved by the model.

Both the model and data resolution matrices describe important features of the solutions of parameter identification problems (Menke 1989). It should be noted, however, that $\mathbf{R}_m$ and $\mathbf{R}_d$ are independent from the actual data measurements and model parameters. They are only a function of the properties of **G** and $\mathbf{G}^{-g}$ and thus of the experimental setting (e.g., geometry of data acquisition system) and on the adopted inversion algorithm. Also, possible a

priori information is reflected in $\mathbf{G}^{-g}$ and thus in $\mathbf{R}_m$ and $\mathbf{R}_d$. For this reason, a careful assessment of $\mathbf{R}_m$ and $\mathbf{R}_d$ may be useful in designing the experiment. With reference to surface wave testing, this would mean calculating $\mathbf{R}_m$ and $\mathbf{R}_d$ associated with attenuation measurements and thus with Equation 6.3.

The model and data resolution matrices can also be expressed in terms of the SVD of $\mathbf{G}$ and $\mathbf{G}^{-g}$. The result is (Aster et al. 2005)

$$\begin{cases} \mathbf{R}_m = \left(\mathbf{Q}_2\right)_q \cdot \left(\mathbf{Q}_2^T\right)_q \\ \mathbf{R}_d = \left(\mathbf{Q}_1\right)_q \cdot \left(\mathbf{Q}_1^T\right)_q \end{cases} \tag{6.23}$$

If $q = rank(\mathbf{G}) < M$, the model parameters are not fully resolved because $\mathbf{R}_m$ is not the identity matrix. Concerning $\mathbf{R}_d$, if $q < N$, $\mathbf{R}_d$ is not the identity matrix, and the experimental data are not perfectly resolved.

Useful scalar measures to quantify the spread of model and data resolution matrices are represented by the *Dirichlet spread functions* (Menke 1989). They are defined as the square of the Euclidean norm of the difference between $\mathbf{R}_m$ and $\mathbf{R}_d$, and the identity matrix. Formally,

$$\begin{cases} \text{spread}\left(\mathbf{R}_m\right) = \left\|\mathbf{R}_m - \mathbf{1}\right\|_2^2 \\ \text{spread}\left(\mathbf{R}_d\right) = \left\|\mathbf{R}_d - \mathbf{1}\right\|_2^2 \end{cases} \tag{6.24}$$

From this equation, the closer $\mathbf{R}_m$ and $\mathbf{R}_d$ are to the identity matrix, the smaller is the spread of these two resolution matrices.

6.4.3 Nonlinear inverse problem

6.4.3.1 Linearization by transformation of variables

Solving a nonlinear inverse problem represents a rather ordinary situation in geosciences and engineering even within the ambit of linear field and constitutive theories. The actual nonlinearity, in fact, involves the relationship between observable quantities (i.e., the measured experimental data) and a set of model parameters such as in Equation 6.1. This relationship may, of course, be nonlinear even in the context of otherwise fully linear theories. In surface wave testing, this situation occurs with the problem of determining the shear wave velocity profile from the inversion of an experimental dispersion curve in linear elastic half-spaces, as illustrated by Equation 6.2.

There are certain categories of nonlinear inverse problems that can be easily converted into corresponding linear problems through a *transformation of variables*. A typical example occurs in surface wave testing when performing attenuation measurements of particle displacement spectra (see Section 5.2).

In the uncoupled approach where dispersion and attenuation curves are measured independently, the problem of determining the experimental attenuation curve is based on the solution of the following nonlinear regression (Rix et al. 2001)

$$\left|T(\mathbf{r},0,\omega)\right| = \left|T(\mathbf{r},\omega)\right| \frac{C}{\sqrt{\mathbf{r}}} e^{-\alpha_R(\omega)\cdot\mathbf{r}} \tag{6.25}$$

where $\left|T(\mathbf{r},\omega)\right|$ is the amplitude of vertical displacement transfer function between the source and the receiver that is determined experimentally, C is a constant, $\mathbf{r}$ is the vector of receiver offsets, and $\alpha_R(\omega)$ is the unknown Rayleigh attenuation function. In Equation 6.25, higher modes of propagation are neglected.

This equation is nonlinear in the model parameter α_R and for every discrete value of the angular frequency ω_j ($j = 1, nf$), $\alpha_R(\omega_j)$ can be determined through a nonlinear regression of the measured transfer function amplitudes $\left|T(\mathbf{r},\omega_j)\right|$. Figure 6.9a shows an example of such calculation at a real testing site. The goodness of the fitting is excellent, which shows that at the frequency of 69.5 Hz the attenuation response at this site is dominated by the fundamental mode of propagation.

A quick inspection of Equation 6.25 suggests that by applying a logarithm transformation to both sides of this equation, the problem of determining the attenuation curve can be reduced to that of performing a linear regression

$$\log\left(\sqrt{\mathbf{r}}\cdot\left|T(\mathbf{r},\omega)\right|\right) = \log C - \alpha_R(\omega)\cdot\mathbf{r} \tag{6.26}$$

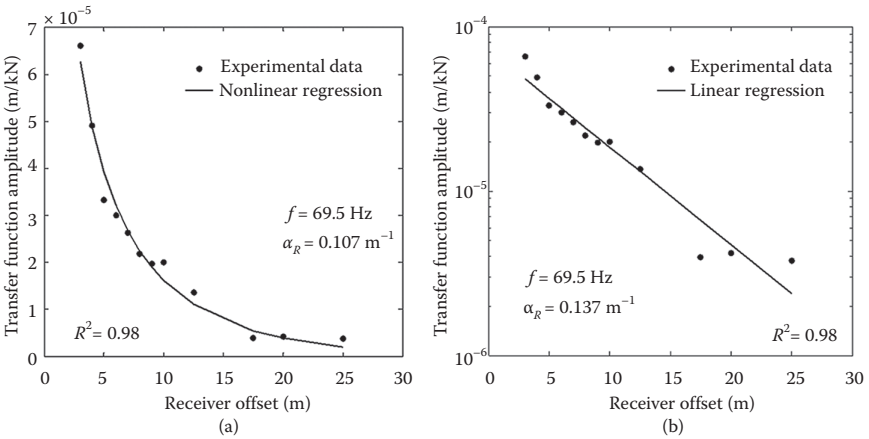

Figure 6.9 (a) Nonlinear and (b) linearly transformed regression analyses in surface wave attenuation measurements.

If r_k denotes the discrete receiver offset (with $k = 1$, np and np the number of receiver positions), Equation 6.26 can be easily recasted like Equation 6.4, which corresponds to the standard linear problem. The equivalence is shown here:

$$\mathbf{G}\cdot\mathbf{m} = \mathbf{d} = \begin{bmatrix} 1 & -r_1 \\ 1 & -r_2 \\ \dots & \dots \\ \dots & -r_i \\ 1 & \dots \\ 1 & -r_{np} \end{bmatrix} \cdot \begin{bmatrix} \log C_j \\ \alpha_R(\omega_j) \end{bmatrix} = \mathbf{d} \tag{6.27}$$

where $\mathbf{d}$ is the data measurement vector and is given by

$$\mathbf{d} = \left[\sqrt{r_1}\cdot\left|T(r_1,\omega_j)\right| \ \sqrt{r_2}\cdot\left|T(r_2,\omega_j)\right| \ \dots \ \sqrt{r_i}\cdot\left|T(r_i,\omega_j)\right| \ \dots \ \sqrt{r_{np}}\cdot\left|T(r_{np},\omega_j)\right|\right]^T \tag{6.28}$$

Figure 6.9b shows the linear regression of the same data illustrated in Figure 6.9a after performing the linearization described by Equations 6.27 and 6.28. The attenuation coefficient obtained with the linear regression is about 30% greater than the one calculated using Equation 6.25. Obviously, the two procedures are not equivalent because they lead to different results. Also the goodness of the fitting is slightly dissimilar.

This example shows that although it sometimes may be convenient to apply the method of transformation of variables to linearize a nonlinear inverse problem, this procedure should be used with caution because it always introduces a bias in the solution (Menke 1989). This is particularly relevant if the uncertainty of the data has to be taken into account (Section 6.5). In linear regressions, the rigorous use of the least-squares method in the presence of uncertainty requires that data measurements are assumed to be uncorrelated, normally distributed (or Gaussian) random variables with uniform variance (Section 6.5.1). If the statistics of the data, represented, for instance, by Equation 6.28, is back-transformed into the original variables due to the nonlinearity of Equation 6.25, the original measurements will no longer be normally distributed. In addition, the fact that the data in Equation 6.28 were assumed characterized by uniform variance implies that the transfer function amplitudes were measured with an accuracy that increased with the distance from the source. This is exactly the opposite of what actually happens in reality. Because the signal-to-noise ratio tends to deteriorate with the increase of r due to the spatial, exponential decay of the displacement amplitude spectra, the accuracy of the measurements progressively decreases with the increase in the distance from the source.

6.4.3.2 LS iterative methods and GS techniques

A natural procedure for solving Equation 6.1 is to expand this equation in a Taylor's series about an initial guess of model parameters $\mathbf{m}_0$

$$\mathbf{G}(\mathbf{m}) = \mathbf{G}(\mathbf{m}_0) + \mathbf{J}(\mathbf{m})_{\mathbf{m}_0} \cdot (\mathbf{m} - \mathbf{m}_0) + o\left\|(\mathbf{m} - \mathbf{m}_0)\right\|_2^2 \tag{6.29}$$

where $\mathbf{G}(\mathbf{m}_0)$ denotes the vector-valued function $\mathbf{G}$ of Equation 6.1 calculated for $\mathbf{m} = \mathbf{m}_0$ and $\mathbf{J}(\mathbf{m})_{\mathbf{m}_0}$ is the N by M *Jacobian matrix* calculated for $\mathbf{m} = \mathbf{m}_0$, and it is defined by

$$\mathbf{J}(\mathbf{m})_{\mathbf{m}_0} = \left[grad\,\mathbf{G}(\mathbf{m})\right]_{\mathbf{m}_0} \tag{6.30}$$

where $i = 1, N$ and $j = 1, M$. By neglecting terms higher than the first order, Equation 6.29 simplifies as follows

$$\mathbf{J}(\mathbf{m})_{\mathbf{m}_0} \cdot \mathbf{m} = \mathbf{J}(\mathbf{m})_{\mathbf{m}_0} \cdot \mathbf{m}_0 + \left[\mathbf{d} - \mathbf{G}(\mathbf{m}_0)\right] \tag{6.31}$$

Equation 6.31 represents a linear problem analogous to Equation 6.4 because all terms but $\mathbf{m}$ are known. Thus, it could be used as a basis for the application of any of the methods discussed in Section 6.4.2 for the solution of the linear inverse problem, including the least-squares algorithm.

The result will be a new estimate $\mathbf{m}_1$ of the unknown vector of model parameters. The procedure can then be repeated, and Equation 6.31 will provide a sequence $\{\mathbf{m}_0, \mathbf{m}_1, \mathbf{m}_2, \ldots \mathbf{m}_k, \ldots \mathbf{m}_n\}$ of successive approximations of the model parameters that under appropriate circumstances and conditions will converge toward the desired vector $\mathbf{m}_{true}$ of *true* model parameters. For a fully nonlinear, overdetermined problem, the recursive equation for the least-squares method can be written as follows

$$\mathbf{m}_{k+1} = \left(\mathbf{J}_k^T \mathbf{J}_k\right)^{-1} \mathbf{J}_k^T \cdot \left\{\mathbf{J}_k \cdot \mathbf{m}_k + \left[\mathbf{d} - \mathbf{G}(\mathbf{m}_k)\right]\right\} \tag{6.32}$$

More refined iterative methods exist to solve the least-squares nonlinear inverse problem including the downhill simplex method and various types of gradient techniques such as the method of steepest descent, conjugate gradient, the Gauss–Newton method, the Levenberg–Marquardt algorithm. The latter is the method of choice for small- to medium-size nonlinear least-squares problems (Aster et al. 2005).

Although a nontrivial issue related to iterative methods has to do with the definition of a criterion for deciding when to terminate the iterations, the two major difficulties associated with the iterative solution of inverse problems are convergence and uniqueness of the solution. For instance,

there is no guarantee that the iterative scheme specified by Equation 6.32 converges, for a sufficiently large k, to the "true" solution. Even if it does, the solution may not be unique even in overdetermined problems.

In linear inverse problems (Section 6.4.3), the situation was far more favorable. First, there was no need to introduce iterative strategies to determine the solution. Second, depending on the degree of indeterminacy of the problem at hand, a unique solution could always have been identified, if needed, by supplementing the problem with a priori information on the solution's simplicity. All these approaches are not guaranteed to work in nonlinear inverse problems. The ill-posedness that characterizes almost any inverse problem becomes severe and sometimes pathological if the problem is nonlinear.

Figure 6.10 shows a conceptual plot of the prediction error, defined according to Equation 6.5, as a function of model parameter for the case of a nonlinear (panel a) and linear (panel b) inverse problem. Solution uniqueness for the linear problem is guaranteed by the existence of a single minimum of the prediction error function the shape of which, in the most general case, is a multidimensional paraboloid.[23] If the problem is nonlinear, the error hypersurface may be nonconvex; thus, it may have multiple minima, and there is a need to distinguish between a *local* minimum and a *global* minimum (Figure 6.10a).

The "true" solution would correspond to the global minimum of the error hypersurface; however, depending on the initial guess introduced in the iterative scheme, the algorithm may actually converge and provide a solution corresponding to a local minimum. Thus, the main problem

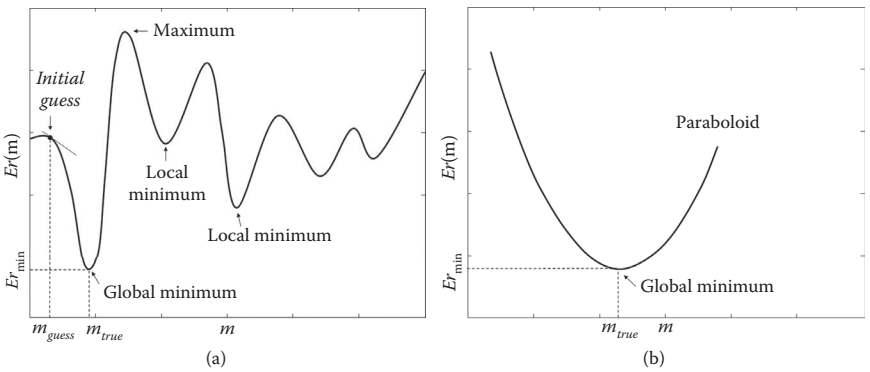

Figure 6.10 Prediction error as a function of model parameter in the solution of a least-squares (a) nonlinear and (b) linear inverse problem.

[23] For ease of graphical representation, in Figure 6.10, the prediction error function has been assumed to be a function of only a single model parameter.

associated with *LS iterative methods* (Figure 6.3) is the choice of an initial guess that is sufficiently close to the global minimum. This practically may be very difficult due to the extension of the solution space coupled with the objective difficulty of anticipating the position of a neighborhood of a global minimum. A wrong initial guess may lead to a failure of the iterative process represented, for instance, by Equation 6.32, and even a local minimum may not be found. As a consequence, the convergence of an LS iterative method will strongly depend on the geometry of the prediction error hypersurface (Menke 1989). Figure 6.11 shows four examples of increasing degrees of complexity that may be encountered in solving nonlinear inverse problems, from a single, well-defined

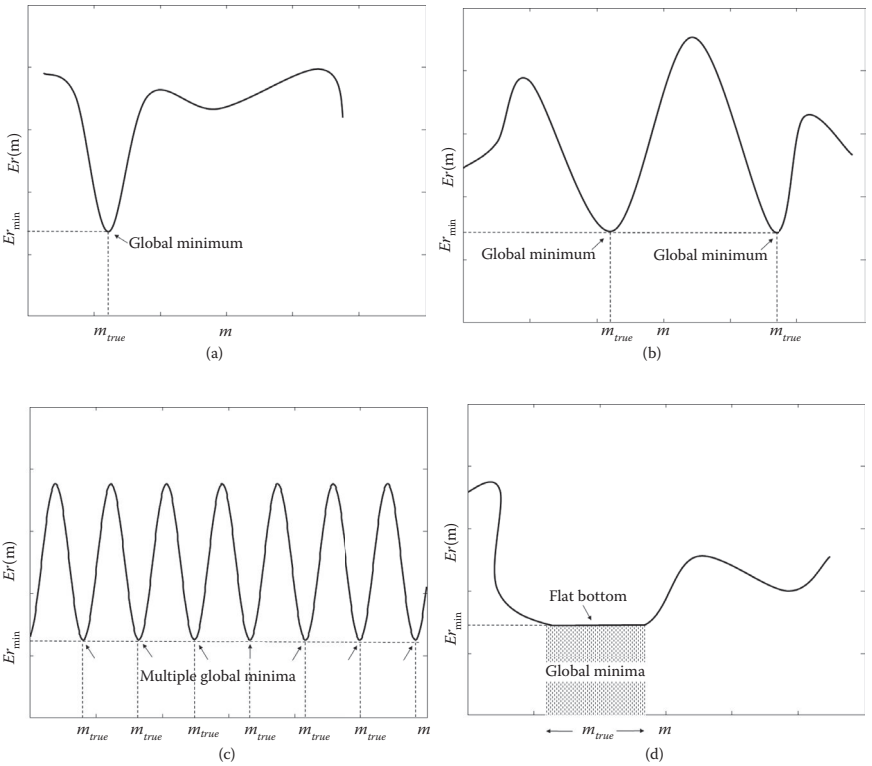

Figure 6.11 Prediction error as a function of model parameter in the solution of a least-squares nonlinear inverse problem. (a) Well-defined single minimum corresponding to a unique solution. (b) Two well-separated minima with lack of uniqueness in the solution. (c) Infinite well-separated countable minima with severe lack of uniqueness in the solution due to periodicity of the prediction error function. (d) "Flat bottom" having an uncountable finite range of solutions with severe lack of uniqueness due to ill-conditioning of the prediction error function.

global minimum (Figure 6.11a) to a "flat bottom" with an uncountable, finite range of solutions (Figure 6.11d).

GS methods (Figure 6.3) have been purposely developed to overcome these situations; in recent years, a great deal of research that is still very active has been carried out in this field. The overall subject falls under the name of *global optimization* (Sen and Stoffa 1995; Horst et al. 2000). Robust deterministic—and for large-size problems, stochastic—algorithms have been created to find the "true" solution of nonlinear inverse problems after scanning the whole error hypersurface in a search for the global minimum.

A conceptually simple GS method is represented by the so-called *multistart strategy* (Aster et al. 2005), which consists of randomly generating a large number of initial guess solutions to any one, of which a LS iterative algorithm such as the Gauss–Newton or the Levenberg–Marquardt is applied. The set of the local minima found by the method is then compared to find the solution corresponding to the global minimum. For the surface wave inverse problem represented by Equation 6.2, a set of initial guess solutions would be given by a series of initial profiles of shear wave velocity.

Multistart methods are efficient GS procedures because they can take advantage of the typical fast converging properties of LS iterative algorithms. Other global optimization techniques include simulated annealing, evolutionary algorithms, fractal inversion, enumerative methods, and, of course, Monte Carlo simulation. Although the "philosophy" adopted in each of these techniques is peculiar and different from each of the others, all aim at finding the global minimum of the prediction error function without being trapped in local minima.

GS methods have already been successfully applied in solving the inverse problem associated with surface waves. Yamanaka and Ishida (1996) have developed a genetic algorithm for the inversion of surface wave dispersion data. Although the application is concerned with the definition of the Earth crust model from the inversion of dispersion curves obtained from observed earthquake data, the algorithm could also be used for near-surface site characterization. By mimicking population genetics and processes such as selection, crossover, and mutation, genetic algorithms can simultaneously search globally and locally for the minimum of the prediction error function in the solution space by using several forward models. The authors of this paper developed a genetic algorithm to invert short- and intermediate-period surface wave dispersion data to obtain a shear wave velocity profile for a deep sedimentary basin.

Direct applications of genetic algorithms (Pezeshk and Zarrabi 2005; Dal Moro et al. 2007), simulated annealing (Beaty et al. 2002), and Monte Carlo simulations (Socco and Boiero 2008; Maraschini and Foti 2010; Bergamo et al. 2011) for near-surface site characterization using surface waves have also been done. It is an active field of research; the contributions are countless and continue to be produced. Only a few of them have been cited here.

Although artificial neural network inversion cannot be classified as a GS method, it also represents an alternative to LS techniques. Meier and Rix (1993) have proposed the use of an artificial neural network as an expeditious alternative to the trial-and-error and least-squares surface wave inversion techniques. About 99,000 synthetic dispersion curves were calculated for randomly generated two-layer shear wave velocity profiles using a theoretical wave propagation algorithm. An artificial neural network was then "taught" to map these dispersion curves back into their respective shear wave velocity profiles. After the network was successfully trained on these synthetic dispersion curves, simulated experimental dispersion curves were inverted by passing them through the neural network. Because the neural network required only a single forward pass of the data, the algorithm performed the inversion much more quickly than the standard iterative procedures. Other applications of the artificial neural network algorithm to solve the surface wave inverse problem include the works by Kim and Xu (2000) and Shirazi et al. (2009).

Similar to linear problems, the nonuniqeness of nonlinear inverse problems may also be due to the underdeterminacy or mixed-determinacy of Equation 6.1. This happens when the number of model parameters increases considerably with respect to the number of the experimental data (Aster et al. 2005). These situations may be resolved by applying regularization methods, specifically the zeroth and higher-order Tikhonov regularization that were introduced for the linear case. Formally, Equations 6.15 and 6.20 remain valid as long as the matrix product $\mathbf{Gm}$ is replaced by the nonlinear vector-valued function operator $\mathbf{G}(\mathbf{m})$. The same occurs for the solutions represented by Equations 6.16 and 6.17 on the condition that they are intended in a recursive sense such as it was in Equation 6.32 for the least-squares method. Once the regularization has been implemented using, for instance, the variational formulation, the actual solution can then be obtained using standard least-squares, LS iterative methods (e.g., the Levenberg–Marquardt algorithm), or GS techniques.

6.4.3.3 Analytical versus numerical Jacobian

Most least-squares, LS iterative methods require the computation of the *Jacobian* of $\mathbf{G}(\mathbf{m})$ with respect to model parameters $\mathbf{m}$ as shown by Equation 6.30. Depending on the problem, the partial derivatives involved in the calculation of the Jacobian may be available in closed-form, that is, analytically, or they must be computed numerically using a finite difference scheme. Numerical differentiation itself is known to be an ill-conditioned problem that introduces inaccuracies and instabilities in the implementation of the iterative scheme represented by equations such as Equation 6.32.

A critical issue in the calculation of finite difference partial derivatives is the correct choice for the step of the perturbed model parameters $\delta\mathbf{m}$. It must be not too small to avoid round-off errors in the term at the numerator $\mathbf{G}(\mathbf{m} + \delta\mathbf{m}) - \mathbf{G}(\mathbf{m})$ of the finite difference, and simultaneously, it must not be too large or the calculation may become too inaccurate. A guiding principle is to set $\delta m_j = \sqrt{\varepsilon}$ ($j = 1, M$), where ε is the accuracy in the evaluations of $\mathbf{G}(\mathbf{m})$ (Aster et al. 2005). Instability and lack of convergence of an algorithm implementing a nonlinear inversion may be caused by imprecise calculation of the Jacobian matrix of $\mathbf{G}(\mathbf{m})$. The other problem related to computing partial derivatives numerically is that it is computationally very inefficient and time-consuming if compared with using the corresponding explicit formulas.

In light of this, it is obviously preferable when solving nonlinear inverse problems with LS iterative methods to use closed-form, analytical expressions of the partial derivatives of $\mathbf{G}(\mathbf{m})$ with respect to model parameters $\mathbf{m}$ entering the Jacobian matrix. However, they are not always available, and sometimes the recourse to numerical differentiation is compulsory. With regard to the solution of the nonlinear surface wave problem represented by Equation 6.2, the situation is favorable because exact formulas for computing the partial derivatives of $\mathbf{G}(\mathbf{V}_S)$ with respect to $\mathbf{V}_S$ do exist. They can be derived from the variational principles of Love and Rayleigh waves (Aki and Richards 2002), and results have been obtained for computing the Jacobian of modal and apparent surface wave phase velocity with respect to medium parameters $\mathbf{V}_S$ and $\mathbf{V}_P$. For a stratified medium composed of a finite number of homogeneous layers overlaying a homogeneous half-space, the partial derivatives of the Rayleigh phase velocity V_R with respect to changes of $(\mathrm{V}_S)_i$ and $(\mathrm{V}_P)_i$ of the ith layer ($i = 1, nl$) are given by the following relations

$$\begin{cases} \left(\dfrac{\partial V_R}{\partial V_S}\right)_i = \dfrac{(\rho V_S)_i}{2k^2 U_R I_R} \cdot \displaystyle\int_{x_2^i}^{x_2^{i+1}} \left[\left(kr_2 - \dfrac{dr_1}{dx_2}\right)^2 - 4k\, r_1 \dfrac{dr_2}{dx_2}\right] dx_2 \\ \left(\dfrac{\partial V_R}{\partial V_P}\right)_i = \dfrac{(\rho V_P)_i}{2k^2 U_R I_R} \cdot \displaystyle\int_{x_2^i}^{x_2^{i+1}} \left(kr_1 + \dfrac{dr_2}{dx_2}\right)^2 dx_2 \end{cases} \tag{6.33}$$

where the brackets with a subscript i denote quantities evaluated in the ith layer. The limits of integration indicate the lower and upper bounds of the depth variable x_2 for the ith layer. See Chapter 2 for the definitions of the other terms appearing in Equation 6.33. The derivatives of the eigenfunctions r_1 and r_2 with respect to depth, appearing inside the integral sign, can be computed directly from the knowledge of the eigenfunctions (see Equation 2.66). Details for the derivation of Equation 6.33 are reported in Aki and Richards (2002) and Lai (2005).

The remarkable feature of these equations, which makes them so important in the solution of the Rayleigh inverse problem, is that the partial derivatives of V_R with respect to medium parameters can be computed using Rayleigh wave parameters referred to the original and not the perturbed V_S and V_P profiles. Conversely, it would be very expensive to compute these partial derivatives numerically with, say, a four-point central finite difference scheme; a single computation of a derivative would require the solution of four Rayleigh eigenproblems instead of just one eigenproblem using Equation 6.33.

It can be shown (Lee and Solomon 1979; Ben-Menahem and Singh 2000) from a detailed assessment of Equation 6.33 that the phase velocity of Rayleigh waves is relatively insensitive to changes in the parameter V_P; thus, the corresponding partial derivative is small if compared to the derivative of V_R with respect to V_S. This result is important for the solution of the Rayleigh inverse problem because it implies that inverting an equation analogous to Equation 6.2 for V_P would be a severe ill-posed problem. In fact, relatively large changes in V_P would correspond to small changes in V_R and, thus, in the prediction error function of Equation 6.5, a situation graphically described by the "flat bottom" shown in Figure 6.11d. For this reason, either the V_P profile or the Poisson ratio profile at a site is typically assumed as known in the inversion of an experimental dispersion curve.

Equation 6.33 has been derived and, therefore, is valid for a stratified medium, that is, for a discrete model of the subsoil. Similar relations have been derived by Rix and Lai (2014) for a subsurface model in which the material properties vary continuously with depth. Once the explicit formulas for the partial derivatives of V_R with respect to the medium parameters have been calculated, they can be introduced in Equation 6.30 to form the Jacobian matrix required to solve the inverse Rayleigh problem[24] with any of the LS iterative methods discussed in the previous section.

6.4.3.4 An example of a LS iterative method: Occam's algorithm

This section will describe in detail the application of Tikhonov regularization for the inversion of an experimental dispersion and attenuation curve to obtain the unknown shear wave velocity and shear damping profiles at a site. The procedure is known in the literature as the *Occam's algorithm*, and it has been introduced in geophysics by Constable et al. (1987) for the inversion of electromagnetic sounding data.

The name of the algorithm refers to William of Ockham, a fourteenth-century English philosopher considered the father of a philosophical

[24] Explicit formulas analogous to Equation 6.33 also hold for the Love waves. For details, see Aki and Richards (2002).

principle called the *Occam's razor* according to which parsimony and conciseness should be the guiding principles to be adopted in the interpretation of natural phenomena and in the formulation of new theories. He wrote that "it is vain to do with more what can be done with fewer" (Russell 1946). Occam's razor also has become an inspiring principle in today's science because the assumptions of a theory should not be unnecessarily numerous and complicated.

Constable et al. (1987) have adopted this principle to formulate what in essence is a constrained, damped least-squares, LS iterative method for the solution of a nonlinear inverse problem. This algorithm will now be applied to the inversion of Equation 6.2 written in this form

$$\mathbf{G}^*\left(\mathbf{V}_S^*\right) = \mathbf{V}_R^* \tag{6.34}$$

where the "asterisk" notation refers to the complex-valued nature of the variables involved. It was shown in Section 2.5.3 that in linear dissipative media, phase velocity and attenuation of viscoelastic waves are not independent; furthermore, they are frequency dependent as a result of material dispersion. Thus, a correct inversion procedure of surface wave data should determine these two parameters jointly, at the same frequency of excitation and for arbitrary values of material damping ratio (Lai et al. 2002).

A natural way to construct an inversion algorithm satisfying these requirements is to start from the solution of the forward problem of surface waves in strongly dissipative, layered media illustrated in Section 2.5.3. In the frequency domain, a linear viscoelastic, arbitrary dissipative, isotropic material has its constitutive parameters completely defined by the complex-valued phase velocity of bulk P- and S-waves given by Equation 2.125 and rewritten here for convenience

$$V_\chi^*(\omega) = \frac{V_\chi(\omega)}{\sqrt{[1+4D_\chi^2(\omega)]}} \cdot \left[\frac{1+\sqrt{[1+4D_\chi^2(\omega)]}}{2} + i \cdot D_\chi\right] \tag{6.35}$$

with $\chi = P,S$ and by the Kramers–Krönig relations (Equations 2.130, 2.131, or 2.132) specifying the functional dependence between material damping ratio $D_\chi(\omega)$ and phase velocity $V_\chi(\omega)$. In Equation 6.35, $i = \sqrt{-1}$ is the imaginary unit. The formalism of complex-valued bulk wave velocities allows two fundamental model parameters of the medium (the shear wave velocity and the shear damping ratio) to group together in a single parameter.

An algorithm for the strongly coupled, joint inversion of Rayleigh dispersion and attenuation data may be constructed by viewing the complex-valued Rayleigh phase velocity as a *holomorphic* (or *analytic*) *mapping* (Remmert 1997) of the complex-valued speed of propagation of shear waves of the medium. Formally, this mapping is

represented by Equation 6.34, where $\mathbf{V}_S^* = \left[\left(V_S^*\right)_1, \left(V_S^*\right)_2, \ldots \left(V_S^*\right)_i \ldots \left(V_S^*\right)_{nl}\right]$ and $\mathbf{V}_R^* = \left[\left(V_R^*\right)_1, \left(V_R^*\right)_2, \ldots \left(V_R^*\right)_j \ldots \left(V_R^*\right)_{nf}\right]$. Again the symbols *nl* and *nf* denote the number of layers (including the half-space) of a stratified medium and the set of experimental frequencies at which the Rayleigh phase velocity has been measured, respectively.

Equation 6.34 represents the Rayleigh forward problem in linear visco-elastic media. The complex-valued Rayleigh phase velocity vector $\mathbf{V}_R^*$ jointly encloses the dispersion and attenuation (discrete) curves. It is computed by combining Equation 6.35 and the following relation (Lai et al. 2002)

$$D_R(\omega) = \left[\frac{\dfrac{\alpha_R \cdot V_R}{\omega}}{1 - \left(\dfrac{\alpha_R \cdot V_R}{\omega}\right)^2}\right] \tag{6.36}$$

where $\alpha_R = \alpha_R(\omega)$ is the frequency-dependent Rayleigh attenuation coefficient.

For the solution of the Rayleigh inverse problem, Equation 6.34 can be expanded in a *Taylor series* about an initial guess of the model parameter vector $\mathbf{V}_{S_0}^*$, thereby obtaining

$$\mathbf{G}^*\left(\mathbf{V}_S^*\right) = \mathbf{G}^*\left(\mathbf{V}_{S_0}^*\right) + \mathbf{J}^*\left(\mathbf{V}_S^*\right)_{\mathbf{V}_{S_0}^*} \cdot \left(\mathbf{V}_S^* - \mathbf{V}_{S_0}^*\right) + o\left\|\left(\mathbf{V}_S^* - \mathbf{V}_{S_0}^*\right)\right\|_2^2 \tag{6.37}$$

where... $\|\ldots\|_2$ denotes the Euclidean norm of a complex-valued vector[25] and $\mathbf{G}^*\left(\mathbf{V}_{S_0}^*\right) = \mathbf{V}_{R_0}^*$ is the *nf* by 1 vector of Rayleigh phase velocities corresponding to the solution of Equation 6.34 with model parameter equal to $\mathbf{V}_{S_0}^*$. The term $\mathbf{J}^*\left(\mathbf{V}_S^*\right)_{\mathbf{V}_{S_0}^*}$ represents the *nf* by *nl* complex-valued Jacobian matrix calculated for $\mathbf{V}_S^* = \mathbf{V}_{S_0}^*$, and it is defined by the following expression

$$\mathbf{J}^*\left(\mathbf{V}_S^*\right)_{\mathbf{V}_{S_0}^*} = \left[grad\ \mathbf{G}^*\left(\mathbf{V}_S^*\right)\right]_{\mathbf{V}_{S_0}^*} \tag{6.38}$$

The elements of the Jacobian matrix are defined as follows

$$\left[\left(\mathrm{J}^*\right)_{jk}\right]_{\mathrm{V}_{S_0}^*} = \left[\frac{\partial\left[\mathrm{G}^*\left(\mathrm{V}_S^*\right)\right]_j}{\partial\left(\mathrm{V}_S^*\right)_k}\right]_{\mathrm{V}_{S_0}^*} = \left[\frac{\partial\left(\mathrm{V}_R^*\right)_j}{\partial\left(\mathrm{V}_S^*\right)_k}\right]_{\mathrm{V}_{S_0}^*} \tag{6.39}$$

[25] The Euclidean norm of a complex-valued vector $\mathbf{v}^*$ is given by the square root of the product of vector $\mathbf{v}^*$ by vector $(\mathbf{v}^*)^H$, which is the transpose of the complex conjugate of $\mathbf{v}^*$. The vector $(\mathbf{v}^*)^H$ is also called the *Hermitian* transpose of *vector* $\mathbf{v}^*$.

where $j = 1, nf$ and $k = 1, nl$. The subscripts outside the brackets of Equation 6.39 indicate the value of the model parameter at which the Jacobian matrix is evaluated. The partial derivatives of the Jacobian are calculated using Equation 6.33 extended by *analytic continuation* to complex values of field variables.

Neglecting terms higher than the first order, Equation 6.37 reduces to

$$\begin{aligned} \mathbf{J}^*\left(\mathbf{V}_S^*\right)_{\mathbf{V}_{S_0}^*} \cdot \mathbf{V}_S^* &= \mathbf{J}^*\left(\mathbf{V}_S^*\right)_{\mathbf{V}_{S_0}^*} \cdot \mathbf{V}_{S_0}^* + \left[\mathbf{G}^*\left(\mathbf{V}_S^*\right) - \mathbf{G}^*\left(\mathbf{V}_{S_0}^*\right)\right] \\ &= \mathbf{J}^*\left(\mathbf{V}_S^*\right)_{\mathbf{V}_{S_0}^*} \cdot \mathbf{V}_{S_0}^* + \left(\mathbf{V}_R^* - \mathbf{V}_{R_0}^*\right) \end{aligned} \tag{6.40}$$

Equation 6.40 could be used as a basis for the implementation of a standard least-squares algorithm to determine the complex-valued shear wave velocity profile of a layered medium possibly subjected, in under- and mixed-determined problems, to the minimum norm constraint. However, as discussed extensively earlier, inversions performed with algorithms where the solution is constrained to have the minimum norm are often inadequate, and they may lead to physically unreasonable profiles of model parameters (Constable et al. 1987). The inadequacy of this class of algorithms often can be attributed to a lack of physical justification for assuming the minimum norm constraint.

A more reasonable approach for constraining the solution is represented by the *Occam's inversion* (Parker 1994), the strategy of which is to find the *smoothest* profile of model parameters subjected to the constraint that the error misfit between observed and predicted data cannot exceed a prescribed value. In the current problem, the profile of model parameters is represented by the vector $\mathbf{V}_S^*$ containing the complex-valued shear wave velocities of the individual layers. Because the number of layers nl is generally assumed, the inverted profile of model parameters will depend on the a priori assumption about nl, and it may contain large discontinuities or other features that are not essential for matching the experimental dispersion and the attenuation curves. By enforcing maximum smoothness and regularity in the solution, it is possible to mitigate its dependence upon the assumed number of layers and at the same time reject solutions that are unnecessarily complicated (Constable et al. 1987).

Implementation of the algorithm requires a quantitative definition of *smoothness* of a profile of model parameters. This definition, actually its opposite *roughness*, has already been introduced in Section 6.4.2 when discussing higher-order Tikhonov regularization methods. Equations 6.18 and 6.19 provide two scalar definitions, R_1 and R_2, of the *roughness* of a discrete profile of model parameters. However, they are applicable only for real-valued model parameters. For complex-valued model parameters

such as $\mathbf{V}_S^*$, roughness may be defined by either one of the two following expressions (Menke 1989; Lai 2005)

$$\begin{cases} R_1 = \left\|\mathbf{L}\mathbf{V}_S^*\right\|_2^2 = \left(\mathbf{L}\mathbf{V}_S^*\right)^H \cdot \left(\mathbf{L}\mathbf{V}_S^*\right) \\ R_2 = \left\|\mathbf{L}\left(\mathbf{L}\mathbf{V}_S^*\right)\right\|_2^2 = \left\|\mathbf{L}^2\mathbf{V}_S^*\right\|_2^2 = \left(\mathbf{L}^2\mathbf{V}_S^*\right)^H \cdot \left(\mathbf{L}^2\mathbf{V}_S^*\right) \end{cases} \tag{6.41}$$

where $\mathbf{L}$ is the *nl* by *nl* real-valued matrix representing the two-point central finite difference operator defined in Equation 6.19, and the symbol $(\ldots)^H$ denotes the Hermitian transpose of a complex-valued matrix.

The prediction error misfit *Er* between the measured and the predicted complex-valued Rayleigh phase velocities may be written as follows

$$Er = \left[\mathbf{W}^*\bar{\mathbf{V}}_\mathrm{R}^* - \mathbf{W}^*\mathbf{G}^*\left(\mathbf{V}_\mathrm{S}^*\right)\right]^H \cdot \left[\mathbf{W}^*\bar{\mathbf{V}}_\mathrm{R}^* - \mathbf{W}^*\mathbf{G}^*\left(\mathbf{V}_\mathrm{S}^*\right)\right] \tag{6.42}$$

where $\bar{\mathbf{V}}_R^*$ is a complex-valued vector of experimental Rayleigh phase velocities, and $\mathbf{W}^*$ is a complex-valued diagonal *nf* by *nf weight matrix* defined by

$$\mathbf{W}^* = \begin{bmatrix} 1/\sigma_1^* & \ldots & 0 & \ldots & \ldots & 0 \\ 0 & 1/\sigma_2^* & \ldots & 0 & \ldots & 0 \\ 0 & \ldots & \ldots & \ldots & 0 & 0 \\ 0 & 0 & \ldots & 1/\sigma_j^* & \ldots & 0 \\ 0 & \ldots & 0 & \ldots & \ldots & 0 \\ 0 & \ldots & \ldots & 0 & \ldots & 1/\sigma_{nf}^* \end{bmatrix} \tag{6.43}$$

with σ_j^*, $j = 1, nf$ being the *uncertainties* associated with the experimental data $\bar{\mathbf{V}}_R^*$ (see Section 6.5.2 to learn how they can be estimated). Equation 6.42 defines a *weighted* measure of the error prediction misfit. This definition is able to accommodate the fact that, in real practice, some experimental measurements may be more accurate than others; thus, it would make more sense to assign to these measurements a greater weight in the overall estimate of the error misfit (Menke 1989). If $\mathbf{W}^* = \mathbf{1}$, that is the weight matrix is equal to the identity matrix, then Equation 6.42 reduces to the standard definition of error misfit given by Equation 6.5.

Solution of the *linearized* Rayleigh inverse problem represented by Equation 6.40 with the Occam's algorithm consists of finding a vector $\mathbf{V}_S^*$ that minimizes R_1 and R_2 defined by Equation 6.41 while constraining the residual error to be equal to $\hat{E}r$, an acceptable value in light of the uncertainties. This setting of the problem corresponds to the iterative application of the first- and second-order Tikhonov regularization (Section 6.4.2) to a locally linearized

nonlinear inverse problem. The method of Lagrange multipliers is employed to solve this complex-valued, constrained optimization problem resulting in

$$\mathbf{V}_S^* = \left\{ \mu\left(\mathbf{L}^{\mathrm{T}}\mathbf{L}\right) + \left[\mathbf{W}^* \cdot \left(\mathbf{J}^*\right)_{\mathbf{V}_{S0}^*}\right]^H \cdot \left[\mathbf{W}^* \cdot \left(\mathbf{J}^*\right)_{\mathbf{V}_{S0}^*}\right] \right\}^{-1} \cdot \left[\mathbf{W}^* \cdot \left(\mathbf{J}^*\right)_{\mathbf{V}_{S0}^*}\right]^H \cdot \mathbf{W}^* \cdot \left[\left(\mathbf{J}^*\right)_{\mathbf{V}_{S0}^*} \cdot \mathbf{V}_{S0}^* + \left(\overline{\mathbf{V}}_R^* - \mathbf{V}_{R0}^*\right)\right] \tag{6.44}$$

In Equation 6.44, μ is the Lagrange multiplier, which may be viewed as a *smoothing parameter* and must be determined with the additional constraint that the specified residual error $\hat{E}r$ is matched with a vector $\mathbf{V}_S^*$ composed only of negative imaginary parts. This condition will insure that the shear damping ratio $\mathbf{D}_S$ obtained from the inversion algorithm always turns out to be a positive quantity.

Equation 6.44 is used iteratively to refine the estimated complex-valued shear wave velocity profile $\mathbf{V}_S^*$ until convergence. Once the unknown complex-valued vector $\mathbf{V}_S^*$ is estimated, the real-valued shear wave velocities $(V_S)_k$ and shear damping ratios $(D_S)_k$, $(k = 1, nl)$ of the nl-layer medium can be recovered using the following relations, obtained from Equation 6.35

$$\begin{cases} \left(V_S\right)_k = \left[V_S(x_S, y_S)\right]_k = \left[\dfrac{(x_S^2 + y_S^2)}{x_S}\right]_k \\ \left(D_S\right)_k = \left[D_S(x_S, y_S)\right]_k = \left[\dfrac{x_S \cdot y_S}{(x_S^2 - y_S^2)}\right]_k \end{cases} \tag{6.45}$$

where $\left(x_S\right)_k = \left[\Re\left(V_S^*\right)\right]_k$ and $\left(y_S\right)_k = \left[\Im\left(V_S^*\right)\right]_k$ are the real and the imaginary parts of $\left(V_S^*\right)_k$, respectively, which is the complex-valued shear wave velocity of the kth layer.

In this derivation, no specific assumption was made about the frequency-dependence law of the complex-valued vector $\mathbf{V}_S^*$. A standard technique used to account for material dispersion is to carry out the inversion procedure at a prescribed *reference frequency* often assumed equal to $\omega_{ref} = 2\pi$ (Lee and Solomon 1979; Herrmann 2007). The procedure can then be used to perform a *causal inversion* of the Rayleigh dispersion and attenuation curves provided that the partial derivatives appearing in Equation 6.39 are computed with respect to $\left(V_{S(ref)}^*\right)_k$ and the vector $\mathbf{V}_S^*$ in Equation 6.44 is replaced by $\mathbf{V}_{S(ref)}^* = \mathbf{V}_S^*\left(\omega_{ref}\right)$.

Figures 6.12 and 6.13 show the results obtained from the application of the Occam's algorithm to a pair of synthetic dispersion and attenuation curves generated using the medium parameters of Table 6.1 (Lai 2005).

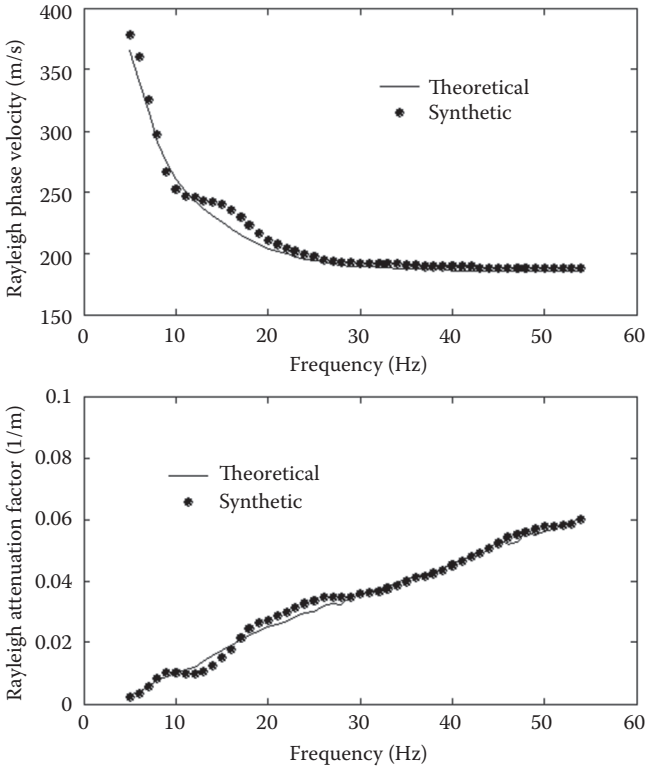

Figure 6.12 Comparison between synthetic Rayleigh dispersion and attenuation curves generated using the medium parameters of Table 6.1 and the corresponding theoretical curves (fundamental mode of propagation) obtained by employing the constrained, damped least-squares Occam's algorithm. (From Lai, C. G., Surface waves in dissipative media: Forward and inverse modelling, *Surface Waves in Geomechanics—Direct and Inverse Modelling for Soil and Rocks*, Springer-Verlag, New York, 2005.).

The synthetic response functions were computed by solving the Rayleigh inhomogeneous forward problem for a vertical, time-harmonic point source applied at the free boundary of the layered medium of Table 6.1. The displacement field was calculated at 24 receiver locations, with the closest receiver located at a distance equal to one wavelength from the source. The frequency range considered in the analysis varied from 5 to 54 Hz. The techniques used to determine dispersion and attenuation curves from the synthetic displacement field are not discussed here because they have been thoroughly explained in Chapters 4 and 5.

The algorithm converged after only three iterations. Figure 6.12 shows the agreement between *synthetic* and *theoretical* Rayleigh dispersion and

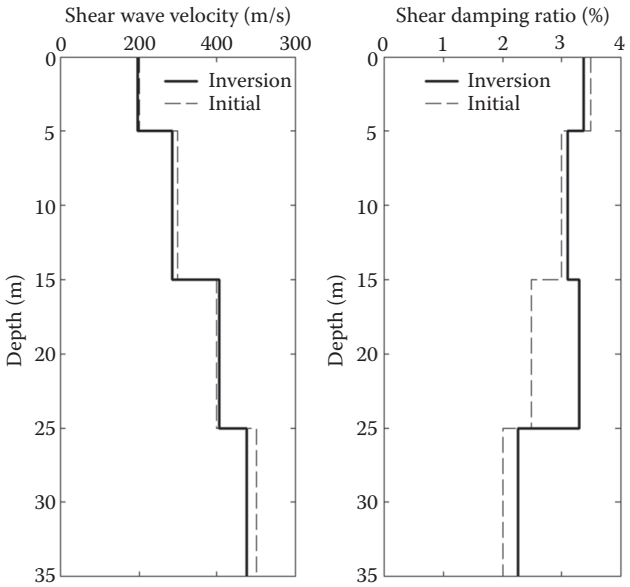

Figure 6.13 Shear wave velocity and shear damping ratio profiles obtained from the joint inversion of Rayleigh synthetic dispersion and the attenuation curves of Figure 6.12 using the constrained, damped least-squares Occam's algorithm (boldface line). The dashed lines denote the V_S and D_S profiles of Table 6.1. (From Lai, C. G., Surface waves in dissipative media: Forward and inverse modelling, *Surface Waves in Geomechanics—Direct and Inverse Modelling for Soil and Rocks*, Springer-Verlag, New York, 2005.)

Table 6.1 Medium parameters used to generate Rayleigh synthetic dispersion and attenuation curves*

Layer	*Thickness (m)*	V_P *(m/s)*	V_S *(m/s)*	D_P *(–)*	D_S *(–)*	ρ *(Mg/m³)*
1	5.0	400	200	0.020	0.035	1.7
2	10.0	600	300	0.015	0.030	1.8
3	10.0	800	400	0.010	0.025	1.8
Half-space	∞	1000	500	0.010	0.020	1.8

* Jointly inverted by employing the constrained, damped least-squares Occam's algorithm (Lai 2005).

attenuation curves at the last iteration. The theoretical curves were computed with reference to the *fundamental* mode of propagation, and the good agreement with the synthetic curves can be attributed to the fact that the layered medium of Table 6.1 is *normally dispersive*.

Figure 6.13 illustrates the shear wave velocity V_S and shear damping ratio D_S profiles resulting from the *joint* inversion of Rayleigh synthetic

dispersion and attenuation curves. They correspond to the theoretical curves of Figure 6.12. The comparison between the values of V_S and D_S predicted by the inversion algorithm (boldface lines) and the V_S and D_S profiles of Table 6.1 used to generate the synthetic dispersion and attenuation curves (dashed lines) is satisfactory.

The joint inversion of Rayleigh dispersion and attenuation curves is superior to the corresponding uncoupled analysis because it takes the inherent coupling existing between phase velocities and attenuation of viscoelastic waves as a consequence of material dispersion implicitly into account. However, there is another, perhaps more subtle, reason why a coupled analysis is preferred. An uncoupled inversion of dispersion and attenuation data requires the solution of two inverse problems for a total of $2 \cdot nl$ unknown model parameters, which are the shear wave velocities $(V_S)_k$ and shear damping ratios $(D_S)_k$, (k = 1, nl) of the nl-layer medium. The solution of these two inverse problems is not completely independent because the V_S profile obtained from the nonlinear inversion of the Rayleigh dispersion curve is subsequently used as input in the linear inversion of the attenuation curve to obtain the D_S profile (Section 2.5.3). Therefore, the amplification of the uncertainties resulting from inversion of the dispersion curve will carry over to the inversion of the attenuation curve, a process that will eventually add its own uncertainty. In other words, the uncoupled inversion suffers from a negative cross-coupling effect resulting from the solution of two inverse problems where the input data of one problem comes from the solution of the other.

Conversely, the joint inversion of both dispersion and attenuation curves is freed from this negative cross-coupling because both sets of experimental data are inverted simultaneously using the elegant formalism of complex variable theory. Furthermore, the coupled analysis of surface wave data takes advantage of a built-in *internal constraint* embedded in the algorithm that makes the inversion a *better-posed* mathematical problem. This internal constraint is represented by the *Cauchy–Riemann equations* that are satisfied by the Rayleigh phase velocity V_R^* if viewed as an *holomorphic* function $V_R^* = G^*\left(V_S^*\right)$ of the complex-valued shear wave velocity (Remmert 1997). In summary, application of complex variable theory for the joint inversion of surface wave data is not only an elegant procedure to account for the inherent coupling between phase velocities and dissipative properties of viscoelastic media, but it also improves the well-posedness of the associated inverse problem. An example of joint inversion of experimental data is presented in Chapter 7.

6.4.4 A priori information in surface wave inversion

The importance of a priori information as a means of mitigating the ill-posedness of linear and nonlinear inverse problems has been emphasized

throughout the whole chapter. Of particular significance is the need to alleviate the problems arising from the nonuniqeness of the solution emerging from the interpretation of the experimental measurements.

In geotechnical site investigation, surface wave tests are typically conducted in association with numerous other surveys including drilling of boreholes, penetration tests, other types of geophysical prospecting, and so on. The extension of an exploration program at a construction site is usually redundant, and a considerable amount of data is available for the definition of a geological–geotechnical model of the subsurface. This situation is ideal for making good use of cross-correlations among the results of various tests and of a priori information for the inversion of geophysical measurements including surface wave data. At a larger spatial scale, detailed geological information may also be useful to constrain for instance, the position of the bedrock or to define the geometries of the contacts among different geological formations or lithostratigraphic units.

6.4.4.1 Borehole logs

If borehole data and lithostratigraphic information are available, layer thicknesses may be assumed as known a priori information to reduce the number of unknowns in setting up a 1D geotechnical model at a site. If these data are not available, a good rule of thumb is to assume layer thicknesses increasing with depth to comply with the decrease of model parameter resolution with depth, which is an intrinsic shortcoming of non-invasive geophysical methods including surface wave testing. Obviously, this modeling assumption reduces the ability of surface wave tests to locate the spatial position of layer interfaces.

6.4.4.2 P-wave refraction survey

Testing configurations for MASW testing and seismic P-wave refraction survey are basically the same, so that with a minimum effort in ensuring data acquisition compatibility between the two tests, useful information from the refraction survey may be exploited for the inversion of surface wave data (Foti et al. 2003). Two typical situations in which refraction data are particularly valuable for the inversion of surface wave data are the presence of either a shallow bedrock interface or of a shallow water table.

Furthermore, P-wave refraction can also be useful for recognizing the presence of dipping layers or other lateral variations that cannot be resolved with 1D modeling adopted in the standard interpretation of surface wave data.

In the example reported in Figure 6.14, the depth of the bedrock roof evaluated with seismic refraction has been used as a constraint in

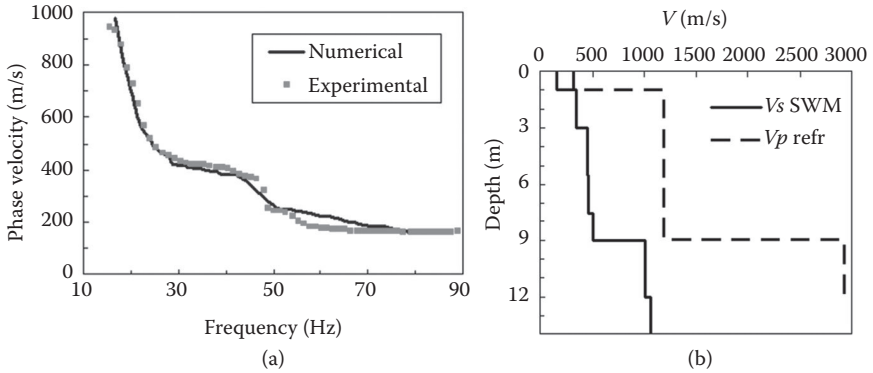

Figure 6.14 Inversion of surface wave data with constraints from P-wave refraction (a) theoretical and experimental dispersion curves. (b) Shear wave velocity profiles. (From Foti, S. et al. *Near Surface Geophys*, 1(1), 119–129, 2003.)

the inversion of surface wave data, and the final result is in good agreement with independent borehole logs at the same site.

The role that the spatial position of the water table plays in surface wave inversion is subtle, yet quite relevant. Although the presence of the water table does not have a great influence on the values of the measured shear wave velocity, which is the main goal of surface wave survey, it significantly affects the measured value of the speed of propagation of P-waves and, in turn, of the resulting Poisson ratio. Even though the dispersion curve's sensitivity to changes in the Poisson ratio is low compared to the sensitivity to changes in the shear modulus profile and layer thicknesses, the role played by the Poisson ratio during the inversion is not always negligible (Foti and Strobbia 2002). The transition from dry to wet soil, which is sharp for coarse-grained materials, is always associated with an abrupt change of Poisson ratio and mass density of the soil. If this transition is not properly recognized, it may lead to erroneous results.

The examples shown in Chapter 7 demonstrate with real data the importance of a priori information concerning the position of the water table when interpreting surface wave data. Although this information may be obtained through different types of exploration methods (e.g., electrical surveys, borehole drilling, piezometer measurements), P-wave refraction appears to be an effective and economically convenient option. When no information is available, it is necessary to take into account the influence that the uncertainty in defining the position of the water table may have on the results of surface wave measurements. A possible strategy in this regard could be to perform a few trials assuming reasonable guesses for the

position of the water table and then use *engineering judgment* to appreciate the differences, if any, among the results obtained from the inversions of surface wave data.

6.4.4.3 Joint inversion of geophysical data

Synergies of surface wave testing with other geophysical exploration methods can be effective for improving the reliability of the investigation campaign and for obtaining a more consistent soil model. Hayashi et al. (2005) performed a joint analysis of surface wave data and microgravity measurements aimed at detecting the presence of underground voids. Another appealing possibility is represented by the joint inversion of surface wave measurements and vertical electrical sounding (VES) data (Hering et al. 1995; Misiek et al. 1997; Comina et al. 2002; Wisén and Christiansen 2005). Indeed, despite the marked differences between these two geophysical techniques, the interpretation of data measurements presents rather strong similarities. VES data analysis is based on the inversion of field measurements consisting of an apparent resistivity curve, which can be considered analogous of the dispersion curve for surface wave data. Moreover, the inversion of the resistivity curve is typically based on assuming a 1D layered model similar to the one adopted in surface wave testing; however, in this case, the unknown parameters are the electrical resistivities of the layers.

Assuming that layer interfaces are the same for the seismic and the electrical model, it is possible to use a joint inversion strategy, simultaneously inverting the experimental dispersion and apparent resistivity curves to obtain thickness, shear wave velocity, and resistivity of each layer. In practice, $3nl-1$ unknowns are determined from the joint inversion of the two experimental datasets, whereas each uncoupled inversion aims at determining $2nl-1$ parameters from a single dataset. From a mathematical point of view, the solution of the joint inverse problem is better-posed than the solution of two single inverse problems.

The main limitation of this approach is demonstrated by situations in which variations in electrical properties are not associated with variations in seismic velocity or vice versa. Indeed, there are some circumstances in which a discontinuity in one of the two model parameters does not correspond to a discontinuity in the other. Another potential shortcoming of these types of joint inversions is computational. In fact, not all the partial derivatives of the response functions with respect to the model parameters required to construct the Jacobian matrix can be computed analytically using explicit formulas. Thus, the recourse to inefficient and unstable finite difference calculations and numerical differentiation is unavoidable (Section 6.4.3).

The general principles and further applications of joint inversion with other geophysical data are discussed in Section 8.3.

6.5 UNCERTAINTY

6.5.1 Inverse problems and measurement errors

6.5.1.1 Linear problems with Gaussian data errors

An essential aspect of any measurement technique is the ability to (a) estimate the uncertainty of the measured data and (b) to determine how this uncertainty is projected onto the model parameters derived from them. This is particularly important in geophysical prospecting where the parameters of interest are often inferred from complex inversion procedures. In surface wave testing, the measured data are represented by the Rayleigh (or Love) dispersion and attenuation curves, whereas the derived parameters are the variation with depth of the shear modulus (or equivalently, the shear wave velocity) and shear damping ratio at small-strain levels.

The statistics of normally distributed experimental data enjoy peculiar properties such as being completely described by the expected value and variance. Furthermore, a basic theorem of probability theory states that any linear combination of normally distributed (i.e., Gaussian) random variables is normally distributed (Harr 1996). Therefore, if the relationship linking the experimental data and the model parameters is linear and, for instance, is represented by Equation 6.12 and, furthermore, if the data errors are assumed normally distributed, the model parameters obtained from the inversion are also normally distributed. Moreover, if $\mathbf{X}$ is a multivariate, normal, random vector with expected value $E(\mathbf{X})$ and covariance matrix $\mathrm{Cov}(\mathbf{X})$, and $\mathbf{Y} = \mathbf{AX}$ is a multivariate random vector linearly related through matrix $\mathbf{A}$ to the random variable $\mathbf{X}$, $\mathbf{Y}$ is also multivariate normal with expected value and covariance given by the following relations (Aster et al. 2005)

$$\begin{cases} E(\mathbf{Y}) = \mathbf{A}E(\mathbf{X}) \\ \mathrm{Cov}(\mathbf{Y}) = \mathbf{A}\mathrm{Cov}(\mathbf{X})\mathbf{A}^T \end{cases} \tag{6.46}$$

Equation 6.46 is important because, in a linear inverse problem, it allows us to estimate, with a simple calculation, mean and covariance of model parameters directly from the mean and covariance of experimental measurements. This relation clearly shows that the way the uncertainty of data errors is mapped into uncertainty of model parameters is intimately connected to the algorithm adopted for the inversion represented here by matrix $\mathbf{A}$. For instance, for the standard least-squares problem solved in an overdetermined situation, from Equation 6.6, matrix $\mathbf{A}$ would then be equal to $\mathbf{A} = (\mathbf{G}^T\mathbf{G})^{-1}\,\mathbf{G}^T$.

In general, if the linear inverse problem is resolved through the formalism of the Moore–Penrose generalized inverse matrix $\mathbf{G}^{-g}$ (Section 6.4.2) as shown by Equation 6.12. Equation 6.46 can be specialized as follows

$$\begin{cases} E(\mathbf{m}) = \mathbf{G}^{-g} E(\mathbf{d}) \\ \mathrm{Cov}(\mathbf{m}) = \mathbf{G}^{-g} \mathrm{Cov}(\mathbf{d}) \left(\mathbf{G}^{-g}\right)^T \end{cases} \tag{6.47}$$

where $E(\mathbf{d})$ and $\mathrm{Cov}(\mathbf{d})$ are, respectively, the expected value and covariance matrix of the experimental measurements viewed as a multivariate, normal, random vector. In the very special, yet relevant, case in which $\mathrm{Cov}(\mathbf{d}) = \sigma^2 \cdot \mathbf{1}$, where σ^2 is the variance of data vector and $\mathbf{1}$ is the identity matrix, Equation 6.47 simplifies to

$$\begin{cases} E(\mathbf{m}) = \mathbf{G}^{-g} E(\mathbf{d}) \\ \mathrm{Cov}(\mathbf{m}) = \sigma^2 \cdot \mathbf{G}^{-g} \cdot \left(\mathbf{G}^{-g}\right)^T \end{cases} \tag{6.48}$$

6.5.1.2 Normality assessment

The validity of Equation 6.47 rests on the assumption that the data errors are normally distributed. Prior to its adoption, it is therefore of primary importance to assess the validity of this hypothesis with respect to multivariate vector $\mathbf{d}$ of experimental measurements. A standard test that can be performed to validate the assumption that a set of experimental data follows a Gaussian distribution (also called a *normality test*) is the χ^2*-test* ("*chi-squared test*"). Given a random variable of interest, say x, the test is based on grouping a set of discrete, observed realizations x_j ($j = 1, Nc$) of the variable into Nc classes followed by the calculation of the absolute observed f_j^O and expected frequency f_j^E for each class. The latter is computed as if the sample came from a Gaussian distribution with mean and variance determined from the observed data. The following quantity is finally calculated

$$X^2 = \sum_{j=1}^{Nc} \frac{\left(f_j^O - f_j^E\right)^2}{f_j^E} \tag{6.49}$$

It can be shown that the random variable X^2 defined in Equation 6.49 approximately approaches the χ^2 distribution, the probability density function of which is given by the following relation (Bain and Engelhardt 1992):

$$f_{\chi^2}(x|\nu) = \frac{x^{\left(\frac{\nu-2}{2}\right)} \cdot e^{-x/2}}{2^{\nu/2}\Gamma(\nu/2)} \tag{6.50}$$

where ν denotes the degrees of freedom of the χ^2 distribution and for the Gaussian distribution is equal to $\nu = Nc-3$; Γ(...) defines the gamma function. An assessment can be made about the goodness of the assumption of normal distribution for x by comparing the value of X^2 in Equation 6.49 and the value of χ^2 given by Equation 6.50; χ^2 is small if the observed and the expected frequencies are nearly the same. Thus, a small value of χ^2 is indicative of a good correspondence between measured and normal probability distributions. It is standard practice to assume that the data do not follow a normal distribution if values greater or equal than X^2 occur with a frequency of less than 5% (Menke 1989). The threshold upper bound of X^2 is established from the χ^2 distribution.

Figure 6.15 shows an example of the relative frequency distribution of Rayleigh phase velocities computed using the results of an actual surface wave test performed at a site in Italy (Lai et al. 2005). The relative and cumulative frequency distributions of the experimental phase velocities were computed for each frequency and were arranged in five homogeneous classes. The elements of each class were then compared with those obtained for a Gaussian distribution with mean and standard deviation

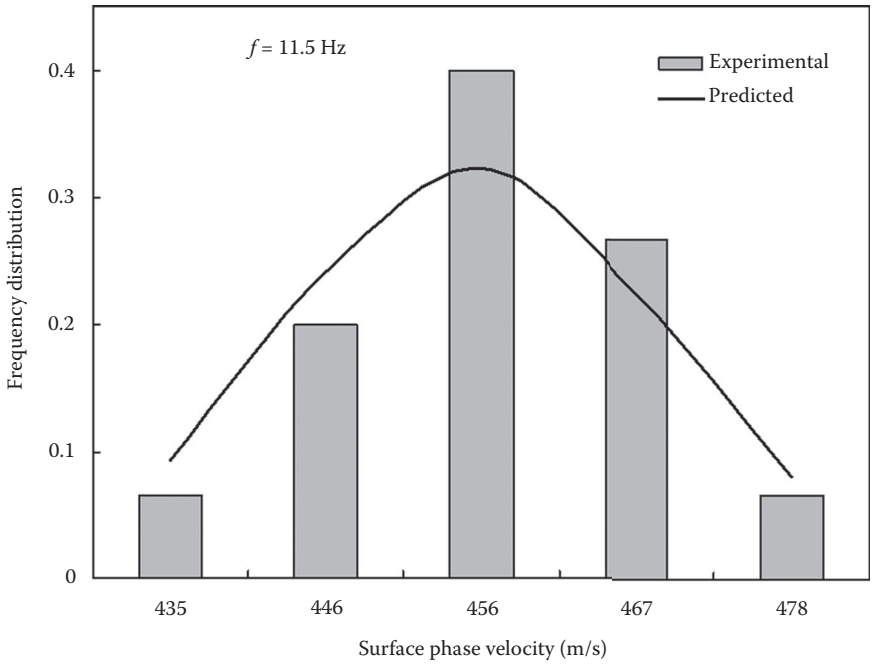

Figure 6.15 Relative frequency distribution of experimental phase velocities measured in a real surface wave test. Comparison with the values predicted by a Gaussian distribution. (From Lai, C. G. et al., *J Environ Eng Geophys*, 10, 219–228, 2005.)

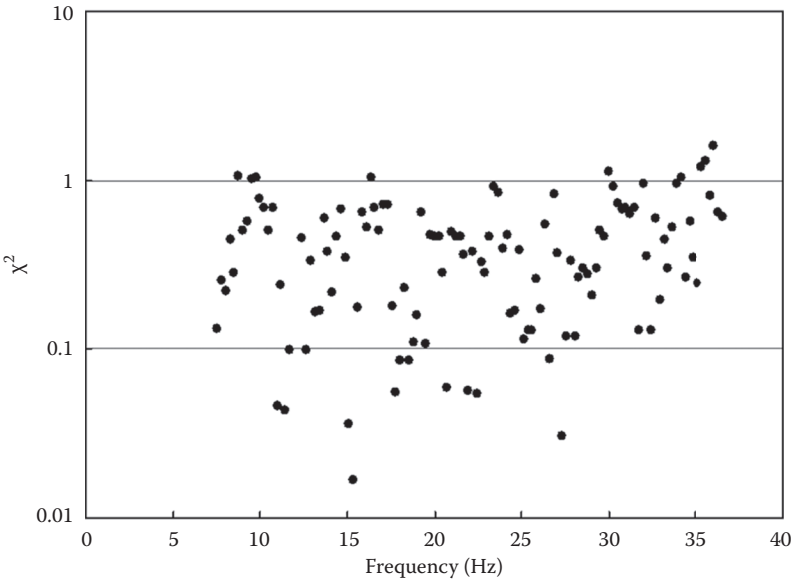

Figure 6.16 Results of χ^2-test using frequency distribution of surface phase velocities measured in a real surface wave test over the whole frequency range. (From Lai, C. G. et al., *J Environ Eng Geophys*, 10, 219–228, 2005.)

derived from the experimental data. The comparison between the experimental and the theoretical distributions appears reasonable. For the same example, Figure 6.16 shows the plot of the quantity X^2 computed with the experimental dataset as a function of frequency. From the plot, it can be noted that all the points fall below $\chi^2_{0.05}$ confirming that Gaussian distribution is a reasonable assumption for this dataset over the whole frequency range of interest.

In addition to the χ^2-test, there are a number of other statistical trials that can be used for normality assessment including the Kolmogorov–Smirnov test, the Anderson–Darling test, the Lilliefors test, and the Q–Q plot (Bain and Engelhardt 1992; Aster et al. 2005).

6.5.1.3 Nonlinear problems with Gaussian data errors

It can be shown that if an inverse problem is nonlinear, a Gaussian distribution of the errors in the data generally will be mapped into a non-Gaussian distribution of the errors in the model parameters (Tarantola 2005). In principle, this fact poses serious difficulties in the estimation of the uncertainty of model parameters derived from nonlinear inversion of data affected by measurement errors. First, in non-Gaussian probability distributions, central

estimators such as the *expected value* and the *maximum likelihood point* do not coincide and, more importantly, they do not necessarily represent the most sensible estimator of that distribution (Menke 1989). The same can be said about the covariance as an estimator of dispersion. Another critical problem in dealing with non-Gaussian distributions is that, in general, there are no analytical expressions for the solution and the associated uncertainty. In this case, a rigorous approach for solving the problem would be a thorough exploration of the space of model parameters, either systematically by defining a grid or randomly as in Monte Carlo methods (Tarantola 2005). The main shortcoming of this approach is that it is computationally time-consuming (Lai et al. 2005).

However, if the inverse problem is not too nonlinear, especially around the point of maximum likelihood in the model space (i.e., the probability distribution of model parameters around the point of maximum likelihood is not too far from a Gaussian function), it is possible to use a simplified approach to estimate the uncertainty of the model parameters. Specifically, while solving equation $\mathbf{m} = \mathbf{G}^{-g}\mathbf{d}$, it is assumed that

a. The uncertainty of the multivariate data vector **d** follows a Gaussian distribution that, from an experimental viewpoint, is often a reasonable assumption.
b. The inverse problem $\mathbf{m} = \mathbf{G}^{-g}\mathbf{d}$ is only moderately nonlinear around its solution.
c. Relation $\mathbf{G}(\mathbf{m}) = \mathbf{d}$ is inverted using the *method of maximum likelihood*.

then the uncertainty associated with the expected value of model parameter vector **m** can be approximately calculated using the following formula (Menke 1989)

$$\begin{aligned}\mathrm{Cov}[\mathbf{m}] \approx &\left[\left(\mathbf{J}^{\mathrm{T}}\left(\mathrm{Cov}[\mathbf{d}]\right)^{-1}\mathbf{J}\right)^{-1}\mathbf{J}^{\mathrm{T}}\left(\mathrm{Cov}[\mathbf{d}]\right)^{-1}\right]_{\text{last \#}} \\ &\cdot\mathrm{Cov}[\mathbf{d}]\cdot\left[\left(\mathbf{J}^{\mathrm{T}}\left(\mathrm{Cov}[\mathbf{d}]\right)^{-1}\mathbf{J}\right)^{-1}\mathbf{J}^{\mathrm{T}}\left(\mathrm{Cov}[\mathbf{d}]\right)^{-1}\right]^{\mathrm{T}}_{\text{last \#}}\end{aligned} \tag{6.51}$$

where the covariance of data vector Cov(**d**) has been assumed to be a diagonal matrix composed of *statistically uncorrelated* measurements and **J** is the Jacobian matrix calculated from *grad* **G**(**m**). The subscript last # outside the brackets of Equation 6.51 denotes that the terms inside the parentheses (essentially the Jacobian matrix) have to be computed with respect to the *last iteration* in the solution of the nonlinear problem $\mathbf{G}(\mathbf{m}) = \mathbf{d}$ using a LS recursive technique (for instance, the standard least-squares method).

Because Equation 6.51 is not exact, the " = " sign has been replaced by "≈" to denote the approximated nature of this relationship. For most problems, if the nonlinearity of $\mathbf{G}(\mathbf{m}) = \mathbf{d}$ is not too severe at the point near the optimal values of the model parameter where the function $\mathbf{G}(\mathbf{m})$ has been replaced by the Jacobian linearization, this approximation turns out to be adequate unless data measurements are exceptionally noisy (Aster et al. 2005). Finally, from Equation 6.51, it is noted that the availability of the analytical Jacobian would be beneficial not only for the solution of the nonlinear inverse problem itself (Section 6.4.3) but also for the estimate of the uncertainty of the calculated model parameters.

6.5.2 Uncertainty in surface wave measurements

Uncertainties in any experimental test are related to a number of different sources. In geophysical measurements such as surface wave tests, it is important to distinguish between uncertainties in the measurements (also called *aleatory*) and uncertainties related to the model used for the interpretation of the experimental results (also called *epistemic*). The latter are caused, for instance, by near-field effects or by the inadequacy of the horizontally layered model to capture lateral heterogeneities of the soil deposit. Strobbia and Foti (2006) proposed a statistical procedure to detect the presence of epistemic uncertainty based on the regression of phase versus offset. This section quantifies aleatory or data uncertainties in surface wave measurements.

Aleatory uncertainties are mainly, although not exclusively, due to noise in the recorded signals and to geometrical uncertainties related to the configuration and possible tilting of the receivers. The influence of several sources of uncertainty occurring in surface wave testing has been studied by O'Neill (2003) using Monte Carlo numerical simulations and repeated field tests. The author reports minimal influence from geophone tilting and ground coupling, although positional errors, static shift, and additive Gaussian noise introduce larger uncertainties in the measured experimental dispersion curve. Overall, he found *coefficients of variation*[26] for Rayleigh wave phase velocities increasing nonlinearly with the decrease of frequency and proposed that the errors be described by a *Cauchy–Lorentz* distribution for the low-frequency range. Uncertainties in recorded signals are associated with coherent and uncorrelated noise. The latter is externally generated noise (i.e., environmental noise) and can be studied via the statistical distribution of the recorded signals if many repetitions of the test in a given configuration are available (Lai et al. 2005). Coherent noise may be due, for instance, to events generated by the seismic source (e.g., near-field effects, lateral variations, higher modes; see Chapter 3).

[26] The coefficient of variation of a random variable is the ratio between the square root of the variance (i.e., the standard deviation) and the expected value (i.e., the mean).

In general, the estimation of the uncertainty associated with surface wave measurements is not a trivial exercise also because it is strongly dependent upon the particular technique that is adopted to calculate the experimental dispersion and attenuation curves (Chapters 4 and 5). Technical difficulties arise in properly quantifying the influence that inevitable errors affecting the measured quantities propagate through a series of complicated steps of data-processing and curve-fitting procedures, from the acquisition of the signals in the time-offset domain, to the calculation of dispersion and attenuation curves. (These errors may be due to factors such as the inaccuracies of the instrumentation, testing configuration, modality of data acquisition, and the even subjective behavior of the operator running the test.) The following sections will describe simplified procedures that can be used to quantify the uncertainty in surface wave measurements.

6.5.2.1 Experimental dispersion curve

6.5.2.1.1 Spectral analysis of surface waves

In conventional SASW measurements, the experimental dispersion curve is determined using the *two-station method* (see Sections 3.4.3.2 and 4.3). The data are collected with two receivers located at distances x_1 and x_2 from the source. Typically, data are recorded in the frequency domain, and the Rayleigh phase velocity is computed from Equation 4.7

$$V_R(\omega) = \frac{\omega(x_2 - x_1)}{\arg\left[S_{12}(\omega)\right]} \tag{6.52}$$

where arg $[S_{12}]$ is the phase of the particle velocity *cross-power spectrum* of the signals detected by the two receivers (see Section 4.3). This is the quantity measured experimentally. By making the simplifying, though reasonable, assumption that the experimental data are normally distributed, the measurements represent an estimate of the expected value of S_{12}, hereinafter denoted by $E[S_{12}(\omega)]$. The *variance* of S_{12} can be computed using the following relations (Bendat and Piersol 2010)

$$\begin{cases} \mathrm{Var}\left[\left|S_{12}(\omega)\right|\right] = \dfrac{\left|S_{12}(\omega)\right|^2}{n_d \cdot \gamma_{12}^2} \\ \mathrm{Var}\left[\arg\left\{S_{12}(\omega)\right\}\right] \approx \dfrac{\left(1-\gamma_{12}^2\right)}{2\gamma_{12}^2} \end{cases} \tag{6.53}$$

where n_d is the number of independent averages used to estimate $S_{12}(\omega)$ and $\gamma_{12}^2(\omega)$ is the *ordinary coherence function* of the signals (see Section 4.3).

Equation 6.53 allows us to compute the uncertainty of the measured cross-power spectrum in a *single* realization of a test. It represents the uncertainty associated with the spectral estimate of arg $[S_{12}(\omega)]$ inferred from the signals detected by the two receivers.

A rapid inspection of Equation 6.52 also shows that $V_R(\omega)$ is a nonlinear function of $\arg\left[S_{12}(\omega)\right]$; thus, $V_R(\omega)$ is generally non-Gaussian, even with Gaussian data. However, applying the FOSM method (Harr 1996; Baecher and Christian 2003) to Equation 6.52 yields the following relations for the expected value and variance of $V_R(\omega)$

$$\begin{cases} E\left[V_R(\omega)\right] \approx \dfrac{\omega(x_2 - x_1)}{E\left[\arg\left(S_{12}(\omega)\right)\right]} \\ Var\left[V_R(\omega)\right] \approx \dfrac{Var\left[\arg\left(S_{12}(\omega)\right)\right] \cdot \left[\omega(x_2 - x_1)\right]^2}{\left\{E\left[\arg\left(S_{12}(\omega)\right)\right]\right\}^4} \end{cases} \tag{6.54}$$

Because the experimental dispersion curve is constructed by averaging Rayleigh phase velocities $[V_R(\omega)]_j$ ($j = 1, n_{SP}$) obtained over n_{SP} different receiver spacings (see Section 4.3), and because every linear combination of normal distributions is itself a normal distribution, the expected value and variance of the *composite* experimental dispersion curve $V_R(\omega)$ are given by

$$\begin{cases} E\left[V_R(\omega)\right] = \dfrac{1}{n_{SP}} \displaystyle\sum_{j=1}^{n_{SP}} E\left[V_R(\omega)\right]_j \\ Var\left[V_R(\omega)\right] = \dfrac{1}{n_{SP}^2} \displaystyle\sum_{j=1}^{n_{SP}} Var\left[V_R(\omega)\right]_j \end{cases} \tag{6.55}$$

In Equation 6.55, the variance of $V_R(\omega)$ has been computed under the assumption that the individual dispersion curves $[V_R(\omega)]_j$ are statistically uncorrelated.

Equations 6.54 and 6.55 can equivalently be used in either a *single* or a *multiple* realization of a SASW test. It all depends on how the quantity $Var[\arg\{S_{12}(\omega)\}]$ is defined. If it is calculated by means of Equation 6.53, then Equations 6.54 and 6.55 allow us to determine the effects yielded onto the experimental dispersion curve by the uncertainty of $\arg[S_{12}(\omega)]$ in a single realization of a test. Alternatively, $Var\{\arg[S_{12}(\omega)]\}$ can be determined from the inherent variability of results in a multiple realization of a test; in that case, Equations 6.54 and 6.55 allow us to estimate how this

uncertainty is projected onto uncertainty of the experimental dispersion curve (Tuomi and Hiltunen 1996).

6.5.2.1.2 Multichannel analysis of surface waves

In recent years, the number of investigators that have adopted multichannel techniques to determine the experimental dispersion curve has grown considerably (Gabriels et al. 1987; Tokimatsu 1995; Tselentis and Delis 1998; Park et al. 1999; Foti 2000). These techniques use linear arrays of multiple receivers for active measurements and a 2D array for passive measurements. Multichannel methods comprise a variety of techniques depending on how the signals detected at the receivers are processed to obtain the dispersion curve (Chapter 4). However, they share the common feature that the signals detected at the receiver array are processed jointly as an ensemble. As a result, the surface wave dispersion curve determined with multichannel methods is generally smoother and more regular than that obtained with the conventional two-station method.

The simplest procedure for determining the apparent dispersion curve from the signals recorded by an array of receivers is to perform a linear regression of the displacement (or particle velocity, acceleration, or transfer function) phases measured at each receiver location (Lai 1998; Strobbia and Foti 2006). The method (see also Section 4.4) consists of determining the Rayleigh wavenumber $k_R(\omega)$ from a linear regression involving the experimental displacement phases for each of the experimental frequencies

$$k_R(\omega) \cdot \mathbf{r} = -\arg[\mathbf{u}_2(\mathbf{r},0,\omega)] \tag{6.56}$$

where $\arg[\mathbf{u}_2(\mathbf{r},0,\omega)] = \arg[\mathbf{u}_2(\mathbf{r},\omega)]$ is the phase of the vertical component of particle displacement and $\mathbf{r}$ is the vector of receiver offsets. The Rayleigh phase velocity is then obtained from $V_R(\omega) = \omega / k_R(\omega)$. By estimating $E[k_R(\omega)]$ using a standard least-squares algorithm, it is a straightforward matter to calculate the uncertainty associated with $k_R(\omega)$ using Equation 6.47, with $\mathbf{G}^{-g} = (\mathbf{G}^T\mathbf{G})^{-1}\mathbf{G}^T$ and $\mathrm{Cov}(\mathbf{d}) = -\arg[\mathbf{u}_2(\mathbf{r},\omega)]$ assuming the data $\arg[\mathbf{u}_2(\mathbf{r},\omega)]$ to be uncorrelated. The term $\mathrm{Cov}(\mathbf{m})$ in Equation 6.47 denotes a 2 by 2 diagonal matrix whose two nonzero elements are the variances of the intercept and the slope (i.e., $Var[k_R(\omega)]$) of the linear regression. Finally, matrix $\mathbf{G}$ has the following expression

$$\mathbf{G} = \begin{bmatrix} r_1 & r_2 & \dots & r_{np} \\ 1 & 1 & \dots & 1 \end{bmatrix}^T \tag{6.57}$$

where np defines the number of receiver positions. Having defined $E[k_R(\omega)]$ and $Var[k_R(\omega)]$, the expected value and variance of $V_R(\omega) = \omega/k_R(\omega)$

can be determined using the FOSM method (Harr 1996; Baecher and Christian 2003), which consists of expanding $V_R(\omega) = \omega/k_R(\omega)$ in a Taylor's series[27] about $E[k_R(\omega)]$ and truncating it to first-order terms only. If it is further assumed that $Var[\omega] = Var[\mathbf{r}] = 0$ (i.e., frequency of excitation and receiver offsets are considered deterministic rather than random variables), using Equation 6.47 it is possible to obtain the following result

$$\begin{cases} E\left[V_R(\omega)\right] \approx \dfrac{\omega}{E\left[k_R(\omega)\right]} \\ Var\left[V_R(\omega)\right] \approx \dfrac{\omega^2 Var\left[k_R(\omega)\right]}{\left\{E\left[k_R(\omega)\right]\right\}^4} \end{cases} \tag{6.58}$$

Other techniques for determining the apparent dispersion curve include the so-called *transform-based methods* that are based on transforming the data measured in the time-offset domain into a different domain where the identification of the loci of points defining the dispersion curves is simpler (Section 4.6).

The estimation of the uncertainty becomes more involved when the experimental dispersion curve is determined using transform-based procedures such as the f–k method. The problem arises from the difficulties of properly quantifying the propagation of the uncertainty in each step of the rather complicated sequence of data processing operations leading to the calculation of the dispersion curve. It is for this reason that the uncertainty associated with the experimental dispersion curve determined in MASW testing with transform-based methods is more easily calculated by a direct measurement of the statistical distribution of primitive and derived surface wave data. Even though this approach requires considerably more time and effort to be implemented, it does not involve simplifying assumptions (e.g., small variance of the raw data); furthermore, it is exempt from major technical difficulties (Lai et al. 2005).

6.5.2.2 Experimental attenuation curve

The experimental setting adopted in attenuation measurements of surface waves is that associated with the multichannel technique. Both controlled harmonic sources (e.g., hydraulic vibrators and electromechanical shakers)

[27] Because $V_R(\omega)$ is a nonlinear function of $k_R(\omega)$, $V_R(\omega)$ generally is *non-Gaussian* distributed. However, assuming that the experimental data, that is, $\arg[\boldsymbol{u}_2(\mathbf{r},\omega)]$ are characterized by small variances, it is possible to linearize $V_R(\omega) = \omega/k_R(\omega)$ about the expected value of $k_R(\omega)$ and to use Equation 6.47 without committing large errors.

and transient impulsive sources (e.g., sledgehammers and dropped weights) are used. The source term can be measured using controlled sources. The Rayleigh attenuation coefficients $\alpha_R(\omega)$ are obtained from measurements of the vertical displacement amplitudes $|u_2(\mathbf{r},\omega)|$ at several receiver offsets over a prescribed frequency range (see Section 5.2).

The relevant spectral quantity is now the particle velocity autopower spectrum $S_{rr}(\omega)$ calculated at each receiver spacing. Because ambient noise may be important, particularly at large receiver offsets, the experimental particle velocity spectra are corrected to account for the noise effects (Rix et al. 2000)

$$S_{rr}(\omega) = \tilde{\gamma}_{sr}^2(\omega) \cdot \tilde{S}_{rr}(\omega) \tag{6.59}$$

where $\tilde{S}_{rr}(\omega)$ is the measured autopower spectrum that is presumed to contain noncoherent noise. The quantity $\tilde{\gamma}_{sr}^2(\omega)$ is the ordinary coherence function between a harmonic source and the measured vertical particle velocity at the receiver. Computation of $\tilde{\gamma}_{sr}^2(\omega)$ requires the use of an accelerometer at the source to monitor the motion of the harmonic oscillator. From Equation 6.59, the experimental vertical particle displacement spectrum is computed from the following relation

$$|u_2(\mathrm{r},0,\omega)| = |u_2(\mathrm{r},\omega)| = \frac{|\mathrm{v}_2(\mathrm{r},\omega)|}{\omega \cdot C(\omega)} = \frac{\sqrt{S_{rr}(\omega)}}{\omega \cdot C(\omega)} \tag{6.60}$$

where $C(\omega)$ is a frequency-dependent *calibration factor* that converts the output of the velocity transducer (volts) into engineering units (e.g., cm/s); v_2 (r, ω) is the Fourier transform of the vertical particle velocity recorded at the receivers located at distance r from the source. Using the FOSM method illustrated here for $V_R(\omega)$, namely expanding Equation 6.60 in a truncated Taylor's series about $E[S_{rr}]$ and assuming $Var[C(\omega)] = 0$, it is possible to obtain

$$\begin{cases} E\left[\,|u_2(\mathrm{r},\omega)|\,\right] \approx \dfrac{\sqrt{E\left[S_{rr}(\omega)\right]}}{\omega \cdot C(\omega)} \\[2ex] Var\left[\,|u_2(\mathrm{r},\omega)|\,\right] \approx \dfrac{Var\left[S_{rr}(\omega)\right]}{4E\left[S_{rr}(\omega)\right] \cdot \left[\omega \cdot C(\omega)\right]^2} \end{cases} \tag{6.61}$$

Once the vertical displacement amplitudes $|\boldsymbol{u}_2(\mathbf{r},\omega)|$ have been calculated, the Rayleigh attenuation coefficients $\alpha_R(\omega)$ can be determined from

$$|\mathbf{u}_2(\mathrm{r},0,\omega)| = |\mathbf{u}_2(\mathbf{r},\omega)| = F \cdot \;_2(\mathbf{r},0,\omega) \cdot e^{-\alpha_R(\omega)\cdot \mathrm{r}} \tag{6.62}$$

where $\Upsilon_2(\mathbf{r},0,\omega)$ is the vertical component of the Rayleigh geometric attenuation function and F is the amplitude of the harmonic vertical point load (Chapter 2). In homogenous half-spaces, $_2(\mathbf{r},0,\omega) \propto 1/\sqrt{\mathbf{r}}$. Sometimes this may be a reasonable approximation even in heterogeneous geomaterials.[28]

Equation 6.62 forms the basis of a nonlinear regression analysis to determine the frequency-dependent attenuation coefficients $\alpha_R(\omega)$ from the experimental displacement amplitudes $|\mathbf{u}_2(\mathbf{r},\omega)|$. The geometric spreading function $\Upsilon_2(\mathbf{r},0,\omega)$ is known because it is obtained from the solution of the elastic Rayleigh forward problem (Section 2.5.3). Alternatively, Equation 6.62 may be substituted with Equation 6.25 where $|\mathbf{u}_2(\mathbf{r},\omega)|$ has been replaced by the amplitude of vertical displacement transfer function $|\mathbf{T}(\mathbf{r},\omega)|$ between the source and the receiver.

The expected value and variance of the Rayleigh attenuation coefficient $\alpha_R(\omega)$ can be obtained from a regression of either Equation 6.62 or Equation 6.25. Due to the nonlinearity of the relationship between either $|\mathbf{u}_2(\mathbf{r},\omega)|$ or $|\mathbf{T}(\mathbf{r},\omega)|$ and $\alpha_R(\omega)$, only estimates of $E[\alpha_R(\omega)]$ and $Var[\alpha_R(\omega)]$ can be obtained. By solving Equation 6.62 using a standard nonlinear least-squares algorithm, the uncertainty associated with the attenuation coefficient can be approximately calculated with the following relation

$$\mathrm{Cov}\left[\alpha_R(\omega)\right] \approx \left[\left(\mathbf{J}_{\alpha_R}^T \mathbf{J}_{\alpha_R}\right)^{-1} \mathbf{J}_{\alpha_R}^T\right]_{\text{last \#}} \mathrm{Cov}\left[|\mathbf{u}_2(\mathbf{r},\omega)|\right] \left[\left(\mathbf{J}_{\alpha_R}^T \mathbf{J}_{\alpha_R}\right)^{-1} \mathbf{J}_{\alpha_R}^T\right]_{\text{last \#}}^T \tag{6.63}$$

where $\mathrm{Cov}\left[|\mathbf{u}_2(\mathbf{r},\omega)|\right]$ is an np by np matrix containing the covariances of the experimental displacement amplitudes at the np different receiver positions at a given frequency ω. Because it is assumed that the data $|\mathbf{u}_2(\mathbf{r},\omega)|$ are uncorrelated, the data covariance matrix is diagonal with the nonzero elements equal to the variances of $|\mathbf{u}_2(\mathbf{r},\omega)|$ given by Equation 6.61. The Jacobian $\mathbf{J}_{\alpha_R}$ is an np by 1 vector whose components $(\mathrm{J}_{\alpha_R})_k$ (k = 1, np) are obtained from Equation 6.62

$$(\mathrm{J}_{\alpha_R})_k = \left[\mathrm{J}_{\alpha_R}(\omega)\right]_k = \frac{\partial |u_2(r_k,\omega)|}{\partial \alpha_R} = -r_k \cdot |u_2(r_k,\omega)| \tag{6.64}$$

The subscript last # outside the brackets of Equation 6.63 indicates that the terms inside the parentheses, essentially the Jacobian $\mathbf{J}_{\alpha_R}$, refer to the *last iteration* in the solution of the nonlinear regression (Equation 6.62).

[28] This occurs, for instance, in *normally dispersive* soil deposits namely shallow geological formations where the mechanical impedance increases gradually with depth. In this case, the wave field is dominated by the fundamental mode of propagation. As a result, the geometric spreading function $\Upsilon_2(\mathbf{r},0,\omega)$ is well approximated by a term proportional to $1/\sqrt{\mathbf{r}}$ (Chapter 2).

6.5.2.3 Joint experimental dispersion and attenuation curves

In the standard approach of surface wave measurements, experimental dispersion and attenuation curves are determined separately using different source-to-receiver configurations and different interpretation methods. However, Rix et al. (2001) proposed a methodology based on measuring the displacement transfer functions in which dispersion and attenuation curves are determined simultaneously from a single set of measurements using the same source and linear array of receivers. The multichannel simultaneous estimation of velocity and attenuation is also possible (Chapter 5). This approach has been introduced after recognizing that, in dissipative media, Rayleigh phase velocity and attenuation are not independent due to material dispersion. Therefore, a coupled analysis of dispersion and attenuation data is a more consistent and fundamentally correct approach. Also, the coupled measurement is compatible with the joint inversion algorithm that can be used to obtain the shear wave velocity and shear damping ratio profiles of a soil deposit (Section 6.4.3).

In coupled measurements of dispersion and attenuation data, the quantity measured experimentally is the particle velocity transfer function $H(\mathrm{r},\omega)$ between source and receiver and is defined by $H(\mathrm{r},\omega) = \mathrm{v}_2(\mathrm{r},\omega)/F \cdot e^{\mathrm{i}\omega\mathrm{t}} = \mathrm{i}\omega \cdot \mathrm{u}_2(\mathrm{r},\omega)/F \cdot e^{\mathrm{i}\omega\mathrm{t}} = \mathrm{i}\omega\mathrm{T}(\mathrm{r},\omega)$. The expected value and variance of the modulus and phase of $T(\mathrm{r},\omega)$ can be computed from the following relations (Bendat and Piersol 2010)

$$E\left[\left|T(r,\omega)\right|\right] = \omega \cdot E\left[\left|H(r,\omega)\right|\right]; \quad E\left[\arg\left(T(r,\omega)\right)\right] = E\left[\arg\left(H(r,\omega)\right)\right] + \frac{\pi}{2}$$

$$Var\left[\left|T(r,\omega)\right|\right] \approx \omega^2 \cdot \frac{\left(1-\gamma_{sr}^2\right)\cdot\left|\mathrm{H}(r,\omega)\right|^2}{2n_d \cdot \gamma_{sr}^2}; \quad Var\left[\arg\left(T(r,\omega)\right)\right] \approx \frac{\left(1-\gamma_{sr}^2\right)}{2\gamma_{sr}^2} \tag{6.65}$$

where $\gamma_{sr}^2(\omega)$ represents the ordinary coherence function between the harmonic source and the receiver output signal. Because the complex Rayleigh wavenumber

$$k_R^*(\omega) = \left[\frac{\omega}{V_R(\omega)} + i \cdot \alpha_R(\omega)\right] \tag{6.66}$$

is determined from the nonlinear regression

$$\mathrm{T}(\mathrm{r},\omega) = {}_2(\mathrm{r},0,\omega) \cdot e^{-i \cdot k_R^*(\omega)\cdot \mathrm{r}} \tag{6.67}$$

it is first necessary to calculate $E[T(\mathrm{r},\omega)]$ and $Var[T(\mathrm{r},\omega)]$ in order to compute the expected value and variance of $k_R^*(\omega)$. For this purpose, it is convenient to write $T(\mathrm{r},\omega) = [T_1(\mathrm{r},\omega) + i \cdot T_2(\mathrm{r},\omega)]$ from which it is simple to obtain (Lai 1998)

$$\begin{cases} E\left[\left|T_1(r,\omega)\right|\right] \approx E\left[\left|T(r,\omega)\right|\right] \cdot \cos\left\{E\left[\arg\left(T(r,\omega)\right)\right]\right\} \\ E\left[\left|T_2(r,\omega)\right|\right] \approx E\left[\left|T(r,\omega)\right|\right] \cdot \sin\left\{E\left[\arg\left(T(r,\omega)\right)\right]\right\} \\ Var\left[T(r,\omega)\right] = Var\left[T_1(r,\omega)\right] - Var\left[T_2(r,\omega)\right] \end{cases} \tag{6.68}$$

where

$$\begin{cases} Var\left[T_1(r,\omega)\right] \approx \cos^2\left[\arg\left(T(r,\omega)\right)\right] Var\left[\left|T(r,\omega)\right|\right] \\ \qquad + \left|T(r,\omega)\right|^2 \sin^2\left[\arg\left(T(r,\omega)\right)\right] Var\left[\arg\left(T(r,\omega)\right)\right] \\ Var\left[T_2(r,\omega)\right] \approx \sin^2\left[\arg\left(T(r,\omega)\right)\right] Var\left[\left|T(r,\omega)\right|\right] \\ \qquad + \left|T(r,\omega)\right|^2 \cos^2\left[\arg\left(T(r,\omega)\right)\right] Var\left[\arg\left(T(r,\omega)\right)\right] \end{cases} \tag{6.69}$$

The uncertainty of the complex wavenumber $k_R^*(\omega)$ is finally computed from the nonlinear regression represented by Equation 6.67 following a procedure that is formally identical to that used for determining the uncertainty of the Rayleigh attenuation coefficient $\alpha_R(\omega)$. The result is

$$\mathrm{Cov}\left[k_R^*(\omega)\right] \approx \left[\left(\mathbf{J}_{k_R^*}^H \mathbf{J}_{k_R^*}\right)^{-1} \mathbf{J}_{k_R^*}^H\right]_{\text{last \#}} \cdot \mathrm{Cov}\left[\mathbf{T}(\mathbf{r},\omega)\right] \left[\left(\mathbf{J}_{k_R^*}^H \mathbf{J}_{k_R^*}\right)^{-1} \mathbf{J}_{k_R^*}^H\right]_{\text{last \#}}^{\mathrm{H}} \tag{6.70}$$

where $\mathrm{Cov}[\mathbf{T}(\mathbf{r},\omega)]$ is an *np* by *np* matrix representing the covariances, at a given frequency ω, of the experimental displacement transfer functions at the *np* receiver positions. Because it is assumed that the data $\mathbf{T}(\mathbf{r},\omega)$ are uncorrelated, matrix $\mathrm{Cov}[\mathbf{T}(\mathbf{r},\omega)]$ is diagonal with the nonzero elements equal to the variances of $\mathbf{T}(\mathbf{r},\omega)$ and given by Equations 6.68 and 6.69. The term $\mathbf{J}_{k_R^*}$ is an *np* by 1 complex-valued vector the components $\left(\mathrm{J}_{k_R^*}\right)_j$, $j = 1, np$ of which are obtained by differentiating Equation 6.67, and they are equal to

$$\left(\mathrm{J}_{k_R^*}\right)_j = \left[\mathrm{J}_{k_R^*}(\omega)\right]_j = \frac{\partial T\left(r_j,\omega\right)}{\partial k_R^*} = -i \cdot \mathrm{r}_j \cdot T\left(r_j,\omega\right) \tag{6.71}$$

Again the subscript last # outside the brackets of Equation 6.71 indicates that the terms inside the parenthesis refer to the *last iteration* in

solving Equation 6.67. Once $E\left[k_R^*(\omega)\right]$ and $Var\left[k_R^*(\omega)\right]$ are computed, the expected value and variance of the complex-valued Rayleigh phase velocity $V_R^*(\omega)$ are the complex-valued counterparts of Equation 6.58, namely

$$\begin{cases} E\left[V_R^*(\omega)\right] \approx \dfrac{\omega}{E\left[k_R^*(\omega)\right]} \\ Var\left[V_R^*(\omega)\right] \approx \dfrac{\omega^2 \cdot Var\left[k_R^*(\omega)\right]}{\left\{E\left[k_R^*(\omega)\right]\right\}^4} \end{cases} \tag{6.72}$$

Equation 6.72 completes the statistical analysis of surface wave measurements when dispersion and attenuation data are determined simultaneously from the particle velocity transfer functions $H(r,\omega)$. However, it may also be important to compute $E\left[V_R^*(\omega)\right]$ and $Var\left[V_R^*(\omega)\right]$ when surface wave data, namely $V_R(\omega)$ and $\alpha_R(\omega)$, are obtained independently. This happens, for instance, in carrying out a joint inversion of dispersion and attenuation curves to determine the shear wave velocity and shear damping ratio profiles at a site where $V_R(\omega)$ and $\alpha_R(\omega)$ have been measured following an uncoupled procedure.

The solution to the problem is obtained by applying the FOSM method to Equation 2.127. In particular, this equation can be viewed as a mapping[29] that assigns a complex random variable V_R^* to a pair of independent random variables V_R and D_R. If the variables V_R and D_R are normally distributed and the mapping is sufficiently smooth, it may be expanded in a Taylor series about the point $\{E(V_R), E(D_R)\}$. Again, assuming small variances for variables V_R and D_R, this series is truncated to first-order terms only, yielding

$$\begin{cases} E\left[V_R^*(\omega)\right] \approx \dfrac{E\left[V_R(\omega)\right]}{1+E^2\left[D_R(\omega)\right]} \cdot \left(1+i \cdot E\left[D_R(\omega)\right]\right) \\ Var\left[V_R^*(\omega)\right] \approx \left[\dfrac{1+i \cdot E\left[D_R(\omega)\right]}{1+E^2\left[D_R(\omega)\right]}\right]^2 \cdot Var\left[V_R(\omega)\right] \\ \qquad + \left[\dfrac{E\left[V_R(\omega)\right] \cdot \left[2E\left[D_R(\omega)\right]+i\left(1+E^2\left[D_R(\omega)\right]\right)\right]}{\left(1+E^2\left[D_R(\omega)\right]\right)^2}\right]^2 \\ \qquad \cdot Var\left[D_R(\omega)\right] \end{cases} \tag{6.73}$$

[29] The frequency-dependence of $V_R^*(\omega)$ in this mapping does not constitute a major problem in this discussion because it can be considered parametric.

where

$$\begin{cases} E\left[D_R(\omega)\right] \approx \dfrac{E\left[\alpha_R(\omega)\right]E\left[V_R(\omega)\right]}{\omega} \\ Var\left[D_R(\omega)\right] \approx \left[\dfrac{E\left[V_R(\omega)\right]}{\omega}\right]^2 \cdot Var\left[\alpha_R(\omega)\right] \\ \qquad\qquad + \left[\dfrac{E\left[\alpha_R(\omega)\right]}{\omega}\right]^2 \cdot Var\left[V_R(\omega)\right] \end{cases} \tag{6.74}$$

In deriving Equations 6.73 and 6.74, it was assumed that the random variables V_R and D_R are uncorrelated.

In conclusion, despite the fact that the relationship among *directly* measured quantities $S_{rr}(\omega)$, $S_{12}(\omega)$, and $H(\mathrm{r},\omega)$ and *derived* surface wave data, namely $V_R(\omega)$, $\alpha_R(\omega)$, and $V_R^*(\omega)$, is nonlinear in most circumstances, the assumption of small variances allowed us to obtain explicit results for the expected values and variances of $V_R(\omega)$, $\alpha_R(\omega)$, and $V_R^*(\omega)$.

6.5.3 Estimate of variance of model parameters

Once the uncertainty of the experimental dispersion and attenuation curves has been estimated, the remaining goal is to determine the projection of this uncertainty into the expected values of the model parameters, which are the shear wave velocity and shear damping ratio profiles at a site. The solution of this problem is not unequivocal because it depends on the specific algorithm adopted for the inversion. Furthermore, whereas determining the shear wave velocity is a nonlinear inverse problem, the calculation of the shear damping ratio is a linear problem, as is shown by Equations 6.2 and 6.3, respectively. Of course, in the case of a joint inversion, the problem is nonlinear and also complex valued (Section 6.4.3).

If the dispersion curve is represented by an *nf* by *1* array $\mathbf{V}_R$ of Rayleigh phase velocities associated with *nf* discrete frequencies and it is viewed as a multivariate, normal, random vector[30] with expected value $E(\mathbf{V}_R)$ and covariance *nf* by *nf* matrix $\mathrm{Cov}(\mathbf{V}_R)$, then Equation 6.51 may be used to estimate $\mathrm{Cov}(\mathbf{V}_S)$, where $\mathbf{V}_S$ is an *nl* by *1* vector, the components of which

[30] It is further assumed that the components of data vector $\mathbf{V}_R$ are statistically uncorrelated; thus, the matrix $\mathbf{Cov}(\mathbf{V}_R)$ is diagonal with the nonzero elements equal to the variances of $\mathbf{V}_R$.

are the unknown shear wave velocities of a stratified soil deposit having *nl* layers (including the half-space). The result is

$$\begin{aligned}\mathrm{Cov}[\mathbf{V}_S] \approx & \left[\left(\mathbf{J}_{\mathbf{V}_S}^T\left(\mathrm{Cov}[\mathbf{V}_R]\right)^{-1}\mathbf{J}_{\mathbf{V}_S}\right)^{-1}\mathbf{J}_{\mathbf{V}_S}^T\left(\mathrm{Cov}[\mathbf{V}_R]\right)^{-1}\right]_{\text{last \#}} \\ & \cdot\mathrm{Cov}[\mathbf{V}_R]\cdot\left[\left(\mathbf{J}_{\mathbf{V}_S}^T\left(\mathrm{Cov}[\mathbf{V}_R]\right)^{-1}\mathbf{J}_{\mathbf{V}_S}\right)^{-1}\mathbf{J}_{\mathbf{V}_S}^T\left(\mathrm{Cov}[\mathbf{V}_R]\right)^{-1}\right]_{\text{last \#}}^{\mathrm{T}}\end{aligned} \tag{6.75}$$

where $\mathbf{J}_{\mathbf{V}_S}$ is the *nf* by *nl* Jacobian matrix calculated from *grad* $\mathbf{G}(\mathbf{V}_S)$ in Equation 6.2. It can be computed analytically using Equation 6.33. The subscript last # outside the brackets of Equation 6.75 denotes again that the terms inside the parentheses are computed with respect to the final iteration when solving Equation 6.2 by using, for instance, the standard least-squares method. Finally, $\mathrm{Cov}(\mathbf{V}_S)$ is an *nl* by *nl* diagonal matrix, the elements of which are the variances of the estimated shear wave velocities of the *nl* layers.

If Equation 6.2 is inverted using the Occam's algorithm (Section 6.4.3), it is possible to show that the uncertainty associated with the estimated profile of shear wave velocities can be computed with the following relationship (Lai et al. 2005)

$$\begin{aligned}\mathrm{Cov}[\mathbf{V}_S] \approx & \left[\left(\mu\mathbf{L}^T\mathbf{L}+\left(\mathbf{W}\mathbf{J}_{\mathbf{V}_S}\right)^T\mathbf{W}\mathbf{J}_{\mathbf{V}_S}\right)^{-1}\left(\mathbf{W}\mathbf{J}_{\mathbf{V}_S}\right)^T\mathbf{W}\right]_{\text{last \#}} \\ & \cdot\mathrm{Cov}[\mathbf{V}_R]\cdot\left[\left(\mu\mathbf{L}^T\mathbf{L}+\left(\mathbf{W}\mathbf{J}_{\mathbf{V}_S}\right)^T\mathbf{W}\mathbf{J}_{\mathbf{V}_S}\right)^{-1}\left(\mathbf{W}\mathbf{J}_{\mathbf{V}_S}\right)^T\mathbf{W}\right]_{\text{last \#}}^{\mathrm{T}}\end{aligned} \tag{6.76}$$

where μ is the Lagrange multiplier, **W** is a diagonal *nf* by nf *weight matrix* defined by Equation 6.43 where the uncertainties are now real valued, and **L** is the *nl* by *nl* real-valued matrix representing the two-point central finite difference operator defined in Equation 6.19.

Figure 6.17 shows the shear wave velocity profile and associated uncertainty obtained from the inversion of an experimental dispersion curve measured at a real site in Italy.

Figure 6.18 illustrates the comparison between the experimental dispersion curve and the corresponding theoretical curve (fundamental mode) obtained using the Occam's inversion algorithm after the ninth and final iteration (Lai et al. 2005). The error bars in Figure 6.17 attached to the V_S profile represent the standard deviation of the expected shear wave velocity profile with a coefficient of variation ranging from 0.2% to about 4%. The experimental dispersion curve of Figure 6.18 appears to be split into two regions separated by a threshold frequency of about 11 Hz: a low-frequency region

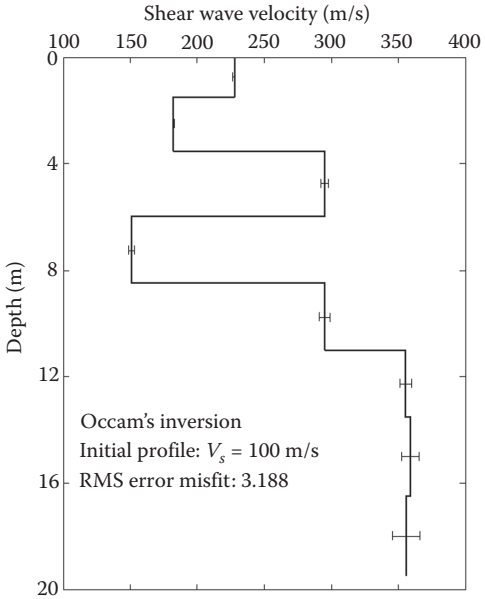

Figure 6.17 Expected value and standard deviation of the shear wave velocity profile from the inversion of the experimental dispersion curve shown in Figure 6.18, obtained at a real testing site in Italy. (From Lai, C. G. et al., *J Environ Eng Geophys*, 10, 219–228, 2005.)

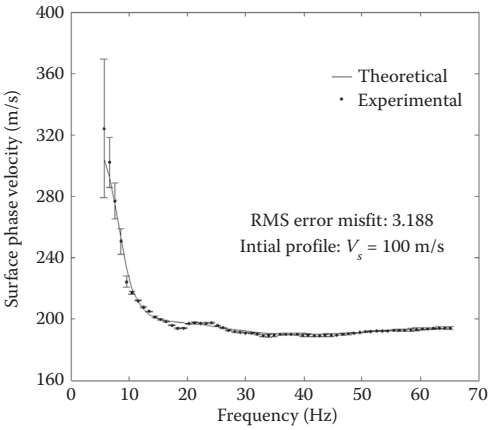

Figure 6.18 Experimental dispersion curve measured at a real testing site in Italy. Expected value and standard deviation of surface phase velocity versus frequency. The continuous line represents the theoretical dispersion curve (fundamental mode) at the final iteration of the inversion algorithm. (From Lai, C. G. et al., *J Environ Eng Geophys*, 10, 219–228, 2005.)

characterized by higher values of coefficient of variation (up to about 14%) and a high-frequency region characterized by lower values of uncertainty.

This example demonstrates quite clearly the stability of the Occam's algorithm in mapping the uncertainty of measured data into a covariance of model parameter. This is mainly due to the damping effect produced by the smoothing parameter appearing in Equation 6.76. The larger uncertainty of the experimental dispersion curves at low frequencies contributes to the observed increase of uncertainty with depth of the expected shear wave velocity profile shown in Figure 6.17.

The low values of the standard deviation obtained for the shear wave velocity profile shown in Figure 6.17 are not to be credited to the Occam's algorithm only. As a matter of fact, it has been demonstrated by other researchers (Xia et al. 2002; Marosi and Hiltunen 2004a, 2004b; Moss 2008; Cox and Wood 2011) that surface wave methods (e.g., SASW and MASW) are rather accurate experimental techniques with a coefficient of variation ranging on the order of 5%–10% even in the presence of strong ambient uncorrelated noise. This result is of particular interest for applications of these methods in urban areas where other geophysical tests may encounter difficulties.

That being said, it is important at this point to emphasize the difference between *accuracy* and *reliability* or *precision* in experimental measurements. Accuracy may be defined as the probability that the expected value of a measurement, in this case $E(\mathbf{V}_S)$, determined with a certain technique is equal to the *true* value. Accuracy of a measurement may be affected by bias and systematic errors (Tuomi and Hiltunen 1996). Figure 6.19 shows a classical

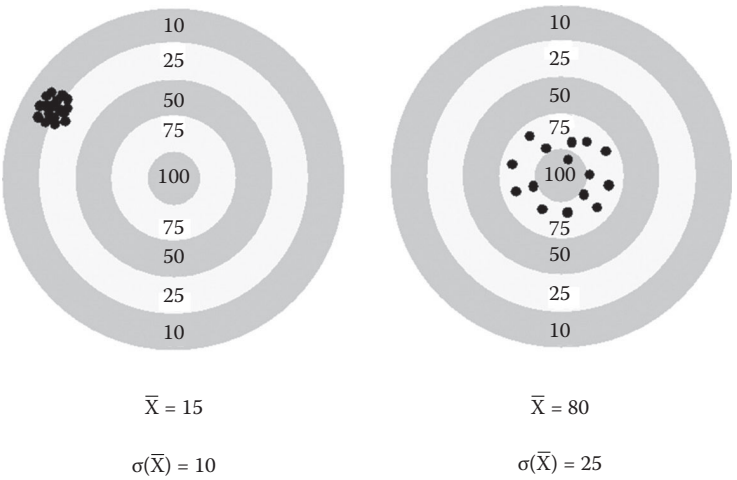

Figure 6.19 Difference between *accuracy* and *reliability* of two sets of experimental measurements. The symbol $\overline{X}$ denotes the expected value of the measurement, whereas $\sigma\left(\overline{X}\right)$ represents its standard deviation.

example that is often used to explain the difference between reliability and accuracy of the outcome of an experiment.

The set of measurements on the left is characterized by a small spread around the mean (i.e., low standard deviation). However, if 100 represents the *true* value of the parameter (target), the mean is off by a factor almost equal to 7. The set of measurements on the right has a mean that is equal to 80, so relatively close to the true value, yet the measurements are grossly scattered around the mean and, as a result, the standard deviation is large. Figure 6.19 can be qualified by saying that the set of measurements on the left is *reliable* but not *accurate*, whereas the set on the right is *accurate* but not *reliable*.

Sometimes, it may be difficult to assess accuracy because the true value of a property or a physical parameter may not be known with certainty. The present discussion concerns with the reliability of surface wave measurements that represent an estimate of the scatter of V_S around its expected value and can be calculated by using Equation 6.76. The accuracy of surface wave methods in predicting the "true" V_S profile at a site may be addressed by comparing the results of the inversion with measurements of V_S profile obtained with other geophysical techniques (see Chapter 7).

The determination of the uncertainty associated with the estimated *damping ratio profile* is simpler than determining $\mathrm{Cov}(\mathbf{V}_S)$ because the inversion of the experimental attenuation curve $\alpha_R(\omega)$ to obtain the damping ratio profile is *linear* as shown by Equation 6.3. If the attenuation curve is represented by an *nf* by *1* array $\boldsymbol{\alpha}_R$ of Rayleigh attenuation coefficients associated with *nf* discrete frequencies and it is viewed as a multivariate, normal, random vector with expected value $E(\boldsymbol{\alpha}_R)$ and covariance *nf* by *nf* matrix $\mathrm{Cov}(\boldsymbol{\alpha}_R)$,[31] then Equation 6.47 may be used to estimate $\mathrm{Cov}(\mathbf{D}_S)$, where $\mathbf{D}_S$ is an *nl* by *1* vector, the components of which are the unknown shear damping ratios of a stratified soil deposit having *nl* layers (including the half-space). If a standard least-squares algorithm is used, the result is

$$\mathrm{Cov}(\mathbf{D}_S) = \left(\mathbf{G}^T\mathbf{G}\right)^{-1}\mathbf{G}^T\mathrm{Cov}(\boldsymbol{\alpha}_R)\left[\left(\mathbf{G}^T\mathbf{G}\right)^{-1}\mathbf{G}^T\right]^T \tag{6.77}$$

where $\mathbf{G}$ is a matrix composed by terms that include the partial derivatives of the modal phase velocity of Rayleigh waves with respect to the model parameters $\mathbf{V}_P$ and $\mathbf{V}_S$ of the various layers of the soil deposit. The precise definition of this matrix can be obtained from Equation 2.133. Finally, in Equation 6.77, $\mathrm{Cov}(\mathbf{D}_S)$ is an *nl* by *nl* diagonal matrix, the elements of which are the variances of the estimated shear damping ratios of the

[31] It is further assumed that the components of data vector $\boldsymbol{\alpha}_R$ are statistically uncorrelated; thus, the matrix $\mathrm{Cov}(\boldsymbol{\alpha}_R)$ is diagonal with the nonzero elements equal to the variances of $\boldsymbol{\alpha}_R$.

nl layers. Of course, other algorithms besides the standard least-squares method can be used to invert the Rayleigh attenuation curve $\alpha_R(\omega)$, including the Occam's algorithm (Rix et al. 2000).

The shear damping ratio profile $\mathbf{D}_S$ can also be obtained through a joint, nonlinear inversion of dispersion and attenuation curves (Section 6.4.3). The procedure for estimating the uncertainty associated with the inverted complex-valued shear wave velocity profile $\mathbf{V}_S^*$ is identical to that described for the uncoupled inversion to obtain $\mathrm{Cov}(\mathbf{V}_S)$, and thus Equations 6.75 and 6.76 are still valid. The only difference is that the computation of $\mathrm{Cov}\left(\mathbf{V}_S^*\right)$ requires a systematic use of the formalism of complex variables.

6.5.4 Trade-off between model resolution and uncertainty

As shown in the previous section, the mode in which the uncertainty associated with an experimental dispersion or attenuation curve is projected into uncertainty of model parameters that are the shear wave velocity, and shear damping ratio profiles at a site generally will depend upon the specific algorithm adopted for the inversion. Inspection of Equations 6.75 and 6.76 makes this statement obvious. However, in addition to the algorithm, there is another factor affecting the amplification/deamplification of data errors. This factor is the spatial resolution of the geometric model assumed for the soil deposit. Suppose that at a construction site the sequence of layers and corresponding thicknesses are well-defined from independent borehole logs until a depth of 20 m. Suppose also that a surface wave test has been executed at the site and the low-frequency range of the measured dispersion curve covers depths up to 20 m. A variety of different geometrical models could be adopted for the inversion of the experimental dispersion curve not only because sharp interfaces among the layers may not exist at all but also because the low-strain shear modulus typically increases with depth due to the increase of overburden stress, even in a homogeneous soil deposit (Santamarina et al. 2001). Thus, a natural question arises regarding what would be the ideal geometrical model to be used for the inversion of the dispersion curve. Would a five-layer model be adequate? Or would a 10-layer model would be preferable? What should be the guiding principle or the rationale for making a correct choice with reference to the number of layers? The question is well-posed and relevant in virtually any field of exploration geophysics and seismology. In seismic tomography, what would be the ideal coarseness of the grid? Of course, the finer the grid, the more details may be resolved. However, a small cell (i,j) is likely to be illuminated by few seismic rays; thus, the corresponding model parameter [e.g., $(V_S)_{ij}$ and/or $(V_P)_{ij}$] determined by inverting the travel times will be characterized by large uncertainty.

In very fine grids, there could be cells that are completely at "dark" (that is, they are not sampled by any ray). In such situations, no algorithm can resolve the model parameters of these cells no matter how sophisticated it is. Again, quoting Lanczos (1961) "A lack of information in the solution of a problem cannot be remedied by any mathematical trickery." Of course, this does not mean that very fine grids should always be avoided. It all depends on the objectives of the geophysical investigation. If a high-resolution tomogram is desired, then the tomography should be carried out using large numbers of shots, and the seismograms should be detected by a large number of closely spaced receivers. The wavelength of the generated waves should be short enough to guarantee a sufficient physical resolution. In short, every cell of the spatial grid should be illuminated by a large number of rays, making sure that there is a redundancy of data measurements so that the problem of inverting the travel times becomes *overdetermined,* Of course, this is costly and not always justifiable. With large cells, the spatial resolution decreases; however, the chances of having a sufficient number of rays illuminating the cell increase even with a limited number of shots and receivers. In these cells, the uncertainty of the model parameters is small (Menke 1989). In summary, the mapping of data errors into uncertainty of model parameters is generally dependent on

- Magnitude of experimental errors
- Algorithm used to invert the experimental data
- Spatial resolution of the model

The issue of model resolution has already been discussed in Section 6.4.2 in conjunction with accuracy and data resolution. Model resolution matrix $\mathbf{R}_m$ and Dirichlet spread function $\|\mathbf{R}_m - \mathbf{1}\|_2^2$ have been introduced and specified by Equations 6.21 and 6.24, respectively. In fully overdetermined problems, $\mathbf{R}_m = \mathbf{1}$ and the model parameters are perfectly resolved by the algorithm. However, for an under- or mixed-determined problem, in general, the model resolution matrix is different from the identity matrix. Because $\mathbf{R}_m = \mathbf{G}^{-g}\mathbf{G}$, the structure of $\mathbf{R}_m$ will depend on the structure of the particular generalized inverse matrix adopted to solve the inverse problem.

The discussion in Section 6.4.2 was conducted under the assumption that the experimental data were exempt from errors. If noise is introduced in the measurements, the treatment of the problem becomes more complicated. A milestone contribution in this regard is presented by the Backus and Gilbert (1970) paper. After introducing a new definition of *resolution spread function* that better captures the spread of model resolution matrix if compared with Equation 6.24, these researchers calculated a generalized inverse matrix by minimizing the spread of the model resolution function in the presence of uncorrelated, Gaussian data errors. They found out that a unique solution to the inverse problem does not exist because resolution

spread and variance of model parameters are not two independent variables. For instance, model resolution can only be improved at the expense of increasing the uncertainty of model parameters (Parker 1977). The notion of the *trade-off curve* helps to explain the nature of the problem, as shown in Figure 6.20. The above qualitative discussion has been substantiated quantitatively by Backus and Gilbert (1970). The ideal solution does not exist, and the acceptable values of model resolution and precision should be set case by case depending on the characteristics of the specific problem at hand. This also answers the initial question posed at the beginning of this section about what is the ideal geometrical model that should be used in surface wave testing for the inversion of an experimental dispersion curve.

Although the Backus and Gilbert method is rigorous only for linear inverse problems, it can be extended to a certain degree to nonlinear problems as long as the nonlinearity is not too severe (Parker 1977).

Figure 6.21 shows the results of an inversion of a dispersion curve conducted after assuming two different geometrical models. Figure 6.21a has been obtained adopting a five-layer model, whereas a ten-layer model was used in producing Figure 6.21b. The overall thickness of the two models is 20 m. Apart from the different number of layers, all the other parameters of the two models are the same. The experimental dispersion

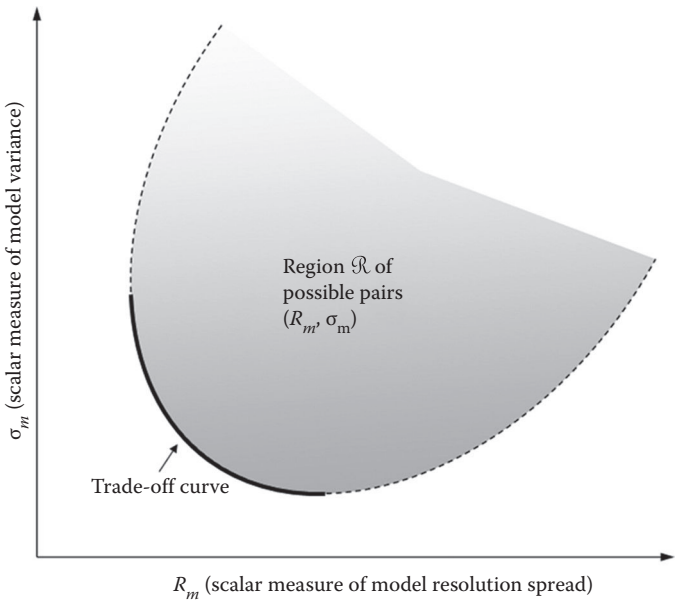

Figure 6.20 Conceptual diagram showing the trade-off curve of model resolution spread and variance in the solution of an inverse problem. (Modified from Parker, R. L., *Ann Rev Earth Planet Sci* 5, 35–64, 1977.)

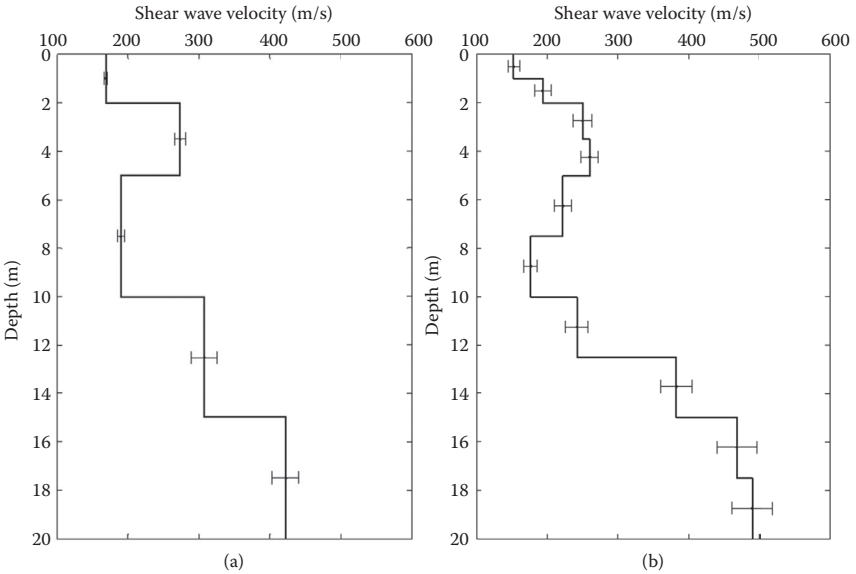

Figure 6.21 Expected value and standard deviation of the shear wave velocity profile obtained using two geometric models with different number of layers with all other parameters being the same: (a) five-layer model; (b) ten-layer model.

curve was assumed to be contaminated with uncorrelated, Gaussian noise characterized by a coefficient of variation equal to 3%, independent from frequency. Figure 6.22 shows the experimental and theoretical dispersion curves (fundamental mode) at the final iteration as obtained from the inversion algorithm. They correspond to the shear wave velocity profiles illustrated in Figure 6.21. The theoretical dispersion curves of the five- and ten-layer model are almost identical as also the error misfit. The general trend of the corresponding shear wave velocity profiles of Figure 6.21 is similar, although there are some differences.

However, Figure 6.21 clearly shows that the uncertainty associated with the values of shear wave velocity calculated for the ten-layer model is greater than the corresponding uncertainty of the five-layer model. This is consistent with the expectations of Figure 6.20 that the greater the desired spatial resolution (i.e., the greater the number of layers), the greater the variance will be. As expected, although the model parameters of the shallowest layers are estimated with greater accuracy with respect to deeper layers in both models, in the shallower layers, the uncertainty of V_S in the ten-layer model is larger than the corresponding uncertainty of the five-layer model. Recently, Xia et al. (2010) conducted an interesting study concerned with the trade-off between model resolution and covariance, specifically devoted to surface wave inversion.

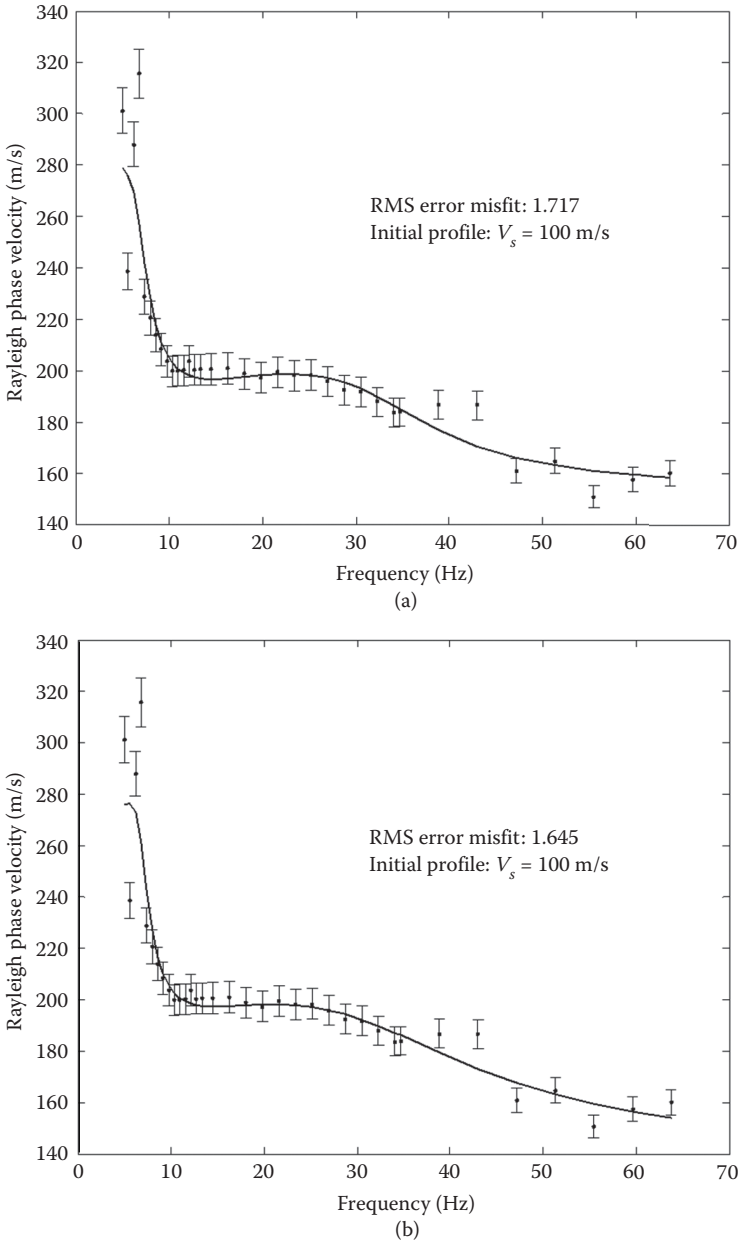

Figure 6.22 Experimental (with uncorrelated Gaussian noise) and theoretical dispersion curves (fundamental mode) at the final iteration of the inversion algorithm corresponding to the shear wave velocity profiles of Figure 6.21: (a) five-layer model; (b) ten-layer model.

6.5.5 Bayesian approach

This section briefly describes an alternative approach to the solution of inverse problems that is known in the literature as the Bayesian probability approach. The name comes from Thomas Bayes, an eighteenth-century British mathematician, who proposed a different interpretation of probability theory based on a theorem of conditional probability that he proposed. Although the theorem has a general validity, two different schools of statistics have originated from it. They are the *frequentist* (classical) and the *Bayesian* schools, and they differ in the interpretation given to the Bayes' theorem. The use of Bayesian statistics in various fields of science and engineering has been growing, particularly in recent years. This is not exempt from controversies raised by the followers of the frequentist school who claim that Bayesian probability introduces a degree of subjectivity and arbitrariness in the interpretation of the postulates of probability theory. However, it has been shown that when Bayesian probability theory is applied to inverse problems, it often leads to similar solutions (Aster et al. 2005). The monograph by Tarantola (2005) is perhaps the most comprehensive and rigorous work currently available on the application of Bayesian probability theory to the solution of parameter estimation and inverse problems.

In using the Bayesian approach to solve an inverse problem, the model parameters are assumed as random variables (more precisely, multivariate random vectors). On the contrary, in the classical approach, model parameters are viewed as coefficients to be determined from the inversion algorithm. Possible errors in the data are accounted for by projecting them into uncertainty of the model parameters. In the Bayesian approach, the solution to a parameter estimation problem consists of determining the probability distribution of the model parameters. This is a radically different method, and it constitutes an essential feature of the Bayesian approach. Its practical implementation requires an *a priori* definition of the probabilistic distribution of model parameters; this is controversial because this choice is subjective, and selecting different distributions will in general result in obtaining different outcomes.

In the Bayesian approach, the result of an inversion is the *a posteriori* probabilistic distribution of model parameters. If no information is available, then the "white" or uniform probability distribution is adopted where any value of a model parameter is equally likely, perhaps within a certain prescribed range (if known a priori). In the Bayesian vocabulary, this distribution is also known as *uninformative*, although some researchers find this terminology controversial (Tarantola 2005). Alternative probability distributions can be adopted if this is justifiable on the basis of availability of a priori information. In this sense, the Bayesian approach allows us to naturally incorporate a priori information about the model parameters (Aster et al. 2005). The maximum entropy method (Rietch 1977) can be used to

properly define the a priori distribution of model parameters. This method is based on the assumption that the "best" distribution is the one with the largest information (or entropy) content defined according to the so-called *Shannon's measure* (Tarantola 2005).

Bayesian theory allows computing the a posteriori probabilistic distribution of model parameters from knowing the a priori distribution and the experimental data. From the a posteriori distribution, one can then compute standard moments such as expected value and variance. In the most general situation, the implementation of the Bayesian approach in solving an inverse problem is computationally demanding because it requires the numerical evaluation of integrals with large dimensions. It can be shown that if the Bayesian approach is used to solve a nonlinear inverse problem with Gaussian, uncorrelated data errors associated with an a priori uninformative distribution of model parameters $\mathbf{m}$, the a posteriori largest value of $\mathbf{m}$ corresponds to the nonlinear classical least-squares solution of the problem (Aster et al. 2005).

Applications of Bayesian statistics to the inversion of surface wave data have been recently proposed by Bodin et al. (2012) and Shen et al. (2012).